Recursive Leverage in Yield-Bearing Stablecoins: Proof-of-Concept and Empirical Analysis

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Abstract. This project empirically studies looping strategies among large holders of yield-bearing stablecoins on Ethereum mainnet. Leverage means using borrowed funds to increase exposure beyond one's own capital. Looping is a specific form of leverage in which a holder deposits yield-bearing collateral, borrows against it, purchases more of the same collateral, and repeats; this strategy is also called recursive leverage. We develop a proof-of-concept bot demonstrating looping mechanics on Morpho Blue, build a data pipeline to identify looper addresses across Aave v3 and Morpho Blue, and analyse strategy archetypes using machine learning. We find that lending against yield-bearing stablecoins is concentrated among a few large holders who run leverage close to the liquidation threshold: most holders never deploy their tokens as collateral, but among those who do, 13.1% of Aave positions in the critical and high-risk health-factor bands account for 97.0% of the value held in those bands. Clustering recovers four recurring management modes rather than one strategy per wallet, leverage is adjusted on only 7.2% of looper-days and is best predicted by recent LTV volatility, and stressed states are predicted almost entirely by yield and borrowing-rate conditions rather than by leverage itself. No liquidation occurred in the sample, yet a 3 to 5% decline in the sUSDe price would raise liquidatable Aave collateral from $40M to over $510M. Our contributions include a working implementation of flashloan-based and iterative looping strategies, a comprehensive dataset of looper activity, and insights into risk management behaviours that inform DeFi protocol design and systemic risk assessment.

Keywords: DeFi · Yield-Bearing Tokens · Recursive Leverage · Ethereum · Machine Learning

1 Introduction

Yield-bearing tokens (YBTs) are ERC-20 tokens whose value accrues over time relative to an underlying asset through staking, lending, or vault returns. Unlike static stablecoins such as USDC or USDT, they compound yield into the token's

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exchange rate while remaining pledgeable as collateral. The ERC-4626 standard provides a uniform interface for these tokenised vaults, defining methods for deposit, withdrawal, and share conversion. As of Q1 2026, major USD-denominated YBTs include sUSDe (Ethena, approximately $1.4B market cap), sUSDS (Sky, approximately $4.7B), and syrupUSDC (Maple Finance, approximately $1B), generating yield through staking rewards, savings rate mechanisms, or institutional lending, typically at 4-8% APY.

Looping, also termed recursive leverage, exploits the dual property of a YBT serving as both a yield-bearing asset and pledgeable collateral: a holder deposits a YBT as collateral, borrows a stablecoin against it, acquires further exposure to the same token, and repeats the cycle, thereby amplifying effective carry at the cost of proportionally higher liquidation risk. Its profitability depends on the carry spread, the difference between the YBT yield and the borrowing cost. This paper studies empirically how large holders construct and manage such positions across Ethereum mainnet lending protocols.

1.1 Motivation

Existing empirical work on DeFi leverage stops short of characterising the behaviour this paper studies. Heimbach and Huang find wallet-level leverage typically ranging from 1.4 to 1.9, with the largest borrowers running higher leverage [3]. Sun et al. study 550 "mega-whales" across five lending protocols and recover four behavioural archetypes via wallet-level clustering, including single-protocol leveraged position managers [9]. Saengchote documents that the top 100 addresses on Compound hold 75% of deposits and 78% of loans [8]. Bartoletti et al. and Gogol et al. systematise DeFi lending mechanics [1, 2], while Perez et al. and Qin et al. study the resulting liquidation mechanisms [5, 6].

This work shows large holders are disproportionately leveraged and behave heterogeneously, but leaves two gaps. It aggregates across collateral types, missing that a yield-bearing token's exchange rate compounds upward rather than holding a fixed peg, so the relevant downside risk differs. And existing typologies assign one archetype per wallet over the full sample, which cannot express strategy switching or how a position is built and unwound through looping's deposit-borrow-redeposit cycle; that requires clustering at the position-day level.

This matters because systemic risk depends on how leverage is built and rebalanced, not just its level; protocol calibration rests on collateral assumptions a compounding token violates; and monitoring requires identifying positions that, as this paper shows, sit inside the protocol and are largely invisible in wallet balances. This paper fills that gap using on-chain data from Aave v3 and Morpho Blue to characterise looping among large holders of sUSDe, sUSDS, and syrupUSDC.

We restrict the study to these three tokens because they are among the largest USD-denominated yield-bearing stablecoins by market capitalisation, generate yield through structurally different mechanisms, staking-derived carry, a savings rate, and institutional lending respectively, and each has sufficient liquidity on

both protocols to identify loopers reliably and support the cross-protocol comparison central to this paper.

1.2 Problem Statement

The primary contribution is an empirical characterisation of strategy archetypes among large YBT holders. A secondary contribution is testing whether on-chain risk indicators can predict short-horizon risk rebalancing and stressed risk states.

1.3 Research Questions

- **RQ1:** What recurring leverage and looping strategy archetypes can be identified among large holders of major yield-bearing stablecoin tokens on Ethereum mainnet?
- **RQ2:** Which observable risk-management characteristics (leverage depth, collateral buffers, and rebalancing behaviour) differentiate these archetypes?
- **RQ3:** To what extent can on-chain state variables predict short-horizon risk rebalancing and stressed risk states?

1.4 Contributions

This work makes the following contributions.

- A proof-of-concept bot demonstrating looping on Morpho Blue, implemented both with and without flashloans, which attains 10× leverage from 10 USDC of own capital in a single flashloan-funded transaction.
- A data collection pipeline identifying large YBT holders and their on-chain activity across Aave v3 and Morpho Blue, yielding a panel of 23 looper addresses across sUSDe, sUSDS, and syrupUSDC over the period from March 2024 to June 2026.
- Evidence that holders allocate yield-bearing tokens in a close to binary manner, either retaining the token in the wallet or committing it fully to a lending protocol: wallet-only holders constitute 81 to 97% of each token's base, and mixed allocation remains below 1% for sUSDS and syrupUSDC. Value weighting sharpens the picture, as the Aave-only group rises from 14.8% of sUSDe holders to 44% of sUSDe value.
- A demonstration that deployed collateral is invisible in wallet balances: a looped position resides inside the protocol, so an active looper's wallet reads near zero, which motivates the behavioural identification method used in this work rather than identification by balance.
- Evidence that borrowers lever close to the protocol ceiling and that the resulting risk is concentrated: Morpho borrowers cluster at 5× and above across all three tokens, whereas most Aave borrowers sit below 1.1×, yet 90.5% of stablecoin-collateralised sUSDe value on Aave ($531.1M of $587.1M) still occupies the critical or high-risk health-factor bands. Within those bands, whales account for 22.0% of Morpho positions but 93.1% of their value, and

13.1% of Aave positions but 97.0% of their value. The pattern holds under two markedly different lending designs and therefore appears structural rather than a product of either protocol's mechanics.
- Evidence that flashloan usage separates the two protocols: on Aave, 18.5% of sUSDe whales have used a flashloan, against 4.16% of the holder base, whereas on Morpho none of the 41 whales ever has.
- A clustering analysis recovering four recurring management modes rather than one strategy per wallet: stale low leverage (1,148 looper-days), active looping (512), maximum-leverage hold (305), and outlier transitions (104). The modes separate on leverage level and rebalancing cadence, with levered carry a secondary signal. Twelve of the 23 holders spend at least 80% of their days in a single mode, while the remainder switch between modes, so strategy is better described as movement between shared modes than as a fixed wallet type.
- A decomposition of position activity showing that risk management must be separated from capital movement: activity occurs on 14.8% of looper-days, but leverage changes on only 7.2%, the remainder consisting of capital flows and minor activity. This distinction is what renders the archetypes interpretable.
- Evidence that a stale position is not risky in itself: long intervals since the last leverage change arise in two opposite modes, one conservative and one running a thin buffer, so what warrants monitoring is a stale position held at high leverage rather than staleness alone.
- A predictive model showing that leverage change is a considered rather than reflexive response: risk rebalancing is predicted most strongly by short-window LTV volatility, with carry economics second and contemporaneous leverage contributing little (ROC-AUC 0.901, recall 0.81).
- A predictive model showing that stress tracks yield and rate conditions rather than leverage: the stressed-state model attains ROC-AUC 0.977 and recall 0.90 using only yield, rate, volatility, and price features, with the two components of the carry spread dominant. Stress is therefore better read as the position ceasing to compensate for its risk than as leverage becoming excessive, and 3 of 8 Morpho and 4 of 7 Aave markets sampled carry a negative spread.
- A stress test showing that the liquidation cliff was never triggered in the sample but remains steep: no liquidation was realised anywhere in the panel, the closest approach being a health factor of 0.998. A decline of 3 to 5% in the sUSDe price would nonetheless take cumulative liquidatable Aave collateral from $40M to over $510M, and Morpho syrupUSDC would liquidate over $130M within a 12 to 18% band. The exposure is best characterised as low-probability and high-severity rather than as absent.

1.5 Paper Organisation

Section 2 provides background on yield-bearing tokens and lending protocols. Section 3 describes the proof-of-concept leverage bot. Section 4 details the method-

ology for data collection and looper identification. Section 5 covers feature engineering. Section 6 profiles the full holder base of the three yield-bearing stablecoins. Section 7 presents the machine learning analysis, combining unsupervised strategy archetype clustering (Section 7.1) with supervised prediction of position activity and stress (Section 7.2). Section 8 discusses findings and limitations, and Section 9 concludes.

2 Background

This section introduces the lending protocols where yield-bearing stablecoins are used as collateral, and looping (also known as recursive leverage).

2.1 Lending Protocols

A lending protocol is a set of smart contracts allowing one party to deposit a crypto-asset and earn interest on it, and another to borrow against collateral they have pledged, without either party trusting the other. Because participants are pseudonymous and no legal recourse exists against a defaulting borrower, collateral substitutes for creditworthiness: every loan is overcollateralised. Solvency is maintained by revaluing collateral continuously through a price oracle and permitting anyone to close a position that falls below the required margin.

A borrower pledges collateral and draws a loan up to a fraction of its value, set by the loan-to-value (LTV) parameter. Interest accrues at a rate determined algorithmically from utilisation, the share of supplied liquidity currently borrowed. If collateral value falls, or debt grows through accrued interest, until the position breaches its liquidation threshold, a third party may repay part of the debt and receive collateral at a discount, restoring solvency and earning a fee for the service.

Aave v3 is a pooled protocol: depositors supply into shared liquidity pools and borrowers draw from them, so all borrowers of an asset face the same rate and all suppliers share the same risk. Risk parameters, and the decision to list an asset at all, are set by Aave governance. Since one borrower may pledge several collaterals against several debts, Aave expresses position safety through the health factor.

Morpho Blue uses isolated markets, each defined at creation by an immutable tuple: collateral token, loan token, oracle, interest rate model, and liquidation LTV (LLTV). Anyone may create a market without governance approval, and a shortfall in one cannot propagate to another. This containment permits higher LLTVs on closely correlated pairs than a pooled design could safely offer, which is what makes deep looping feasible on Morpho for a stablecoin borrowed against a yield-bearing stablecoin.

2.2 Looping (Recursive Leverage)

Looping is a specific form of leverage, also called recursive leverage, in which the holder deposits a yield-bearing token as collateral, borrows a stablecoin

Term	Definition
Overcollateralisation	Collateral must exceed the debt drawn against it.
Loan-to-value (LTV)	Maximum fraction of collateral value borrowable at loan opening.
Liquidation threshold	Fraction of collateral value at which a position becomes liquidatable.
Liquidation LTV (LLTV)	Morpho Blue's liquidation threshold, fixed at market creation.
Liquidation	A third party repays part of an unhealthy debt and takes collateral at a discount.
Health factor	Risk-adjusted collateral over debt. At or below 1 the position is liquidatable.
Oracle	The price feed used to value collateral and trigger liquidation.
Utilisation	Share of supplied liquidity currently borrowed; sets interest rates.
Carry spread	Collateral yield minus borrowing cost.
Flashloan	Loan borrowed and repaid within one transaction, requiring no collateral.

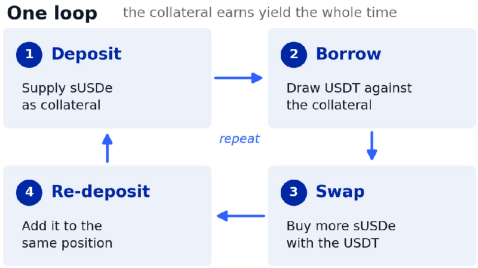


Fig. 1. One pass through the looping cycle. The collateral earns yield throughout.

against it, uses the borrowed stablecoin to purchase more of the same yield-bearing token, and re-deposits it as additional collateral. The collateral earns yield throughout, including on the portion financed by borrowed capital. As shown in Figure 1, one pass through the cycle consists of four steps: deposit the yield-bearing token as collateral, borrow a stablecoin against it, swap the borrowed stablecoin for more of the yield-bearing token, and re-deposit that token into the same position. This cycle can be repeated to amplify effective carry at the cost of proportionally higher liquidation risk.

Each pass increases exposure beyond the holder's own capital, and increases liquidation risk in proportion. Leverage is a property of the position rather than a count of loops: at a fixed LTV, leverage converges geometrically to $1/(1-\text{LTV})$ as the number of loops grows.

2.3 Leverage, Yield and Carry

The quantities below are used throughout the feature panel, the machine learning analysis, and the holder-panel results. Loan-to-value is $\mathrm{LTV} = \mathrm{borrow}_{\mathrm{USD}}/\mathrm{collateral}_{\mathrm{USD}}$.

Leverage. The leverage of an open position is

$$\mathrm{leverage} = \frac{1}{1-\mathrm{LTV}} = \frac{\mathrm{collateral}_{\mathrm{USD}}}{\mathrm{collateral}_{\mathrm{USD}} - \mathrm{borrow}_{\mathrm{USD}}},$$

and is set to 0 when there is no open debt. An LTV of 90% therefore corresponds to 10× leverage. We report this continuous quantity rather than a count of loops, because leverage is a property of the position, not of how it was assembled: a single flashloan can reach 10× in one transaction, whereas several manual loops may reach only 2×.

Liquidation LTV (LLTV) and health factor. Each market (Morpho) or reserve (Aave) has a liquidation LTV, `lltv`, the LTV ceiling at which the protocol is permitted to seize and sell the collateral. Distance to that ceiling is expressed as the health factor,

$$\texttt{health_factor} = \frac{\texttt{lltv}}{\mathrm{LTV}},$$

a value near 1 indicating a position is close to liquidation, and values well above 1 indicating a larger safety margin. On Morpho, `lltv` is a fixed, per-market parameter, so health factor can be computed directly from a position's own LTV. On Aave, collateral is shared across a single pool rather than isolated per market, so health factor instead reflects a wallet's full position across every asset it has supplied and borrowed, and is pulled directly from the protocol (`getUserAccountData`) rather than reconstructed from a single collateral–debt pair. **Liquidation** itself is the event in which the protocol forcibly sells a portion of a position's collateral to repay its debt once health factor falls to or below 1, restoring the position to a safer LTV at the borrower's expense.

Yield and borrow APY. Let `ybt_apy` be the annualised native yield on the collateral token (DeFiLlama) and `borrow_rate` the annualised borrow APY on the debt token (Dune / protocol rate series). A looper deposits YBS as collateral rather than as loan-token supply, so the protocol supply APY does not apply to the looped position and is excluded from the carry calculation.

Carry spread. The carry spread is the difference between the yield earned on collateral and the cost of borrowing,

$$\texttt{carry_spread} = \texttt{ybt_apy} - \texttt{borrow_rate}.$$

A positive spread means the looped position earns more than it costs to fund; a negative spread means it is losing carry. Scaled by leverage,

$$\texttt{carry_spread_levered} = \texttt{carry_spread} \times \texttt{leverage}.$$

Table 1. Exposure per $100 of own capital, at 80% LTV.

	Own capital	Borrowed	Total exposure
Loop 0	$100	$0	1.0×
Loop 1	$100	$80	1.8×
Loop 2	$100	$140	2.4×
Loop 3	$100	$180	3.0×

The profitability of looping depends on the carry spread: the difference between the YBT yield and the borrowing cost. A positive spread means the position earns more than it pays in interest. A negative spread means interest cost exceeds yield, so looping reduces net return.

2.4 Related Work

Prior work on DeFi lending has focused on liquidations rather than on how leverage is built. Qin et al. [6] study liquidations across Aave, Compound, MakerDAO and dYdX, showing that fixed-spread designs sell excessive discounted collateral at the borrower's expense, and Perez et al. [5] document on Compound how positions cluster near the liquidation boundary and liquidate in concentrated bursts. On the structural side, Bartoletti, Chiang and Lluch-Lafuente [1] formalise lending-pool mechanics, while Gogol et al. [2] and Rüetschi et al. [7] place lending protocols within broader taxonomies of DeFi protocol categories and their associated risks.

Closest to this work, Heimbach and Huang [3] measure wallet-level leverage on major lending platforms and find overall values of 1.4 to 1.9, with the largest and most active borrowers consistently the most levered; Saengchote [8] shows that Compound borrowers use debt mainly to fund leveraged investment strategies. Sun, Stasinakis and Sermpinis [9] are the closest methodological parallel, clustering large holders across several lending protocols into behavioural archetypes. We differ in clustering looper–days rather than wallets, so that a holder can occupy different modes over time, and in using HDBSCAN rather than k-means, which labels transitional days as noise instead of forcing every day into a mode.

Looping in yield-bearing stablecoins remains empirically underexplored despite the segment's rapid growth. No existing study identifies looper addresses by behaviour, reconstructs their leverage path day by day, or asks which observable characteristics separate one looping strategy from another.

3 Proof-of-Concept

This section describes the proof-of-concept bot developed to demonstrate and validate looping mechanics on a real blockchain. The bot operates on Base mainnet using the yoUSD/USDC market on Morpho Blue.

3.1 Design Goals

The purpose of the proof-of-concept is to demonstrate looping mechanics on a live chain before analysing looper behaviour on Ethereum mainnet. The bot validates the looping process end-to-end: depositing capital, supplying collateral, borrowing against it, and repeating the cycle to amplify exposure.

We deploy on Base rather than Ethereum mainnet to minimise gas costs during development and testing. Base is an Ethereum Layer-2 with significantly lower transaction fees while maintaining compatibility with mainnet contracts. The empirical analysis in later sections uses Ethereum mainnet data due to higher capital concentration, deeper liquidity, and longer transaction histories.

We selected yoUSD as the yield-bearing token because it is fully permissionless (no whitelist), implements the ERC-4626 vault standard for simple deposit and redemption, offers approximately 8% native APY paid in USDC, and has a confirmed Morpho Blue market on Base with 91.5% liquidation LTV.

Ownership of the deployed contract is transferred to a 1-of-2 Gnosis Safe multisig wallet. This separation of deployment signer and operational signer provides higher security and allows the use of Gnosis Safe's transaction builder for complex operations.

3.2 Strategy 1: Flashloan-Based Leverage

The first strategy uses a Morpho flashloan to build the entire leveraged position atomically in a single transaction. This approach is implemented in the `LeverageBot.sol` smart contract.

The execution flow proceeds as follows:

1. The owner approves 10 USDC and calls `execute()`.
2. The contract pulls 10 USDC from the owner, then requests a 90 USDC flashloan from Morpho.
3. Morpho calls back `onMorphoFlashLoan()`; the contract now holds 100 USDC.
4. The contract deposits all 100 USDC into the yoUSD vault, receiving approximately 93 yoUSD shares.
5. The contract supplies yoUSD as collateral on Morpho in the yoUSD/USDC market.
6. The contract borrows 90 USDC against the yoUSD collateral.
7. The contract repays the 90 USDC flashloan; the transaction completes.

The result is 100 USDC worth of yoUSD collateral with 90 USDC of debt, funded with only 10 USDC of the owner's capital. This represents 10× leverage.

The `unwind()` function reverses this process atomically: it flashloans enough USDC to repay the debt, withdraws all yoUSD collateral, redeems yoUSD back to USDC, repays the flashloan, and sweeps any remaining USDC (original capital plus accrued yield) to the owner.

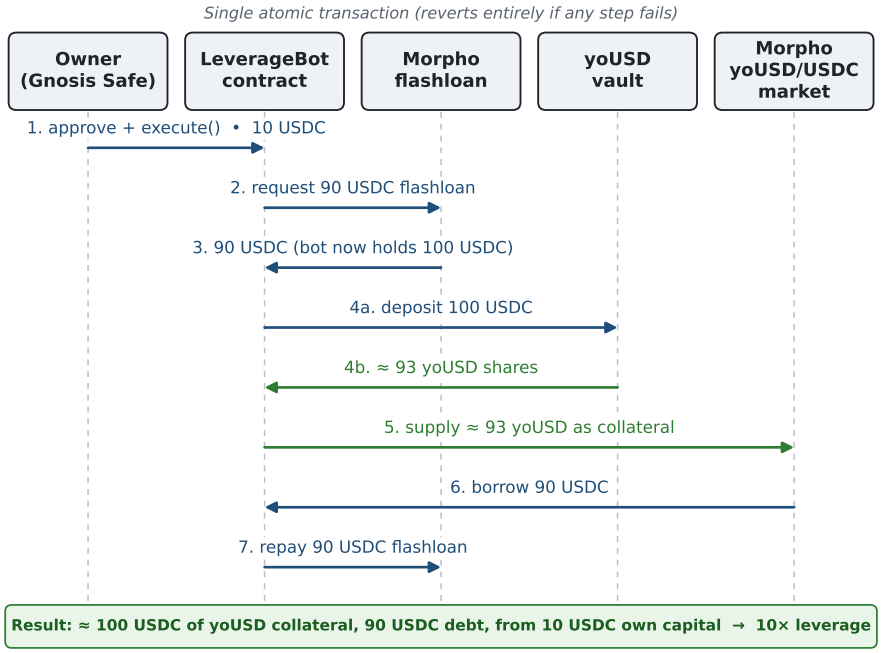


Fig. 2. Execution flow of the flashloan-based leverage strategy. An interactive version of this fund-flow graph is available on Tenderly at `https://dashboard.tenderly.co/public/safe/safe-apps/simulator/7fb53629-e928-45d1-830b-9dc967b76ccc/fund-flow`.

3.3 Strategy 2: Iterative Looping

The second strategy requires no smart contract. It uses a JavaScript script that loops step-by-step across multiple transactions:

1. Deposit USDC into yoUSD vault, receive yoUSD shares.
2. Supply yoUSD as collateral on Morpho.
3. Borrow 88% of the collateral value back as USDC.
4. Repeat with the borrowed USDC.

The script executes three loops by default, building up leverage incrementally. If any loop fails, the script automatically triggers `unwind.js` to clean up the partial position.

This approach is slower (multiple transactions incur multiple gas fees) but simpler to understand and debug. It also demonstrates the manual process that many looper addresses use, as observed in the empirical data.

3.4 Security Considerations

Several security measures are implemented:

Ownership model. The contract uses an `onlyOwner` modifier for privileged functions (`execute`, `unwind`, `rescueToken`). Ownership is transferred to a Gnosis Safe after deployment.

Flashloan callback guard. The `onMorphoFlashLoan` callback is protected by an `onlyMorpho` modifier, ensuring only Morpho can invoke it.

Rescue function. A `rescueToken` function allows the owner to recover any tokens accidentally sent to the contract.

Atomic execution. The flashloan strategy executes entirely within a single transaction. If any step fails, the entire transaction reverts, leaving no partial state.

3.5 Deployment and Testing

The contract is deployed on Base mainnet at address `0xa3B27a...Ecd10d7`, with ownership transferred to a 1-of-2 Gnosis Safe at `0xf9962B...857fCb`. Full addresses are recorded in the deployment artefacts.

Operational workflow. To execute the leverage strategy via the Safe, the holder uses the Gnosis Safe Transaction Builder:

1. Create a new transaction, paste the USDC contract address on Base, and let the interface auto-generate the ABI.
2. Call `approve()` with the bot address as spender and 10 USDC as the amount. This is a one-time approval.
3. Create a second transaction targeting the bot contract and call `execute()`.
4. Optionally simulate the transaction via Tenderly to preview the fund flow before signing.
5. Sign and execute the transaction through the Safe.

The Tenderly simulator provides a visual flowchart animation showing the complete execution path: USDC transfer, flashloan request, yoUSD deposit, collateral supply, borrow, and flashloan repayment (Figure 2).

Testing. Tests run against a real Base fork using Hardhat, with no mocks or stubs. The test suite covers:

- Execution flow and position state verification
- Ownership transfer and access control
- Unwind functionality and USDC recovery
- Flashloan callback guards
- Full lifecycle via Safe-impersonated transactions

All contract addresses are recorded in `deployments/LeverageBot_base_latest.json`. The full contract code is provided in Appendix B.

4 Methodology

We analyse three YBTs, namely sUSDe, sUSDS, and syrupUSDC, on Ethereum mainnet over 2024-03-29 to 2026-06-27. This section describes the data collection pipeline and the methodology for identification of *loopers* and their YBS strategies across lending protocols.

4.1 Whales and Loopers

The paper studies two distinct populations, and because both are commonly lumped together as "large holders" we name and define them separately before any results are reported.

A *whale* is generally understood as an individual, group, or organization holding a massive amount of a specific cryptocurrency, large enough to influence market prices and liquidity within a single trade. For Bitcoin specifically, a holder of 1,000 BTC or more is typically considered a whale [4]. We use the term in exactly this sense, as a statement about *balance*: a whale is an address holding at least 1,000,000 YBS tokens, which because all three tokens are USD-denominated is worth approximately $1,000,000. Balances are reconstructed across the wallet itself, Morpho Blue and Aave v3, so that collateral posted to a lending protocol still counts towards the holding. This is the definition applied to the full holder panel of Section 6.

A *looper* is an address that actively runs the recursive leverage strategy, identified by *behaviour* rather than by balance: it supplies the YBS as collateral and repeatedly borrows against it, or performs the equivalent atomically through flashloans. Section 4.5 states the exact on-chain heuristics used to flag an address as a looper or a flashloaner. The detailed analysis of this paper is run on the *selected loopers*, the subset of flagged addresses whose lifetime cumulative YBS inflow (`total_received`) exceeds $500,000 and which survive the manual review of Section 4.6; they are identified as L-1 to L-23 throughout. A volume-based size criterion is used here rather than a balance one because looped collateral sits inside the lending protocol rather than in the wallet, so the current wallet balance of an active looper is close to zero and a balance test would simply miss these addresses.

Table 2. The two holder populations used in this paper. The terms are not interchangeable: whale is a statement about how much an address holds, looper is a statement about what an address does.

Term	Defined by	Criterion	Used in
Whale	Balance	$\geq$ 1,000,000 YBS tokens ($\approx$ $1M) held across wallet, Morpho Blue and Aave v3	Section 6
Looper	Behaviour	Supplies YBS as collateral and borrows against it repeatedly, or loops atomically via flashloans	Sections 6, 4.5
Selected looper	Behaviour and manual review	Looper with lifetime YBS inflow above $500k, individually vetted; the 23 addresses L-1 to L-23	Sections 4.6–7.2

The two groups overlap without coinciding. Most whales are not loopers: as Section 6 shows, a large share of whale positions, particularly on Aave v3, is passive collateral that never draws debt. Nor is every looper a whale, since an

address can run a deeply levered position off a starting balance well below $1M. Whenever a number in this paper is attached to one of the two words, it refers only to the population defined above under that name.

4.2 Token Selection

Three USD-denominated yield-bearing tokens were selected based on the following criteria: top candidates by Ethereum mainnet circulating value and TVL, minimum on-chain liquidity threshold, minimum active holder count, active integration as collateral on at least one major lending protocol, and activity throughout the sample period.

Table 3. Selected yield-bearing tokens. Average APY is the typical recent range; full history is shown in Figure 3.

Token	Issuer	Inception	Market Cap	Average APY
sUSDe	Ethena	early 2024	$1.4B	4–8%
sUSDS	Sky	Sep 2024	$4.7B	4–6%
syrupUSDC	Maple Finance	May 2024	$1B	5–8%

These tokens represent the largest USD-denominated yield-bearing stablecoins on Ethereum mainnet with active lending protocol integrations.

Figure 3 shows the full APY history of the three tokens. As summarised in Table 3, the tokens launched at different times (sUSDe in early 2024, sUSDS in September 2024, and syrupUSDC in May 2024), but all three run to the present. sUSDe carried by far the highest and most volatile yield, peaking near 55% in early 2024 before falling to 4–5%, whereas sUSDS and syrupUSDC have tracked a steadier 4–9% since launch. Because a looped position only earns a positive carry when the token yield exceeds the borrowing cost, this generally declining yield backdrop is the context in which the stress results of Section 7.2 should be read.

4.3 Protocol Coverage

Leverage analysis covers two lending protocols where these YBTs are used as collateral: Morpho Blue and Aave v3. The collateral set is sUSDe, sUSDS, and syrupUSDC. Position state is reconstructed from four events only: supply, withdraw, borrow, and repay. Table 5 summarises, for each collateral token, which protocols are used and which stablecoins are borrowed in the sample.

Table 2 in Section 4.3 maps collateral tokens to protocols and borrowed stablecoins, but the two protocols do not expose the same underlying event set. Table 4 makes this mapping explicit, since our pipeline (Section 4.4) draws only on a subset of the events each protocol emits.

On Morpho, collateral and plain lending are separate event types; on Aave both use `Supply`/`Withdraw`, with collateral status set by a separate flag.

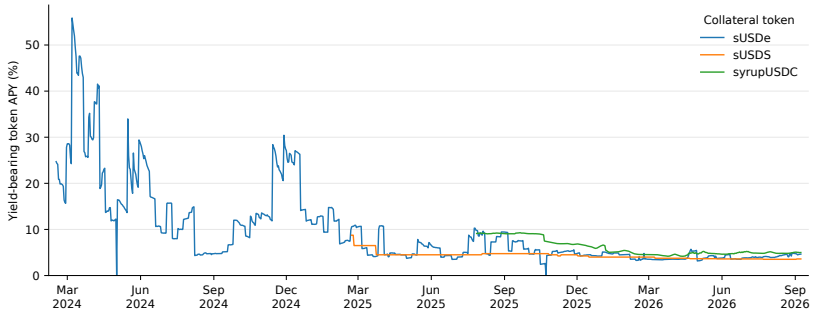


Fig. 3. Historical APY of the three yield-bearing collateral tokens used for looping (sUSDe, sUSDS, and syrupUSDC) on Ethereum mainnet (daily, sourced from DeFiLlama). sUSDe is the most volatile, peaking near 55% in early 2024; sUSDS and syrupUSDC launched later and have tracked roughly 4–9% since.

Table 4. Event mapping between Morpho Blue and Aave v3. Bold rows mark events used in our pipeline (Section 4.4).

Concept	**Morpho Blue**	**Aave v3**
Lender deposit	`Supply` / `Withdraw`	`Supply` / `Withdraw` (same event as collateral)
Collateral	`SupplyCollateral` `WithdrawCollateral`	`Supply` / `Withdraw` (if flagged as collateral)
Borrow / Repay	`Borrow` / `Repay`	`Borrow` / `Repay`
Collateral flag toggle	*n/a*	`ReserveUsedAsCollateral-Enabled/Disabled`
Liquidation	`Liquidate`	`LiquidationCall`

Table 5. Protocol coverage by collateral token. Each non-empty cell lists the stablecoins borrowed against that collateral on the given protocol in our sample.

Collateral	**Aave v3**	**Morpho Blue**
sUSDe	USDT, USDC, USDe	DAI, USDtb, PYUSD
sUSDS	–	USDT
syrupUSDC	–	PYUSD, USDC

4.4 Data Collection

We collect metadata, prices, and protocol events for the three tokens under study. DeFiLlama provides daily TVL and APY series; prices used in the panel are likewise daily. Protocol activity is taken from Ethereum event logs indexed by Dune. Specifically, four event types are extracted: supply, withdraw, borrow, and repay. On Aave v3 these are drawn from the Pool contract (`Pool_evt_Supply`, `Pool_evt_Withdraw`, `Pool_evt_Borrow`, `Pool_evt_Repay`) for sUSDe. On Morpho Blue they are drawn from the MorphoBlue contract (`MorphoBlue_evt_SupplyCollateral`, `WithdrawCollateral`, `Borrow`, `Repay`) for sUSDe, sUSDS, and syrupUSDC. A supply increases the collateral balance; a withdraw decreases it. A borrow opens or increases debt; a repay reduces debt. These event flows are subsequently aggregated into daily position balances.

4.5 Looper Identification in Detailed Analysis

The looper identification pipeline finds the top addresses per token per protocol that have actively engaged in looping or flashloan strategies, ranked by lifetime token volume. **Pipeline:**

1. Per token per protocol, joining ERC-20 Transfer events with protocol-specific supply, borrow, and flashloan events. Ranking uses lifetime cumulative inflow (`total_received`), not a daily timeseries; protocol withdraws are not needed for the looper flag (that flag only requires at least one supply and at least three borrows). Wallet withdrawals appear as outbound ERC-20 transfers and reduce `current_balance`, but they do not change the lifetime-volume rank.
2. Flag each address as EOA or smart contract via `eth_getCode` RPC check.
3. Output ranked looper lists per token as CSV.

Classification logic. The core of step 1 is a per-address classification query. In plain terms, it works as follows:

```
FOR each token T, protocol P:

    transfers = ERC-20 Transfer events WHERE contract = T

    -- Morpho: markets are (collateral, loan) pairs, so filter by market
    -- Aave:   flags are protocol-level; reserve-level attribution
    --         comes from the manual review in Section 4.5
    supplies   = supply events on P   [Morpho: WHERE market.collateral = T]
    borrows    = borrow events on P   [Morpho: WHERE market.collateral = T]
    flashloans = flashloan events on P (any asset)

    FOR each address A:
        supply_count    = COUNT(DISTINCT tx WHERE supplier = A)
        borrow_count    = COUNT(DISTINCT tx WHERE borrower = A)
```

```
        flashloan_count = COUNT(DISTINCT tx WHERE initiator = A)

        IF supply_count >= 1 AND borrow_count >= 3:
            mark A as "looper"
        IF flashloan_count >= 3:
            mark A as "flashloaner"

    candidates = loopers UNION flashloaners

    FOR each candidate A:
        total_received  = SUM(inbound transfers of T to A)
        current_balance = SUM(inbound) - SUM(outbound)

    EXCLUDE the zero address
    EXCLUDE addresses labelled dex / infrastructure / dao /
            institution / cex users

    RANK candidates BY total_received DESCENDING
    OUTPUT top 50 as CSV
```

This is simplified pseudocode; the full Dune SQL implementing this logic is provided in Appendix C.

Strategy definitions:

- **Looper**: supplied the token as collateral at least once and borrowed any asset at least 3 times on the same protocol.
- **Flashloaner**: executed flashloans in at least 3 separate transactions on the respective protocol.

Fields collected per looper:

- `rank`: position by total received volume
- `address`: wallet address
- `current_balance`: current token holdings
- `total_received`: lifetime cumulative inflow
- `distinct_funders`: number of unique addresses that sent tokens to this wallet
- `is_eoa`: True if externally owned account, False if smart contract
- `is_looper`, `is_flashloaner`: strategy flags
- `strategy_type`: looper, flashloaner, or both
- `protocol`: Aave, Morpho

Infrastructure contracts such as the Morpho ETH Bundler are excluded as they aggregate transactions from many end users rather than representing independent strategy actors.

4.6 Selected Loopers

The heuristics of Section 4.5 return a ranked candidate list per token and protocol, but not every flagged address is a usable unit of analysis. What the detailed analysis needs is a single actor whose leverage decisions can be followed day by day, because the archetypes of Section 7.1 and the models of Section 7.2 both interpret a change in LTV as a deliberate choice by one decision maker. Each candidate was therefore reviewed manually on Etherscan and kept only if it met all of the following criteria.

- **One identifiable holder.** The address must act for a single decision maker, an EOA or a Gnosis Safe, rather than a vault, bundler or router that pools many end users. A pooled address reports the average of many unrelated strategies, so its LTV path would not describe anyone's risk management.
- **Enough history to form a panel.** The position must span enough days with an open, actively looped position to estimate the rolling volatility and rebalancing-cadence features of Section 5.
- **One uncontaminated strategy.** The collateral must be the YBS itself rather than a derivative claim on it, and the address must not be running a second, unrelated strategy through the same wallet. Pendle PT looping is the main exclusion: its carry is fixed at entry and its panel breaks at each maturity, so it is not comparable to floating-rate YBS looping (Section 8).
- **Material scale.** Lifetime cumulative YBS inflow above $500,000, the threshold motivated in Section 4.1.

This leaves 23 addresses, 16 on Morpho Blue and 7 on Aave v3, listed in Table 6. Candidates were rejected when their ERC-20 transfer history revealed multi-user aggregation or mixed strategies, including the two PT loopers documented in Section 8.

The result is a purposive rather than a representative sample: these are the addresses on which a clean per-actor panel can be built, not an unbiased draw from the population of everyone who loops. Population-level questions, such as how much of each market is levered and how close to liquidation it sits, are answered instead by the full holder panel of Section 6, which imposes none of these criteria.

4.7 Retrieval of Loopers' Raw Events

For each looper market pair, the four position-changing events (supply, withdraw, borrow, repay) are pulled from the relevant protocol contract, tagged with an `event_type` label, stacked into a single table, and joined once against the token decimals table to convert raw amounts into human-readable units. Complete SQL is reproduced in Appendix C

```
FOR each wallet W, protocol P:
    supplies  = supply events   WHERE user/onBehalfOf = W
    withdraws = withdraw events WHERE user/onBehalfOf = W
```

```
borrows   = borrow events   WHERE user/onBehalfOf = W
repays    = repay events    WHERE user/onBehalfOf = W

events = STACK(supplies, withdraws, borrows, repays)
         WITH event_type LABEL attached to each row

JOIN events WITH token_decimals ON reserve/contract_address
amount_normalized = raw_amount / 10^decimals

SORT events BY timestamp ASCENDING
OUTPUT as raw event timeseries
```

Table 6. Selected loopers for analysis. Addresses are abbreviated and link to Etherscan; full addresses are listed in Appendix A. Lifetime volume is the total received amount used for ranking (Section 4.5); it reflects cumulative token inflow, including amounts cycled through repeated looping, and should not be read as a snapshot of position size. Active days count the days on which the looper held an open, actively looped position.

ID	Address	Protocol	Collateral	Borrow	Lifetime Vol. (USD)	Active Days	Looping Style
L-1	0x080c...fb0	Morpho	sUSDe	USDtb	5,396,757	77	Mixed
L-2	0x97bb...21e	Morpho	sUSDS	USDT	92,034,842	124	Mixed
L-3	0xcb1e...e58	Morpho	sUSDS	USDT	7,082,220	94	Frequent
L-4	0x0d4f...06b	Morpho	sUSDe	PYUSD	73,649,250	26	Heavy
L-5	0x2438...717	Morpho	syrupUSDC	PYUSD	125,818,431	34	Mixed
L-6	0xfafc...e13	Morpho	sUSDS	USDT	1,984,275	38	Mixed
L-7	0x32f0...7ed	Morpho	sUSDe	PYUSD	6,470,158	58	Frequent
L-8	0x218b...0aa	Morpho	sUSDe	DAI	4,095,928	19	Heavy
L-9	0x309e...46c	Morpho	sUSDe	DAI	5,161,023	23	Heavy
L-10	0x3363...7d6	Morpho	sUSDe	DAI	7,069,764	253	Heavy
L-11	0xb48a...df2	Morpho	sUSDe	DAI	31,883,101	336	Mixed
L-12	0x9bcb...495	Morpho	sUSDS	USDT	11,035,633	70	Heavy
L-13	0x2067...836	Morpho	sUSDS	USDT	11,911,970	88	Frequent
L-14	0xc058...a1c	Morpho	sUSDS	USDT	31,614,116	110	Frequent
L-15	0x59b6...773	Morpho	syrupUSDC	USDC	1,739,755	8	Mixed
L-16	0xf28d...442	Morpho	syrupUSDC	USDC	6,055,882	20	Frequent
L-17	0x0294...080	Aave v3	sUSDe	USDT	6,268,698	87	Frequent
L-18	0x5D87...836	Aave v3	sUSDe	USDC	656,927	98	Mixed
L-19	0x5D87...836	Aave v3	sUSDe	USDe	656,927	7	Mixed
L-20	0x8526...4db	Aave v3	sUSDe	USDe	866,491	37	Mixed
L-21	0x2420...1E3	Aave v3	sUSDe	USDe	1,446,812	6	Frequent
L-22	0x2327...34F	Aave v3	sUSDe	USDC	1,256,461	8	Heavy
L-23	0xF3Ad...566	Aave v3	sUSDe	USDC	770,106	77	Heavy

The sample covers all three tokens across two protocols. The Morpho loopers (L-1–L-16) span the sUSDe, sUSDS, and syrupUSDC markets, while the Aave v3 loopers (L-17–L-23) all run sUSDe E-Mode looping. L-18 and L-19 are the

same holder running two Aave markets (USDC and USDe borrows), and are tracked as separate looper–market positions.

Lifetime volume ranges from $0.66M for the smallest holder (L-18/L-19) to $125.8M for the largest (L-5), showing that the sample spans small holders as well as very large ones. Every selected address therefore clears the $500k lifetime volume threshold of Section 4.5. This figure counts every token inflow to the address, so for loopers that loop repeatedly it can be larger than the actual capital held at any single point in time, since the same funds pass through the wallet more than once. Current on-chain balances for these addresses are close to zero, since looped positions sit as collateral inside the lending protocol rather than in the wallet itself, so lifetime volume is reported here as the best available measure of scale rather than a snapshot of position size.

Loop frequency vs. loop size. Beyond identifying loopers, the sample also separates two distinct looping styles: loopers who loop *frequently* in small increments versus those who loop *infrequently* in large increments. For each looper-market pair, `borrow_count` and `avg_loop_size` are computed from the borrow event history. Across the 23 looper-market observations, the median borrow count is 6 and the median average loop size is $268,162 (both taken as the 12th value in each sorted list). Each looper is compared against these two medians and labeled *frequent* if borrow count exceeds the median while loop size does not, *heavy* if loop size exceeds the median while borrow count does not, and *mixed* otherwise, which covers both loopers that are high on both dimensions and loopers that are low on both.[1] This classification is reported alongside the looper table (Table 6) and is used in Section 7.2 to test whether looping style interacts with stress sensitivity.

4.8 Data Sources and Pipeline

The complete data pipeline uses the following sources: All scrapers are implemented in Python using the `dune-client` library for Dune Analytics queries, `web3.py` for RPC calls, and the `gql` library for GraphQL subgraph queries.

Table 7. Data sources.

Source	Data Provided
DeFiLlama	TVL and APY for each token
Dune Analytics	Historical prices, protocol events, looper identification via SQL queries
Ethereum RPC	EOA verification via `eth_getCode`
GraphQL	On-chain position data queried directly from protocol subgraphs

[1] The median loop size of $268,162 corresponds exactly to L-17's own average loop size, so L-17 sits on the size boundary itself. Ties are resolved in favor of "does not exceed the median," which is why L-17 is classified as frequent rather than heavy.

5 Feature Engineering

This section describes how the raw on-chain events are transformed into the per-looper, per-day feature set used in the machine learning analysis. The pipeline is implemented separately for Morpho Blue and Aave v3 because the two protocols expose different position and rate semantics, but both follow the same logical sequence: replay events into balances, value those balances in USD, derive risk and behavioural metrics, and finally attach yield and rate series to compute the carry spread. The unit of observation throughout is the *looper–day*: one row per selected looper per calendar day over which it holds an open or recently closed position.

5.1 Panel Construction

The first three stages convert raw events into a valued daily panel. The `collect_events` stage reads the Dune CSV exports, filters them to the supply, withdraw, borrow, and repay events belonging to each tracked looper, and normalises them into a single clean event table (`panel_raw_events.csv`). The `build_panel` stage then replays this event stream chronologically, cumulating supply and withdraw events into a running collateral balance and borrow and repay events into a running debt balance, emitting one row per looper per day (`daily_panel.csv`). Finally, `build_prices` fetches historical daily token prices from DeFiLlama and joins them onto the panel, converting the native-unit collateral and debt balances into USD values (`daily_panel_usd.csv`). Pricing both legs in USD is necessary because collateral and debt are denominated in different tokens, and risk metrics such as loan-to-value are only meaningful once both legs share a common unit.

In short, Dune provides raw event flows; the position balances are reconstructed in our feature-engineering pipeline, and the resulting position snapshots are daily. Event logs alone are not always a perfect current-balance source unless edge cases are modelled carefully (for example, interest accrual and protocol accounting index updates).

Gross flows versus net capital. On a looped position, gross deposit and withdrawal figures do not measure the holder's own capital, and the USD value withdrawn over a position's life can legitimately exceed the USD value deposited. Three mechanical effects cause this. First, most deposits are funded by borrowed stablecoins recycled into collateral, so gross flows scale with the levered position rather than with equity: L-1 supplied 8.41M sUSDe across 39 events against $1M of own capital. Second, the collateral token is yield-bearing, so identical token amounts are worth more on exit than at entry; over L-1's position the sUSDe price rose from 1.218 to 1.228, meaning the exact same 8.41M tokens it supplied and later withdrew carried about 0.8% more USD value on the way out. Third, debt repayments exceed borrows by the accrued interest (L-1 repaid 9.67M USDtb against 9.65M borrowed). Withdrawing more value than was deposited is therefore the normal signature of a profitable loop, not an accounting anomaly. For this reason the analysis measures capital by equity, $E = C - D$, and classifies behaviour from changes in equity and leverage rather than from gross flows.

5.2 Risk Features

The risk features characterise how aggressively each position is geared and how close it sits to liquidation on a given day. They are computed by `build_features` from the valued panel.

Loan-to-value. The loan-to-value ratio is the central risk measure,

$$\mathrm{LTV} = \frac{\mathrm{borrow}_{\mathrm{USD}}}{\mathrm{collateral}_{\mathrm{USD}}}.$$

Health factor. The health factor expresses the same relationship relative to the market's liquidation threshold ℓ, $\mathrm{HF} = (\mathrm{collateral}_{\mathrm{USD}} \cdot \ell)/\mathrm{borrow}_{\mathrm{USD}}$. A health factor below 1.0 indicates that the position is eligible for liquidation. The liquidation threshold is fixed per market from the protocol's on-chain configuration (Table 9).

Distance to liquidation. The distance to liquidation, $\ell - \mathrm{LTV}$, measures how many percentage points of LTV buffer remain before the position becomes liquidatable. A positive value indicates a safe position and a negative value indicates an underwater one. Compared to the health factor, this linear measure is more directly interpretable as a behavioural target, since holders manage their positions in LTV space.

Leverage. Position leverage is defined in Section 2.3 as $\mathrm{leverage} = 1/(1 - \mathrm{LTV})$, set to 0 when there is no open debt. The feature panel stores this continuous quantity for each looper–day.

5.3 Behavioural Features

The behavioural features capture how actively each holder manages its position over time, providing the temporal context required for the prediction task in RQ3.

Position activity. A binary flag (`position_activity`) marks each day on which any supply, withdraw, borrow, or repay event occurred for the looper, and is zero otherwise. An earlier version labelled every such day as rebalancing. That mixed two different behaviours moving capital in or out of a position, and managing risk by changing leverage so we renamed the broad flag to `position_activity` and reserve `risk_rebalance` for days on which leverage moved materially. `position_activity` therefore records contact with the position only: adding capital, moving capital between protocols, and adjusting risk are indistinguishable at this level. Active days are therefore classified further.

Separating capital movement from risk management. A levered position is described completely by its equity $E = C - D$ (net capital: collateral minus debt, both in USD) and its leverage $\ell = 1/(1 - \text{LTV})$ (collateral per unit of equity), since $C = E\ell$ and $D = E(\ell - 1)$. These are the two independent quantities a holder can change, and they form the axes of the classification. Collateral cannot serve as the size axis, because it moves mechanically whenever leverage moves.

Price movement is removed before the classification. Marking the previous day's token balances at the current day's prices gives `ltv_cf`, the position the holder would have held had they taken no action; the action is whatever differs from that counterfactual. Let Δ_ℓ denote the relative change in leverage ℓ against this counterfactual and Δ_E the net change in equity E, valued at current prices. A change is *material* if its relative magnitude is at least 2%; for equity, the absolute capital flow must additionally exceed $1,000. Otherwise the change is *flat*.

Thresholds are relative rather than absolute because equity across the sample spans $5,000 to $12 million, and because one percentage point of LTV is a 4% change in leverage at LTV = 0.50 but a 29% change at LTV = 0.965. They mark materiality rather than a noise floor: on days without events the measured action is exactly zero, since balances are reconstructed from events. Each active day falls into exactly one of four classes:

- `leverage_adjust`: material Δ_ℓ, flat Δ_E. Leverage changes at constant equity.
- `capital_flow`: material Δ_E, flat Δ_ℓ. Equity changes at constant leverage.
- `levered_scaling`: both material. Equity and leverage both change.
- `minor_activity`: both flat.

The distinction between the two leverage-moving classes matters because a leverage change can itself be caused by a capital movement: withdrawing collateral raises LTV, and injecting collateral lowers it, even when no deliberate risk decision is intended. `leverage_adjust` therefore requires that no material capital moved, so it isolates pure risk management: the holder reshapes the same equity into more or less borrowed exposure. L-1 on 10 February 2026 is a clean example: two atomic transactions each borrow USDtb and immediately supply the proceeds as sUSDe collateral, raising leverage by 14% while equity moves by about $1,300 on a $960,000 position. Any leverage change that is accompanied by a material capital flow is instead `levered_scaling`: when L-16 wound down its position in March 2026, it repeatedly injected $100,000–$800,000 of equity per day to step its LTV from 0.86 to zero, so those days move both axes and classify as scaling, not as pure adjustment.

The union of the two leverage classes is the risk-management signal used for archetype cadence (frequency and recency features) in RQ2 and as the target of Model A in RQ3,

$$\texttt{risk_rebalance} = \texttt{leverage_adjust} \vee \texttt{levered_scaling}.$$

The first observed day of a position has no counterfactual and is classified from what was opened.

Table 8 reports how often each class occurs in the 2,506-day panel. Counts are looper-days (a calendar day on which the flag is one for that looper), not raw on-chain transactions. Every looper records at least one `position_activity` day and at least one `risk_rebalance` day; pure leverage adjusts and capital flows are less universal. Note that full unwinds count as risk rebalances: on the day a position is closed, leverage and equity both move materially, so the day classifies as `levered_scaling`. Such terminal unwind days account for 28 of the 180 risk-rebalance days (about 16%), spread across 18 of the 23 loopers; for holders that never adjusted leverage within an open position, the open and close days can be their only risk-rebalance events.

Table 8. Activity taxonomy counts over the full panel (2,506 looper-days, 23 loopers). A flag equals one on a day with at least one qualifying event; *Days* is the number of such looper-days; *% of panel* is that count divided by 2,506; *Loopers* ≥ 1 is the number of distinct loopers with at least one such day. Counts are looper-days: several on-chain transactions on the same day count as one flagged day, never as multiple events. Consecutive flagged days are counted separately here; Table 16 additionally reports them merged into episodes. *no activity* is the complement of `position_activity`: days with no event at all. The four classes partition active days; `risk_rebalance` is the union of `leverage_adjust` and `levered_scaling`.

Class	Days	% of panel	Loopers ≥ 1
no activity	2,136	85.2	23
`position_activity`	370	14.8	23
`leverage_adjust`	42	1.7	12
`capital_flow`	50	2.0	16
`levered_scaling`	138	5.5	23
`minor_activity`	140	5.6	10
`risk_rebalance`	180	7.2	23

Activity frequency and recency. Rolling counts over trailing 7-day and 30-day windows summarise how busy a holder has been (`activity_freq_7d`, `activity_freq_30d`), how often it risk-rebalances (`risk_rebal_freq_7d`, `risk_rebal_freq_30d`), and how often it moves capital (`capital_flow_freq_7d`, `capital_flow_freq_30d`). The companion recency features `days_since_activity` and `days_since_risk_rebalance` record the gap since each signal last occurred within the open position. Together these distinguish holders that actively manage leverage from those that touch a position only to move capital in or out.

LTV volatility and lags. The rolling 7-day and 14-day standard deviation of LTV (`ltv_vol_7d`, `ltv_vol_14d`) captures how much the position's leverage fluctuates, which reflects both market movement and active management. To give the models short-horizon temporal context for predicting risk rebalances,

the LTV is also lagged by 1, 3, and 7 days (`ltv_lag_1`, `ltv_lag_3`, `ltv_lag_7`).

Token identity. The collateral and borrow tokens are one-hot encoded (`coll_*`, `debt_*`), allowing the models to condition on market identity, since liquidation thresholds and yield characteristics differ by token pair.

5.4 Yield and Carry Features

The profitability of a looping strategy depends on the carry spread between the yield earned on collateral and the cost of borrowing. Computing this spread requires three rate series, joined onto the feature table per looper by market and forward-filled within each looper–market group to cover days without a fresh observation. The `build_rates` script performs this join for both protocols.

Yield-bearing token APY. The annualised yield accruing to the collateral token (`ybt_apy`) is sourced from the corresponding DeFiLlama pool for sUSDe, sUSDS, and syrupUSDC.

Supply rate. A looper deposits YBS as collateral rather than as loan-token supply, so the protocol supply rate does not apply to their position. On Morpho Blue the market's `supplyApy` is paid to loan-token lenders, not to collateral depositors. On Aave v3 the supply rate for sUSDe collateral is effectively zero. We therefore exclude `supply_rate` from the feature set entirely: a constant-zero column has no predictive value, and including it would misrepresent the economics of a looped position.

Borrow rate. The annualised borrow rate paid on the debt token (`borrow_rate`) is taken per market: from Dune exports for the Morpho markets, and as the per-token variable borrow rate (USDT for L-6, USDe for L-7) for the Aave positions.

Carry spread. The carry spread is the difference between the yield earned on collateral and the cost of borrowing,

$$\text{carry_spread} = \text{ybt_apy} - \text{borrow_rate}.$$

A positive spread indicates that each unit of looped exposure earns more than it costs to fund; a negative spread indicates that the position is losing carry. To express the spread at the position's actual leverage, the levered carry spread scales it by leverage,

$$\text{carry_spread_levered} = \text{carry_spread} \times \text{leverage}.$$

5.5 Liquidation Thresholds

Table 9 lists the liquidation thresholds used in the health-factor and leverage calculations, taken from each market's on-chain configuration. The Morpho markets use isolated per-pair thresholds, while the two Aave positions share Aave's sUSDe stablecoin E-Mode threshold.

Table 9. Liquidation thresholds by market.

Protocol	Market	Threshold
Morpho Blue	sUSDe / USDtb	0.915
Morpho Blue	sUSDS / USDT	0.965
Morpho Blue	sUSDe / PYUSD	0.915
Morpho Blue	sUSDe / DAI	0.86
Morpho Blue	syrupUSDC / PYUSD	0.915
Aave v3	sUSDe stablecoin E-Mode	0.920

5.6 Feature Summary

Table 10 summarises the full feature set produced by the pipeline. The resulting `daily_features` table forms the input to the machine learning analysis, where the risk and behavioural features characterise strategy archetypes (RQ1, RQ2) and the lagged features support short-horizon prediction of risk rebalances and stressed risk states (RQ3).

Table 10. Engineered features.

Feature	Description
`ltv`	Borrow USD / collateral USD
`health_factor`	(collateral USD $\times \ell$)/borrow USD; < 1.0 liquidatable
`distance_to_liq`	$\ell -$ LTV; positive safe, negative underwater
`position_activity`	1 if any event occurred that day, else 0
`leverage`	$1/(1 -$ LTV); 0 if no debt
`ltv_cf`	LTV with no action: prior balances at current prices
`delta_leverage_rel`	Signed relative change in leverage against `ltv_cf`
`delta_equity_usd`, `delta_equity_rel`	Signed net capital flow, in USD and relative to prior equity
`leverage_adjust`	Active day: leverage moved, capital flat
`capital_flow`	Active day: capital moved, leverage flat
`levered_scaling`	Active day: both moved (loop built or unwound)
`minor_activity`	Active day: neither equity nor leverage moved materially
`risk_rebalance`	`leverage_adjust` or `levered_scaling`
`ltv_vol_7d`, `ltv_vol_14d`	Rolling 7- and 14-day std. dev. of LTV
`activity_freq_7d/30d`	Rolling 7- and 30-day count of active days
`risk_rebal_freq_7d/30d`	Rolling 7- and 30-day count of risk rebalances
`capital_flow_freq_7d/30d`	Rolling 7- and 30-day count of capital moves
`days_since_activity`	Days since last active day within the open position
`days_since_risk_rebalance`	Days since last risk rebalance within the open position
`ltv_lag_1/3/7`	LTV lagged by 1, 3, and 7 days
`coll_*`, `debt_*`	One-hot identity of collateral and borrow token
`ybt_apy`	Annualised yield on the collateral token (DeFiLlama)
`borrow_rate`	Annualised borrow rate on the debt token
`carry_spread`	ybt_apy $-$ borrow_rate
`carry_spread_levered`	carry_spread $\times$ leverage

6 Full Holder Panel Analysis

This section profiles the entire holder base of the three yield-bearing stablecoins, sUSDe, sUSDS and syrupUSDC, in conjunction with the selected-looper panel of Section 4.6. The goal is to establish where the tokens are held, which addresses are leveraged, and whether that leverage earns a positive carry. Each address's position is reconstructed from on-chain events across the wallet balance, Morpho Blue, and Aave v3. Every single balance, position, and holder count in this section has been recorded as of 14 July 2026 (23:59:59 UTC).

6.0.1 Definitions

- **Whale**: an address holding more than 1,000,000 YBS tokens, worth approximately $1M across the wallet and lending protocols. Whale is a statement about balance only. The loopers of Section 4.6 are a separate, behaviour-defined population; the two should not be conflated (Table 2).
- **Spot hold / Other protocols**: the address holds YBS in its wallet or other protocols and deploys none of that balance to Aave or Morpho.
- **Mixed**: the address both supplies YBS as collateral and retains YBS in its wallet, with the protocol share at most 90% of total YBS exposure. This is a share threshold, not a unit-token threshold: a residual wallet balance below 10% of exposure is classified as fully allocated.
- **Fully allocated**: more than 90% of the address's YBS is deployed to Morpho or Aave, so that the residual wallet balance is negligible. Where the venue is identified, we write Aave-only or Morpho-only.

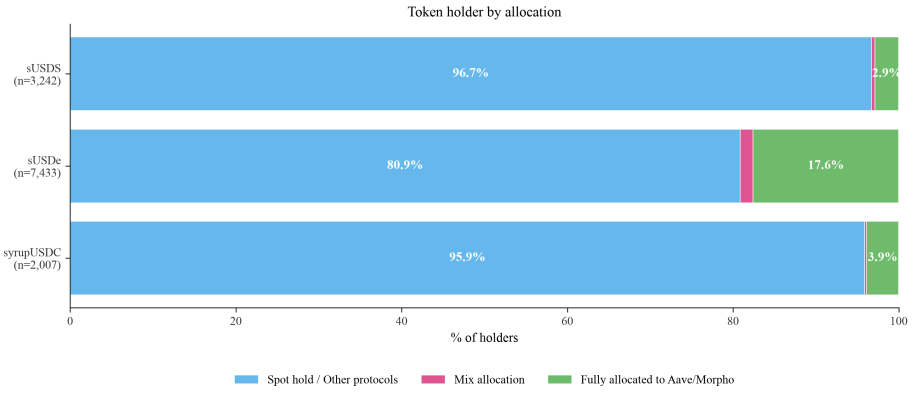


Fig. 4. Holder allocation by token. Snapshot: Ethereum, 14 July 2026. A majority of holders of each token are wallet-only (81–97%); sUSDe has the largest fully allocated share (17.6%).

Figure 4 indicates this allocation for each token. In all three cases a majority of holders are wallet-only (81–97%). Fully allocated shares are 2.9–3.9% for syrupUSDC and sUSDS, and mixed holders remain below 1%. sUSDe has a larger fully allocated share (18.7%), of which 14.8% is Aave-only and 3.9% is Morpho-only, indicating that Aave is the primary venue for deployed sUSDe collateral.

The allocation above counts each holder once. To distinguish a broad pattern from concentration among large holders, the same split is repeated with holders weighted by total YBS exposure.

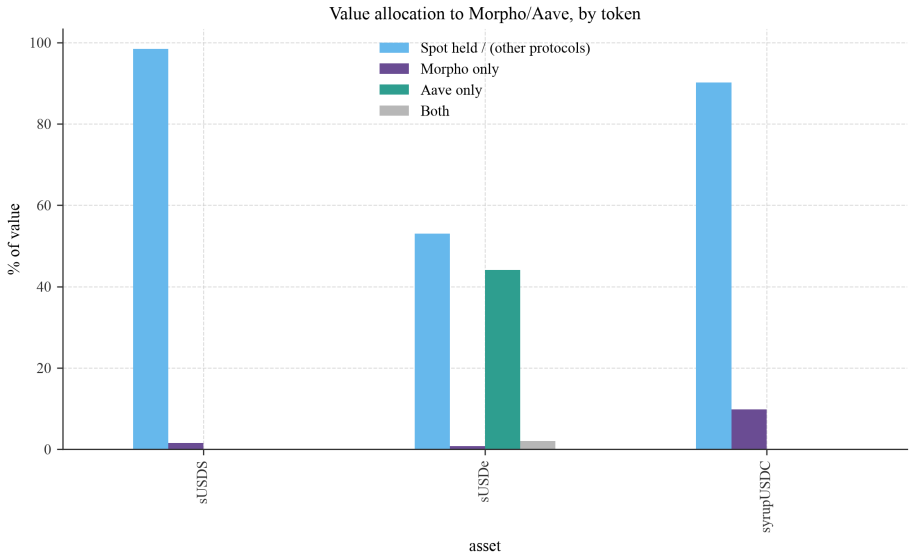


Fig. 5. Value-weighted allocation to Morpho and Aave, by token. Snapshot: Ethereum, 14 July 2026. Each holder is weighted by total YBS exposure. For all three tokens, a majority of value remains undeployed; sUSDe is the only token with substantial value on both Morpho and Aave.

Figure 5 shows that value-weighting changes the picture. For sUSDe, the Aave-only share rises from 14.8% of holders to 44% of value, so a small number of large holders account for a disproportionate share of sUSDe deployed to Aave. For syrupUSDC, the Morpho-only share rises from 2.9% of holders to 10% of value. For sUSDS the Morpho-only share is 2.9% of holders and 2% of value, so Morpho usage of sUSDS is not concentrated among a few large holders.

6.1 Allocation and Concentration

A whale is an address whose total YBS exposure exceeds $1,000,000, summing the wallet balance and any YBS posted as collateral on Morpho or Aave; the

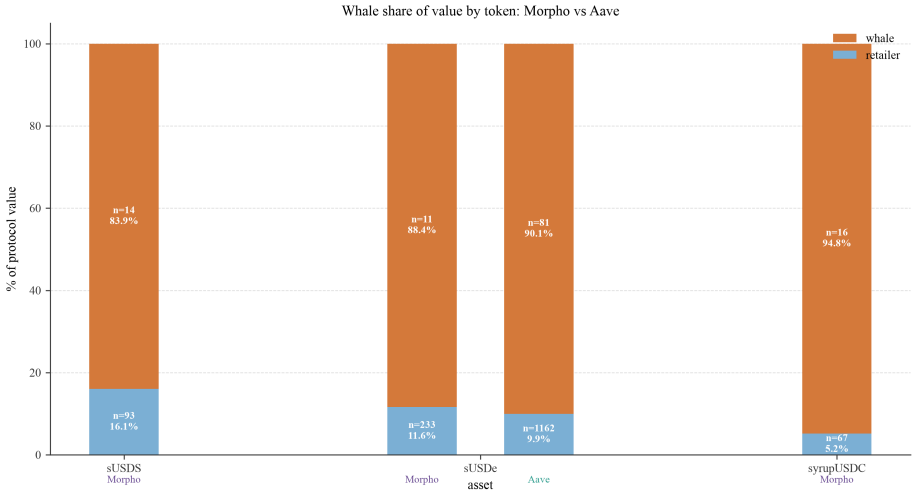


Fig. 6. Whale share of protocol value, by token. Snapshot: Ethereum, 14 July 2026. Whales hold 84–95% of Morpho value and 90% of Aave sUSDe value, while comprising 1.9–6.0% of all holders.

threshold is not a wallet-only balance. By that definition whales are 1.9% of sUSDe holders (139 of 7,433), 2.7% of syrupUSDC holders (55 of 2,007), and 6.0% of sUSDS holders (196 of 3,242), yet they account for most of the value deployed to both protocols (Figure 6). On Morpho they hold 84% of sUSDS value (14 of 107 depositors), 88% of sUSDe value (11 of 244), and 95% of syrupUSDC value (16 of 83). On Aave, which has material exposure only for sUSDe, they hold 90% of value (81 of 1,243 depositors).

As a share of depositors rather than of value, whales remain a minority: 13.1% of Morpho sUSDS depositors, 4.5% of Morpho sUSDe depositors, and 19.3% of Morpho syrupUSDC depositors (Figure 7). On Aave they are 6.5% of sUSDe depositors. These shares, between 4.5% and 19.3%, still correspond to most of the value in Figure 6, so each protocol's collateral base is concentrated.

Table 11 reports total supply, holder count, and top-holder concentration for each token. All three distributions are concentrated. syrupUSDC is the most concentrated: one address holds 43.2% of supply and the ten largest addresses hold 83.1%, on the smallest holder base (2,007). sUSDe is the least concentrated by this measure, with the largest holder base (7,433) and a top-10 share of 62.6%. sUSDS lies between them, with a top-10 share of 71.6%.

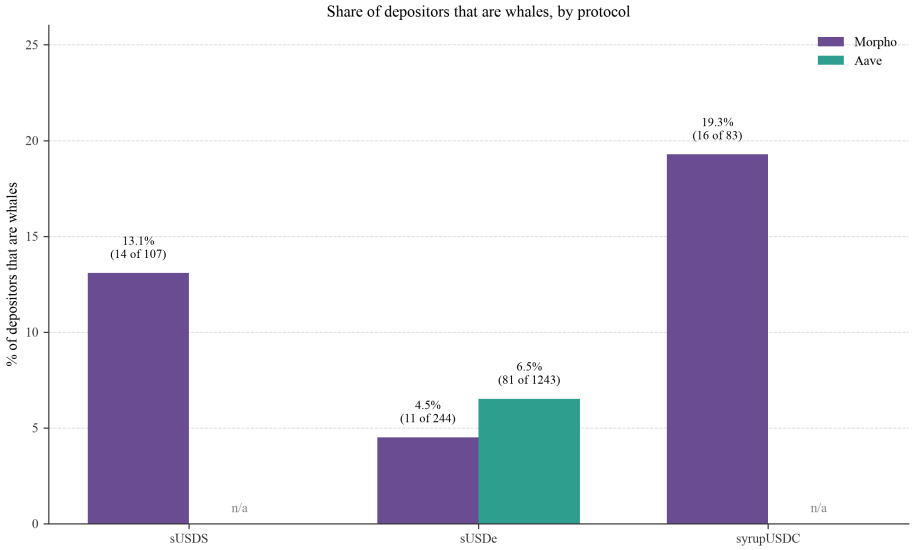


Fig. 7. Whale share of depositors, by protocol. Snapshot: Ethereum, 14 July 2026. A depositor is an address with a positive YBS balance on that protocol; whale status uses the $1M total-exposure threshold. Whales are 13.1% of Morpho sUSDS depositors, 4.5% of Morpho sUSDe depositors, 19.3% of Morpho syrupUSDC depositors, and 6.5% of Aave sUSDe depositors, while accounting for most deployed value (Figure 6).

Token	Total supply	Holders	Whales	Top 1 %	Top 10 %	Gini
syrupUSDC	1.216B	2,007	55	43.2%	83.1%	0.984
sUSDS	4.006B	3,242	196	22.8%	71.6%	0.976
sUSDe	2.121B	7,433	139	23.1%	62.6%	0.987

Table 11. Total supply, holder count, whale count (addresses with total YBS exposure above $1,000,000 across wallet and protocols), and top-holder concentration. Top 1 is the largest address's share of supply; Top 10 is the combined share of the ten largest addresses. The Gini coefficient is 0 when all wallets hold equal amounts and 1 when a single wallet holds the entire supply.

6.2 Which Markets: Morpho vs Aave Collateral Composition

Borrower-level positions identify the market into which deployed YBS collateral is placed. On Morpho, positions are grouped by isolated market, labelled by the debt asset borrowed against the YBS collateral. On Aave, positions are grouped by debt asset in the shared pool. Markets with total collateral below $10,000 are excluded as minor activity.

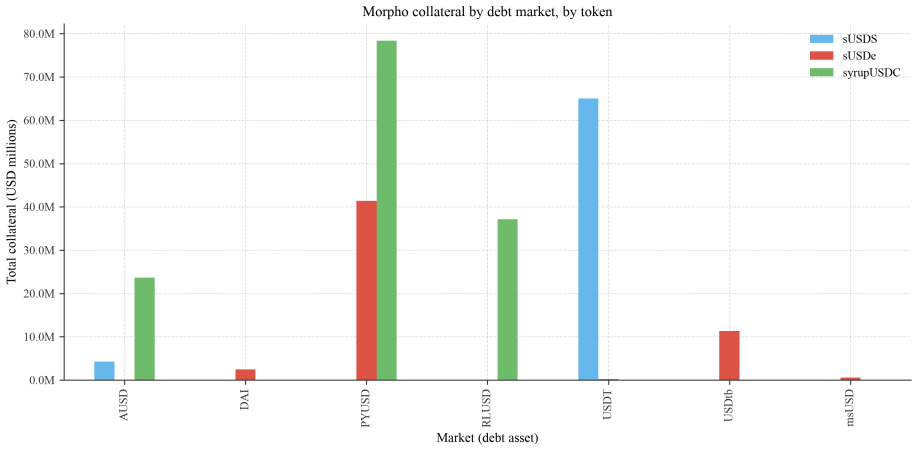


Fig. 8. Morpho collateral by market (debt asset), across all three YBS tokens. Each bar group shows how much of a given token's collateral is deployed against a given debt asset.

Figures 8 and 9 show different market compositions between the two protocols. On Aave, sUSDe collateral is concentrated in a small number of large stablecoin markets: USDT ($985M) and USDC ($780M) are the two largest, followed by USDe ($540M). A second tier of markets, each between $100M and $210M, includes WETH, GHO, USDS, cbBTC, and USDtb, indicating some borrowers use sUSDe collateral to access non-stablecoin debt (particularly WETH) as well as stablecoins. A long tail of smaller markets (weETH, DAI, wstETH, CRV, rETH) each sit in the $20–45M range.

Morpho's market composition is more fragmented and more token-specific. No single debt asset dominates across all three tokens. PYUSD is the largest shared market, used by both sUSDe ($41.3M) and syrupUSDC ($78.3M), but is nearly absent for sUSDS. USDT, by contrast, is the dominant market for sUSDS ($65.0M) but only a minor market for sUSDe ($0.2M) and effectively unused for syrupUSDC. RLUSD ($37.2M) and AUSD ($23.7M) are meaningful only for syrupUSDC, while USDtb ($11.3M) and DAI ($2.5M) appear only for sUSDe. Unlike Aave, where sUSDe borrowers concentrate in a small set of large stablecoin markets and, to a lesser extent, WETH, Morpho borrowers for each

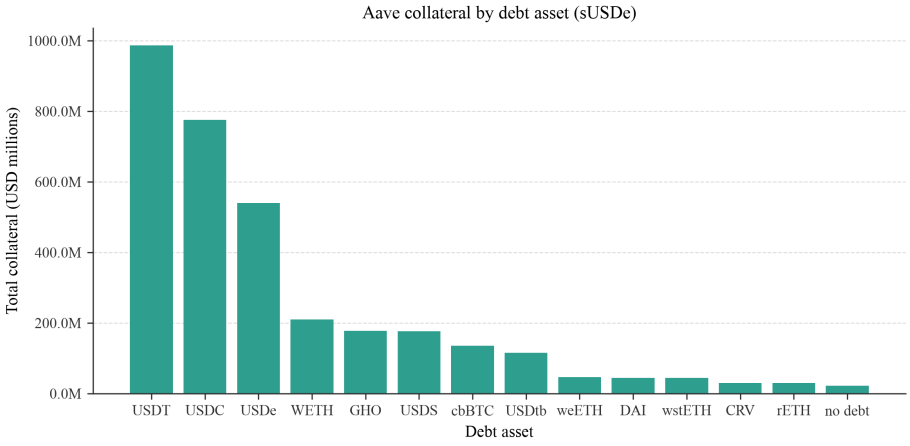


Fig. 9. Aave collateral by debt asset, sUSDe only. Bars show total sUSDe collateral value (USD) backing each debt asset; markets with total collateral below $10M are excluded. Collateral concentrates in USDT ($985M), USDC ($780M), and USDe ($540M).

YBS token cluster around a different, largely non-overlapping set of debt assets, reflecting which isolated markets exist and are liquid for each collateral type.

6.3 Borrowing Behaviour: Morpho vs Aave

Having established which markets each protocol's collateral flows into, we now turn to how holders use that collateral: whether they supply it without borrowing, or borrow against it, and whether this differs by holder size (whale vs retailer, as defined above).

The two protocols show clearly different behaviour patterns. On Morpho (Figure 10), almost every whale position is an active stablecoin borrow: 100% for all three tokens. Retailers borrow at a slightly lower but still very high rate, 82.5% for sUSDe, 93.2% for sUSDS, and 90.7% for syrupUSDC. On Morpho, holding collateral without borrowing against it is rare for both groups, and closer to non-existent for whales specifically.

On Aave (Figure 11), the pattern flips. The majority of both whales (61.0%) and retailers (69.4%) supply sUSDe as collateral without drawing any debt against it, even though supplying collateral earns no yield. Only a minority loop into a stablecoin borrow: 39.0% of whale positions and 28.9% of retailer positions. Non-stable borrowing (for example, against WETH) is rare in both groups, effectively zero for whales and 1.7% for retailers. The supply rate applies to the borrow asset, not the collateral asset. One explanation is that many retailer holders expect to earn a supply yield by depositing YBS as collateral; that expectation is mistaken.

Beyond whether holders borrow at all, the two protocols also differ sharply in how much leverage those who do borrow actually carry. On Morpho (Figure 12),

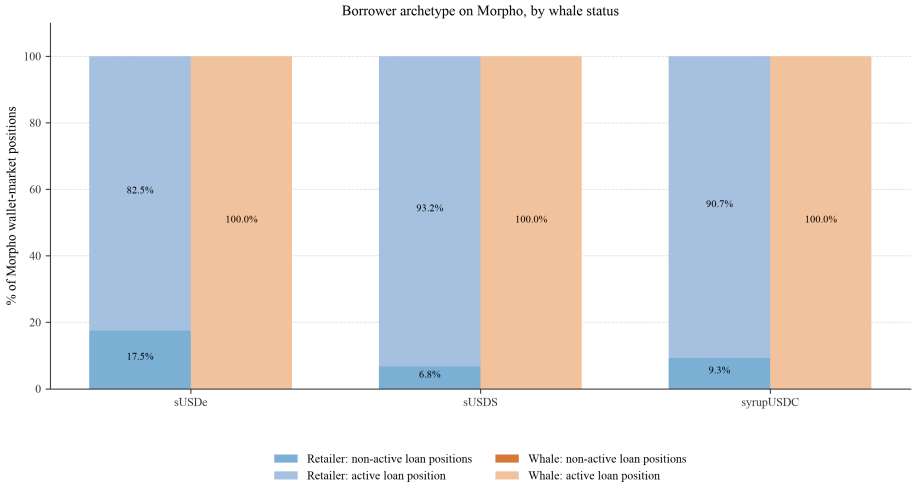


Fig. 10. Morpho: share of whale and retailer wallet-market positions that are collateral only versus stablecoin borrow, by token. Data from Ethereum, 14 July 2026. Almost every whale position is an active stablecoin borrow (100% across all three tokens), and retailers borrow at a similarly high rate (82.5–93.2%), so holding collateral without borrowing is rare on Morpho for both groups.

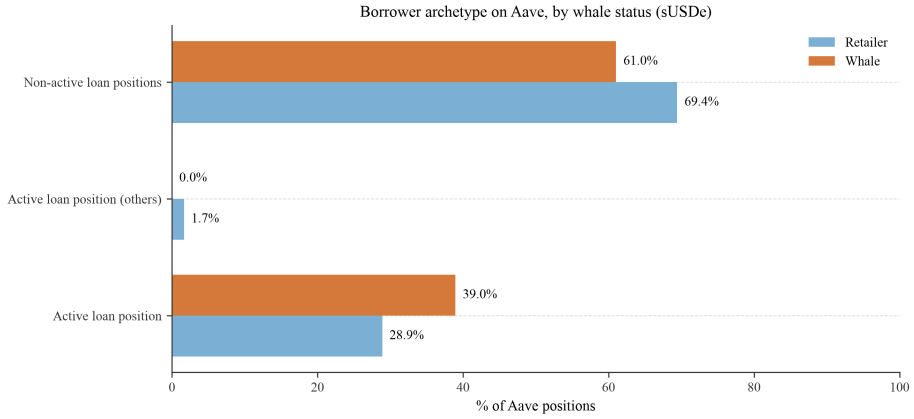


Fig. 11. Aave (sUSDe only): share of whale and retailer wallet positions split into collateral only, non-stable borrow, and stablecoin borrow (loop). Data from Ethereum, 14 July 2026. Stablecoin borrow is treated as a loop, since it lets the holder use borrowed funds to buy more sUSDe. Most whales (61.0%) and retailers (69.4%) supply sUSDe as collateral without drawing debt; 39.0% of whale positions and 28.9% of retailer positions are stablecoin borrows.

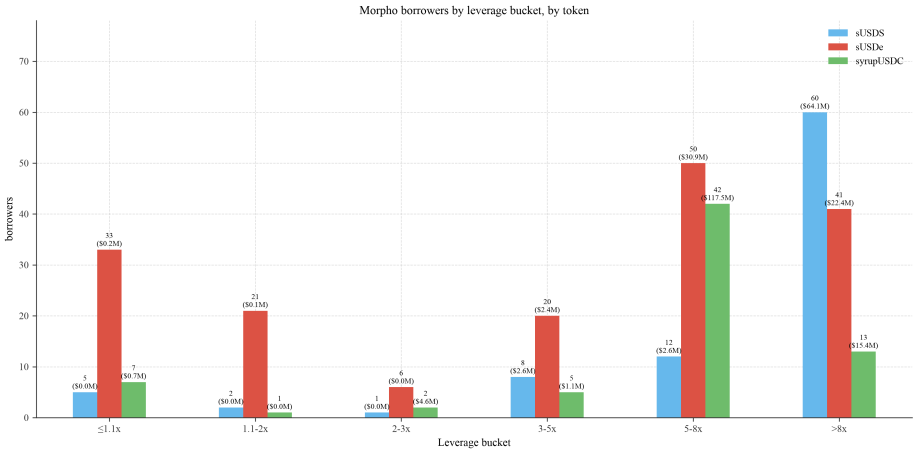


Fig. 12. Morpho borrowers by leverage bucket, across tokens. Leverage is read directly from the `leverage_market` field in the borrower data; labels show borrower count and total collateral value in each bucket. Data from Ethereum, 14 July 2026. Borrowers skew toward high leverage across all three tokens: syrupUSDC concentrates $117.5M in the 5–8x bucket (42 borrowers), sUSDS has its largest count above 8x (60 borrowers, $64.1M), and sUSDe peaks at 5–8x and above 8x combined, with very few borrowers near 1x.

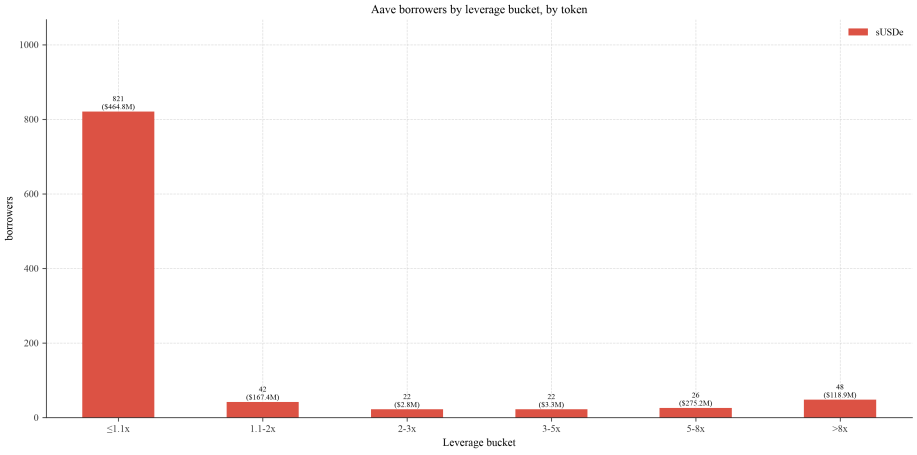


Fig. 13. Aave (sUSDe only) borrowers by leverage bucket, using leverage from stablecoin debt; labels show borrower count and total collateral value in each bucket. Data from Ethereum, 14 July 2026. Most borrowers (821 of 981, $464.8M) sit at or below 1.1x leverage, but a smaller cluster carries far more leverage relative to headcount: 26 borrowers in the 5–8x bucket hold $275.2M, and a further 48 borrowers above 8x hold $118.9M.

borrowers skew toward the high end of the leverage spectrum for all three tokens. syrupUSDC concentrates its largest collateral value, $117.5M, in the 5-8x bucket among just 42 borrowers, while sUSDS has 60 borrowers, more than any other bucket, sitting above 8x with $64.1M behind them. sUSDe is more evenly spread, but still peaks at 5-8x (50 borrowers, $30.9M) and above 8x (41 borrowers, $22.4M). Very few Morpho borrowers sit near 1x, and the dollar amounts there are negligible, reinforcing that a Morpho position is typically not just a borrow, but a levered one.

On Aave (Figure 13), 821 of the 981 borrowers with any stablecoin debt sit at or below 1.1x leverage, together holding $464.8M. This is consistent with the largely collateral-only picture from Figure 11: even among the minority who do borrow, most take on only a token amount of debt rather than meaningfully levering their position. That said, a smaller cluster of borrowers carries far more leverage and far more capital than their headcount would suggest: 26 borrowers sit in the 5–8x bucket but account for $275.2M, more than half the value of the entire <= 1.1x bucket despite being 3.2% of its headcount (26 of 821), and a further 48 borrowers sit above 8x with $118.9M. Aave leverage is therefore rare in headcount terms and concentrated in dollar terms.

The two protocols therefore serve different purposes. Morpho positions are almost always borrow-driven: supplying sUSDe, sUSDS, or syrupUSDC there is nearly the same thing as borrowing against it. Aave is used more often as a passive place to park collateral, with looping showing up as a smaller, secondary behaviour among large holders.

Morpho's isolated markets are often chosen specifically for leverage, while Aave's shared liquidity pool is a default venue for yield-bearing collateral, with borrowing optional. On-chain data cannot establish whether these large Aave holders are idle or waiting to loop again; their debt cycling fully to zero rather than staying open is consistent with deliberate timing.

6.4 Flashloan Activity

This subsection focuses on the whale cohort: the share of sUSDe supply held by each of the 81 Aave whales, and whether these addresses use flashloans. A flashloan can open or unwind a leveraged position in one transaction on the blockchain, without a lasting residual position.

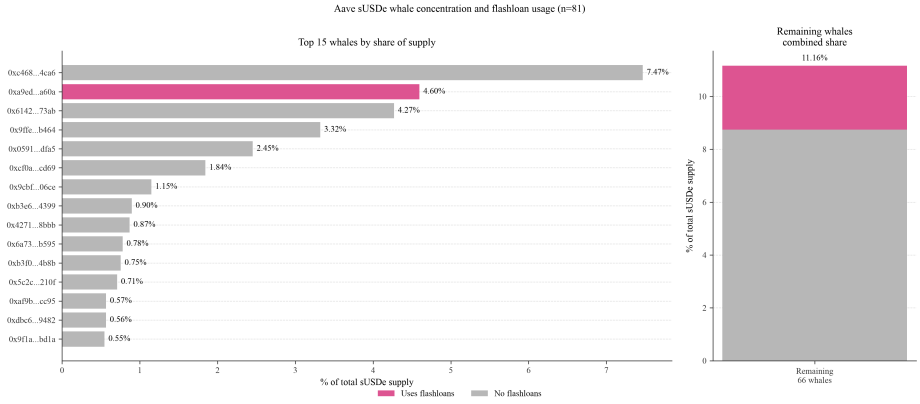


Fig. 14. Aave sUSDe whale size distribution (n=81 whales). Left panel shows the 15 largest whales individually, each bar labeled with its share of total sUSDe supply; right panel shows the combined share held by the remaining 66 whales, split by flashloan usage. Data from Ethereum, 14 July 2026. The largest whale alone holds 7.47% of total supply, more than the combined 66 smallest whales (11.16%); flashloan usage is more common among whales (18.5%) than the overall holder base (4.16%), but is spread across whale sizes rather than concentrated among the very largest.

On Aave, the 81 sUSDe whales are themselves highly unequal in size. The single largest whale holds 7.47% of total sUSDe supply on its own, more than the combined 66 smallest whales, who together hold 11.16%. Across all sUSDe holders, 4.16% (309 of 7,433) have used a flashloan at least once; narrowing to whales specifically, that rate rises to 18.5% (15 of 81). Flashloan usage does not appear concentrated among the very largest holders: the second-largest whale (4.60% of supply) is a flashloan user, but the largest (7.47%) and third-largest (4.27%) are not, suggesting flashloan activity is spread across whale sizes rather than being a signature specifically of the biggest holders.

On Morpho, none of the 41 whales across the three tokens have used a flashloan. This is not confined to whales: across all holders, flashloan usage on Morpho is almost nonexistent, at 0.013% for sUSDe (1 of 7,433), 0.123% for sUSDS (4 of 3,242), and 0.000% for syrupUSDC (0 of 2,007). Flashloan volume on Morpho instead comes from a small, separate population of retailer addresses, and even there it is highly concentrated: for sUSDS, for example, the top three callers alone account for 42.4%, 33.2%, and 20.3% of total flashloan

volume respectively, consistent with a small number of arbitrage, liquidation, or leverage-bootstrapping bots rather than genuine token holders.

Flashloans and whale positions therefore serve different functions on the two protocols. On Aave, a meaningful minority of whales (18.5%) incorporate flashloans into how they manage their position, alongside the mostly passive collateral-only behaviour documented in Figure 11. On Morpho, flashloans and whale activity appear fully decoupled: whales build and hold leveraged positions through ordinary, multi-transaction borrowing, as shown in Figures 10 and 12, while flashloans are used transiently by a distinct, non-holding set of addresses for purposes outside the scope of this holder analysis.

6.5 Loan-to-Value and Liquidation Risk

Having established how much leverage borrowers carry on each protocol, we now turn to how close that leverage sits to forced liquidation, and whose exposure dominates that risk. Two questions drive this subsection: first, how much room is there before a position is liquidated; second, is that risk concentrated among a small number of large holders, or spread broadly across the borrower base. We begin with Morpho, where isolated markets give each position its own liquidation threshold. Aave is treated separately below, since its shared pool does not expose a reliable per-wallet threshold in this data.

6.5.1 Definitions

- **LTV (loan-to-value)**: the ratio of a borrower's debt to their collateral value, expressed as a percentage.
- **Liquidation LTV (`lltv`)**: the LTV ceiling at which the protocol is permitted to seize and sell the collateral.
- **Health factor**: `lltv / ltv`. A health factor near 1 indicates a position is close to its liquidation threshold; values well above 1 indicate a larger safety margin.
- **Risk band**: positions are bucketed by health factor into critical (<1.05), high-risk (1.05–1.2), moderate (1.2–1.5), and safe (>1.5).

6.5.2 Morpho Figure 15 shows that the large majority of active Morpho collateral, across all three tokens, sits in the high-risk band (health factor 1.05–1.2) rather than spread evenly across safer bands. sUSDS is the only token carrying any critical-band exposure, $5.4M across 5 positions, though its high-risk band is far larger still at $61.3M across 68 positions. sUSDe and syrupUSDC have no critical-band positions, but both carry substantial high-risk exposure, $50.6M and $123.7M respectively, with syrupUSDC's high-risk band the largest of any token-band combination in the sample.

Figure 16 shows that the concentration documented earlier in this section (Figure 6) holds specifically within the risky subset of positions, not only across holdings generally. Whale wallets make up 22.0% of positions in the critical or high-risk bands, yet those same positions account for 93.1% of the $241.0M in collateral sitting in those bands. In dollar terms, this means 45 whale positions hold $224.3M of the collateral currently close to liquidation, versus $16.7M spread across the remaining, far more numerous, retailer positions.

Figure 17 shows that at-risk collateral is not spread evenly across Morpho's markets. Three markets, syrupUSDC/PYUSD ($77.7M), sUSDS/USDT ($62.4M), and sUSDe/PYUSD ($41.3M), account for the large majority of collateral currently in the critical or high-risk bands; every other market holds under $24M. Size and leverage do not move together: syrupUSDC/PYUSD, the largest market by dollar exposure, carries a comparatively moderate 7.4x average leverage, while sUSDS/USDT carries the highest average leverage in the sample

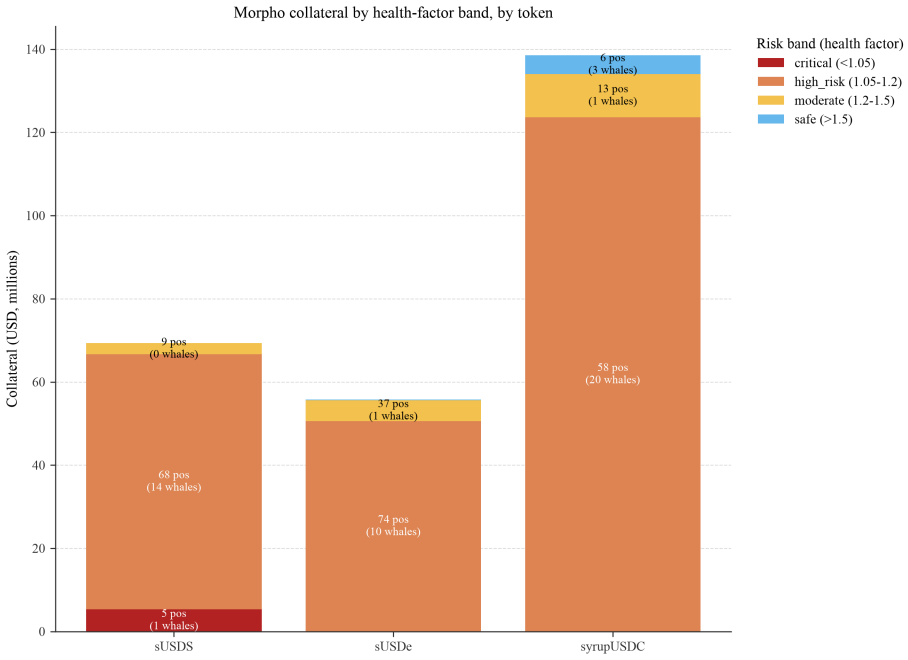


Fig. 15. Morpho collateral by health-factor band, by token. Only active borrow positions (`ltv_percentage` ≥ 0.5) are included. Data from Ethereum, 14 July 2026. Most collateral across all three tokens sits in the high-risk band; sUSDS is the only token with any critical-band exposure.

(13.0x) on the second-largest dollar exposure, making it a market that combines scale with a thin safety margin.

Figure 18 translates the market-level concentration above into a direct stress test: for a given collateral price drop, how much collateral would be forced into liquidation. sUSDS is the most fragile at small shocks, consistent with its high average leverage, beginning to liquidate at drops of only 5%, and reaching $60M liquidated by a 10–11% drop. syrupUSDC behaves differently: it remains largely flat until a 12% drop, at which point a cluster of positions sharing a similar liquidation distance triggers a near-simultaneous liquidation event, taking cumulative liquidated collateral from near zero to over $130M within a 12–18% band. sUSDe rises more gradually across a wider range and plateaus lowest of the three, at $55M. This cliff-edge behaviour in syrupUSDC is the mechanism by which risk to one borrower threatens others: a single market-wide price move can trigger a large, near-simultaneous liquidation event within an isolated market, creating sell pressure that falls on every position in that market regardless of whether it was individually at risk beforehand.

Taken together, these four figures show that Morpho's liquidation risk is both concentrated and, in at least one market, prone to triggering suddenly rather

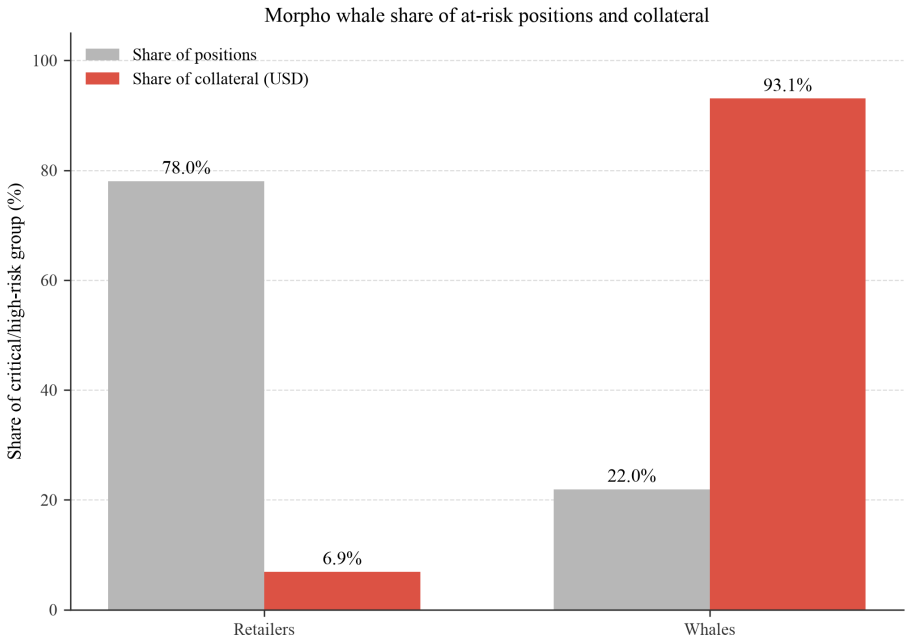


Fig. 16. Share of at-risk Morpho positions versus share of at-risk collateral value, whales versus everyone else, pooled across all three tokens. Data from Ethereum, 14 July 2026. Whales account for only 22.0% of at-risk positions but 93.1% of at-risk collateral value, while everyone else holds the reverse, 78.0% of positions but just 6.9% of the value at risk.

than gradually: whales hold the overwhelming majority of at-risk collateral, that collateral clusters in three markets, and one of those markets would see the bulk of its at-risk collateral liquidate within a narrow price-drop band rather than being absorbed steadily.

6.6 Aave

Aave sUSDe is a shared, cross-collateral pool rather than a set of isolated markets, so health factor cannot be reconstructed from sUSDe collateral and stablecoin debt alone the way it can on Morpho: a wallet's real liquidation distance depends on every asset it has supplied and borrowed, not just its sUSDe position. Rather than approximate this, we pull each wallet's real, contract-computed health factor directly from Aave's Pool contract (`getUserAccountData`), which accounts for a wallet's full position across every reserve. Because health factor is a whole-wallet figure and cannot be decomposed by debt asset, we restrict this subsection to wallets whose debt is entirely stablecoins (dropping any wallet that also carries WETH, cbBTC, weETH, wstETH, CRV, or rETH debt); for this subset, real health factor and stablecoin-specific risk are the same thing, with no

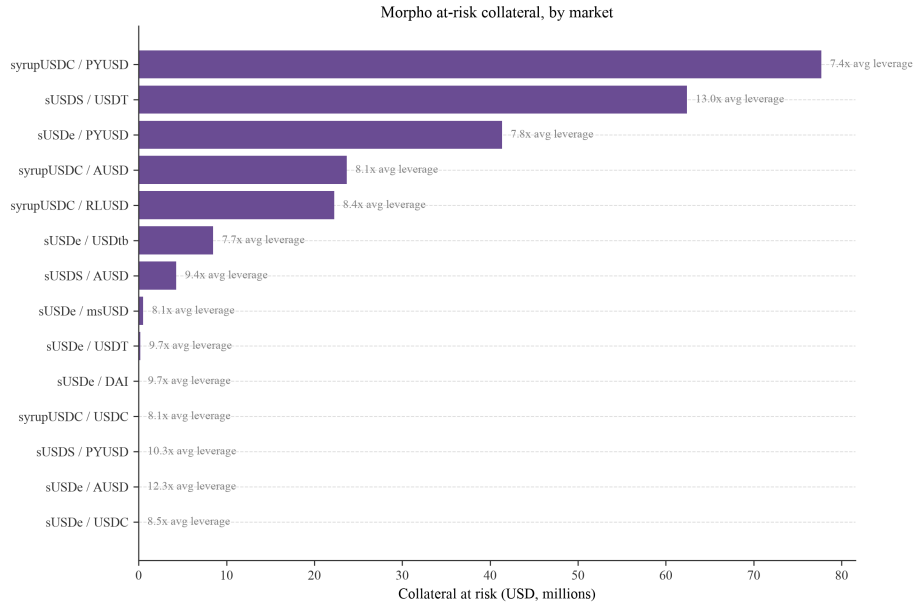


Fig. 17. Collateral in the critical/high-risk bands, by market (debt asset), with average leverage labeled per market. Data from Ethereum, 14 July 2026. Risk concentrates in three markets: syrupUSDC/PYUSD, sUSDS/USDT, and sUSDe/PYUSD. sUSDS/USDT combines the second-largest dollar exposure with the highest average leverage of any market (13.0x).

proxy or assumption required. Of 361 wallets with active stablecoin debt against sUSDe, 345 remain after this filter and after excluding a further 16 wallets with negligible (sub-$1) residual debt, whose health factor is otherwise mechanically inflated to implausible levels by an almost-zero denominator.

Figure 19 shows that critical and high-risk exposure together account for $531.1M, 90.5% of the $587.1M in stablecoin-collateralized Aave sUSDe positions, leaving only $37.0M moderate and $19.0M safe. As on Morpho, dollar value and position count diverge within these bands, though less sharply than under the earlier proxy estimate: the critical band holds both the most positions (95) and the largest dollar amount ($311.5M), while the high-risk band holds fewer positions (81) on a somewhat smaller total ($219.6M). Whale positions appear in both danger bands, 17 in critical and 6 in high-risk, 23 whale positions in total.

Figure 20 shows that whale concentration within the at-risk group is, if anything, tighter on Aave than on Morpho (97.0% of at-risk collateral versus 93.1%), and, notably, this pattern survived the correction from the earlier utilization proxy to real health factor: the underlying risk-band totals shifted substantially (Figure 19 versus the proxy estimate it replaced), yet 23 whale positions, 13.1% of the 176 positions in the critical or high-risk bands, still hold 97.0% of the

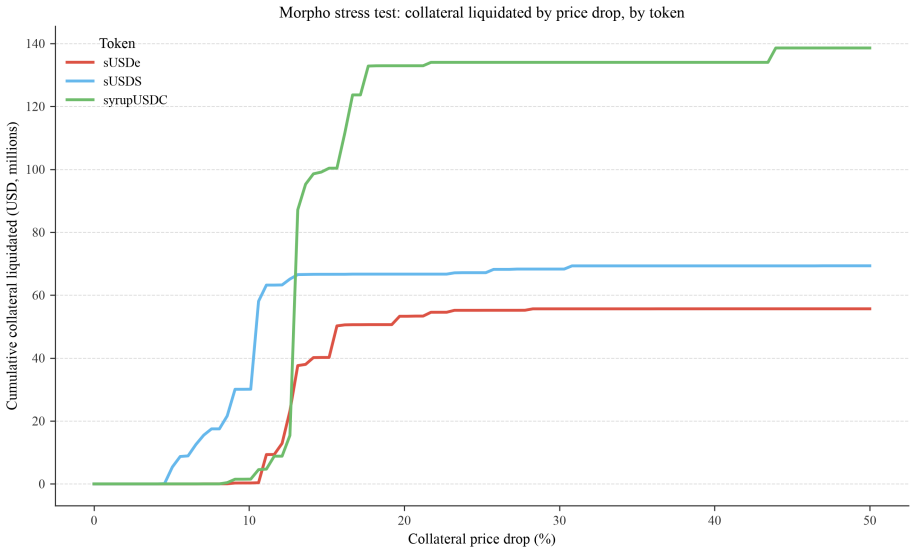


Fig. 18. Stress test: cumulative Morpho collateral liquidated as a function of a hypothetical collateral price drop, by token. Data from Ethereum, 14 July 2026. syrupUSDC shows a cliff-edge pattern, absorbing little liquidation until a 12% price drop, after which over $130M liquidates within a narrow band; sUSDS reacts earliest, beginning to liquidate at price drops as small as 5%.

$531.1M in those bands. Whales do not need to run higher utilization than retail to dominate at-risk dollars, since their positions are large enough that even comparable risk levels translate into a disproportionate share of total value at risk.

Figure 21 shows a different picture from the earlier proxy-based estimate: USDe now carries both the largest whale count (16, more than a third of its 44 at-risk positions) and a dollar total comparable to or exceeding USDC and USDT, rather than being the smallest of the three major markets. USDtb, GHO, USDS, and DAI remain minor, each with few positions. USDe's concentration of whale involvement is 36% of its at-risk positions, versus under 15% for USDC and USDT, so a liquidation event concentrated in the USDe market would involve far fewer, larger counterparties than one centered on USDC or USDT.

Figure 22 shows that, using real health factor rather than the earlier utilization proxy, no Aave sUSDe collateral is already past its liquidation trigger, consistent with real, currently-held positions (a genuinely liquidatable position would not persist in a live snapshot). The danger is instead concentrated in how little room remains: a 3–5% drop in sUSDe's price alone, with every other asset held fixed, is sufficient to push cumulative liquidated collateral from $40M to over $510M, after which the curve is nearly flat until a smaller step near 30%. The plateau, $580M, is consistent with the $531.1M critical-and-high-risk total

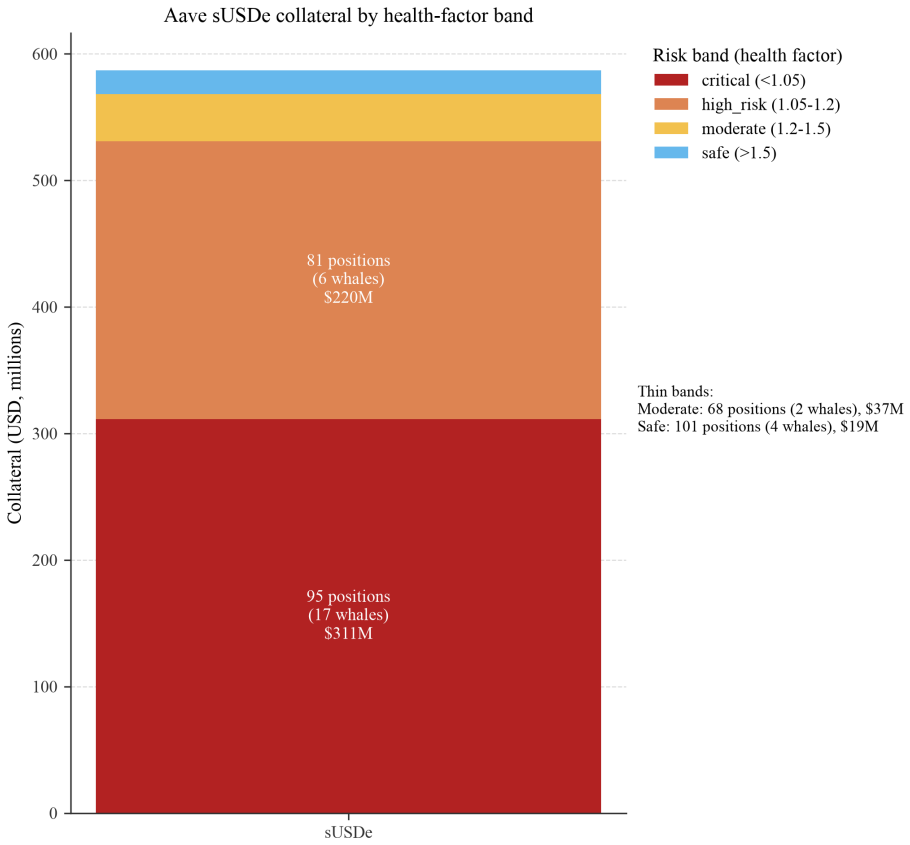


Fig. 19. Aave sUSDe collateral by real health-factor band (stablecoin debt only), with position and whale counts labeled per band. Data from Ethereum, 14 July 2026. $531.1M of the $587.1M total sits in the critical or high-risk bands, split between 95 critical positions (17 whales, $311.5M) and 81 high-risk positions (6 whales, $219.6M).

in Figure 19, plus the moderate-band collateral that this simulation eventually sweeps in as the hypothetical shock grows large. Because health factor here is computed from each wallet's total collateral across all assets while the dollar amounts plotted are sUSDe-specific, this figure should be read as "if sUSDe alone crashes while everything else holds steady," not as a claim about a broader, multi-asset market crash.

Taken together, Aave's liquidation risk, measured with real health factor, is both more concentrated and closer to the edge than the earlier utilization-proxy estimate suggested: 90.5% of stablecoin-collateralized Aave sUSDe value sits in the critical or high-risk bands, and a single-digit percentage price drop in sUSDe alone is sufficient to trigger the large majority of it.

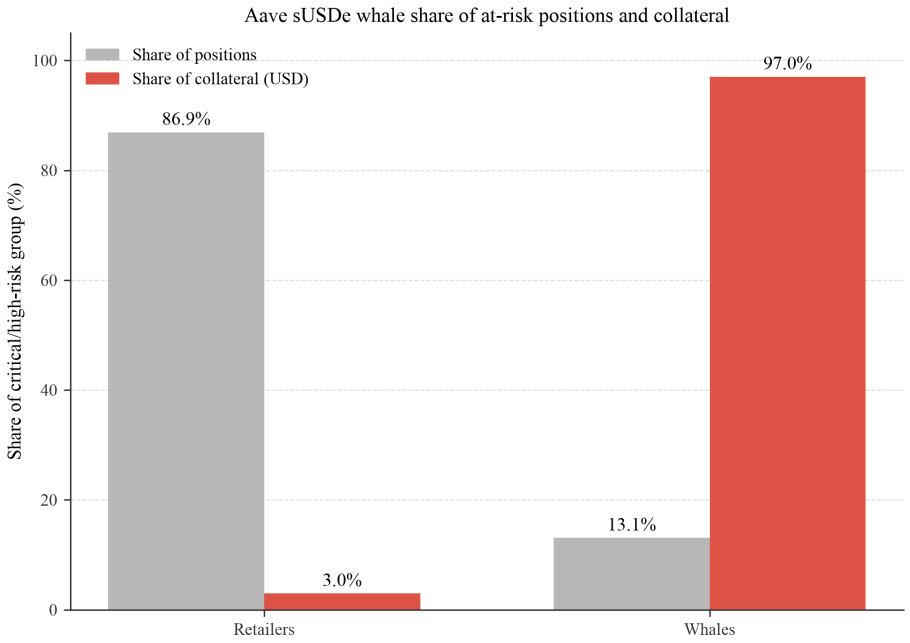


Fig. 20. Share of at-risk Aave sUSDe positions versus share of at-risk collateral value, whales versus everyone else (stablecoin debt only, real health factor). Data from Ethereum, 14 July 2026. Whales account for only 13.1% of at-risk positions but 97.0% of at-risk collateral value, while everyone else holds the reverse, 86.9% of positions but just 3.0% of the value at risk.

6.6.1 Summary Both protocols show liquidation risk heavily concentrated among a minority of large positions: on Morpho, 93.1% of at-risk collateral sits with whale wallets that make up only 22.0% of at-risk positions (Figure 16); on Aave, the concentration is even tighter, 97.0% of at-risk collateral sits with whales that make up just 13.1% of at-risk positions (Figure 20). This concentration appears to be a structural feature of who holds these tokens at all, rather than a byproduct of either protocol's specific borrowing mechanics, since it holds, and if anything strengthens, under both Morpho's leverage-driven, isolated-market design and Aave's largely passive, shared-pool design.

The two protocols differ in how that risk would likely unfold. On Morpho, risk is concentrated in three markets, and one of them, syrupUSDC/PYUSD, shows a cliff-edge pattern only past a 12–18% price drop (Figure 18). On Aave, under real health factor, the cliff arrives earlier and more sharply still: a 3–5% price drop in sUSDe alone is sufficient to trigger the large majority of at-risk liquidations (Figure 22). Within Aave specifically, risk concentrated in the USDe debt market is held by disproportionately more whales than the larger USDC

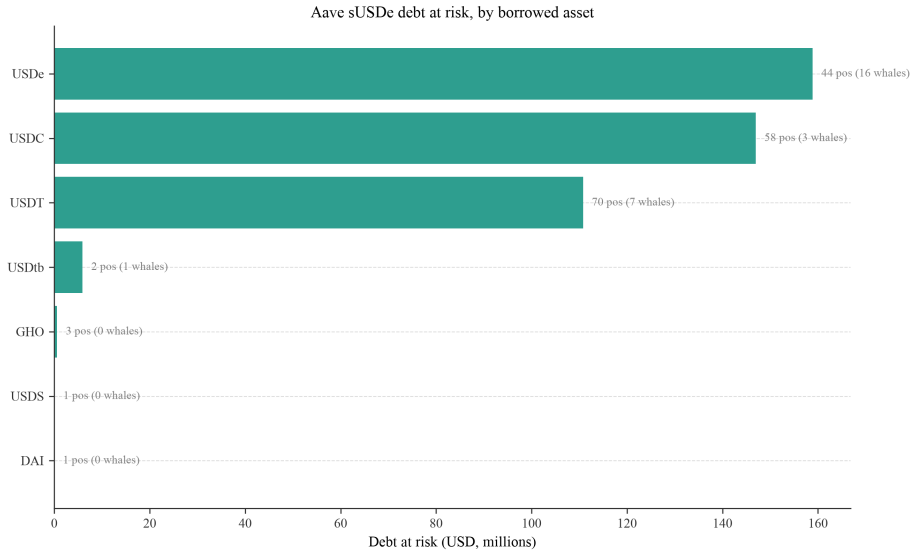


Fig. 21. Debt at risk on Aave, by borrowed stablecoin (stablecoin debt only, real health factor), with position and whale counts labeled per bar. Data from Ethereum, 14 July 2026. USDe (44 positions, 16 whales) is now the largest at-risk market by whale count and among the largest by dollar value, ahead of USDC (58 positions, 3 whales) and USDT (70 positions, 7 whales).

and USDT markets, suggesting a USDe-driven liquidation event would involve fewer, larger counterparties than one centered on USDC or USDT.

These findings should be read alongside the flashloan activity documented a subset of Aave whales already incorporate flashloans into how they manage their positions, a tool that can be used to unwind or rebuild leveraged exposure quickly around exactly the kind of price move that would otherwise trigger the liquidations.

6.7 Loop Profitability

Having established how much leverage borrowers carry and how close that leverage sits to liquidation, we now ask whether that leverage is actually profitable: does looping a YBS token pay at today's rates, and does this differ between Morpho and Aave.

6.7.1 Definitions

– **Native YBS APY**: the yield the collateral token earns on its own, independent of any lending protocol (Ethena's yield for sUSDe, the Sky Savings Rate for sUSDS, Maple's USDC pool yield for syrupUSDC).

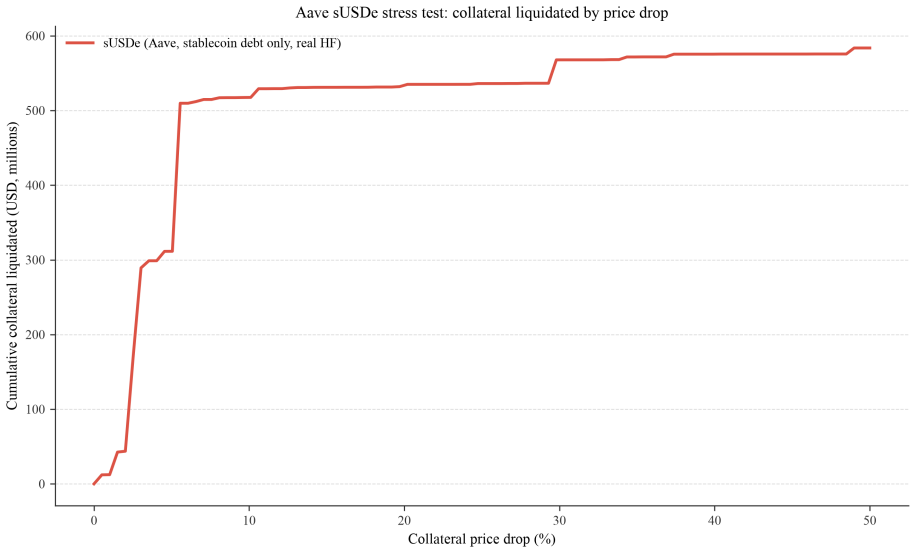


Fig. 22. Stress test: cumulative Aave sUSDe collateral liquidated as a function of a hypothetical sUSDe price drop, holding all other collateral fixed (stablecoin debt only, real health factor). Data from Ethereum, 14 July 2026. No collateral is liquidated at a 0% price drop; a cliff-edge jump from $40M to $510M occurs within a narrow 3–5% price-drop band, after which the curve is nearly flat until a smaller secondary step near 30%, plateauing at $580M.

- **Borrow APY**: the interest rate charged on the borrowed debt asset, set by each protocol's utilization-based rate model and varying by market.
- **Carry spread**: native YBS APY minus borrow APY, the return earned on each additional dollar borrowed and redeployed into more collateral. A positive spread means looping adds to net return; a negative spread means looping subtracts from it.

All rates below are snapshots as of 14 July 2026.

6.7.2 Morpho Figure 23 shows a mix of profitable and unprofitable markets. syrupUSDC/AUSD offers the widest positive spread (+1.40pp), followed by sUSDS/USDT (+1.03pp), sUSDe/PYUSD (+0.76pp), syrupUSDC/PYUSD (+0.43pp), and syrupUSDC/RLUSD (+0.36pp). Three markets carry a negative spread: sUSDS/AUSD is close to flat (-0.02pp), while sUSDe/USDtb (-0.42pp) and sUSDe/msUSD (-1.47pp) are meaningfully unprofitable to loop at these rates.

6.7.3 Aave Figure 24 shows a similar split between profitable and unprofitable markets. sUSDe/USDe carries the widest positive spread (+1.86pp), followed

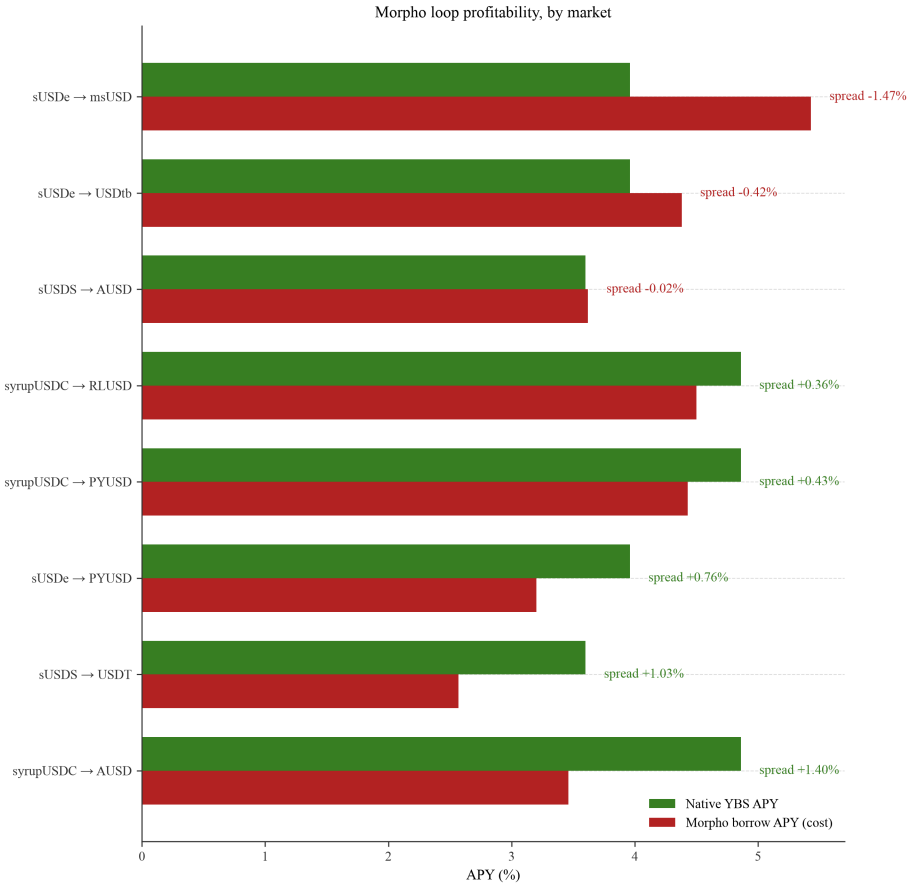


Fig. 23. Native YBS APY versus Morpho borrow APY, by market, sorted by spread. Data from Ethereum, 14 July 2026. Five of eight markets carry a positive spread, led by syrupUSDC/AUSD (+1.40pp) and sUSDS/USDT (+1.03pp); three carry a negative spread, led by sUSDe/msUSD (-1.47pp).

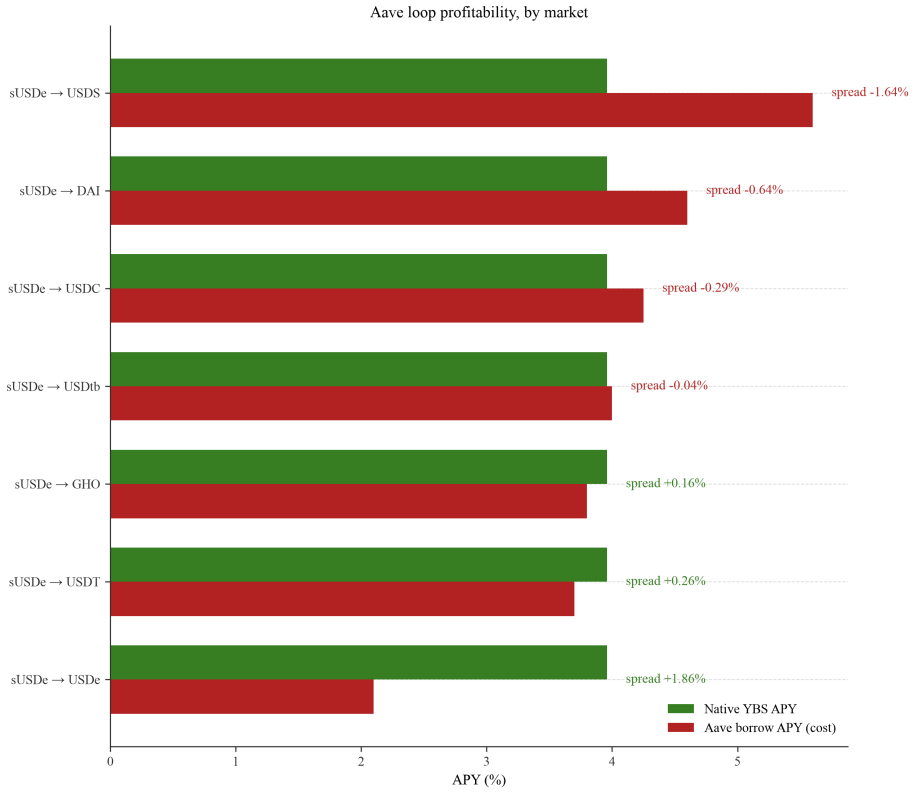


Fig. 24. Native sUSDe APY versus Aave borrow APY, by debt asset, sorted by spread. Data from Ethereum, 14 July 2026. Three of seven markets carry a positive spread, led by sUSDe/USDe (+1.86pp); four carry a negative spread, led by sUSDe/USDS (-1.64pp).

by sUSDe/USDT (+0.26pp) and sUSDe/GHO (+0.16pp). Four markets carry a negative spread: sUSDe/USDtb is close to flat (-0.04pp), sUSDe/USDC is modestly negative (-0.29pp), sUSDe/DAI is more clearly negative (-0.64pp), and sUSDe/USDS carries the largest negative spread of any Aave market sampled (-1.64pp).

6.7.4 Summary Across both protocols, loop profitability is mixed rather than uniformly positive or negative: 5 of 8 sampled Morpho markets and 3 of 7 sampled Aave markets carry a positive spread at these rates. On Morpho, the widest positive spread reaches +1.40pp (syrupUSDC/AUSD); on Aave, the widest reaches +1.86pp (sUSDe/USDe). The largest negative spread on either protocol is sUSDe/msUSD on Morpho (-1.47pp), closely followed by sUSDe/USDS on Aave (-1.64pp).

6.8 Summary of Holder-Panel Findings

Five patterns emerge from the holder panel.

Allocation is all or nothing. Holders either keep the token in the wallet or commit it fully to a lending protocol, with mixed allocation below 1% for sUSDS and syrupUSDC. Value-weighting sharpens rather than changes this, with the Aave-only group rising from 14.8% of sUSDe holders to 44% of sUSDe value.

Deployed collateral is invisible in wallet balances. A looped position sits inside the protocol rather than in the wallet, which is why the looper population is identified by behaviour and ranked by lifetime inflow. The two protocols differ sharply in how that collateral is used: on Morpho every whale position carries an active borrow, whereas on Aave 61.0% of whales and 69.4% of retail holders supply sUSDe and draw no debt at all.

Those who borrow lever close to the ceiling. Morpho borrowers cluster at 5× and above across all three tokens, placing most of their collateral in the high-risk health-factor band. On Aave most borrowers sit below 1.1×, yet 90.5% of stablecoin-collateralised sUSDe value still falls in the critical or high-risk band. Risk concentration holds under both designs: whales are 22.0% of at-risk Morpho positions but 93.1% of at-risk value, and 13.1% of at-risk Aave positions but 97.0% of value.

Liquidation would arrive as a cliff. A 3 to 5% sUSDe decline takes cumulative liquidated Aave collateral from roughly $40M to over $510M, and Morpho's syrupUSDC market liquidates over $130M within a 12 to 18% band.

Flashloan use tracks protocol, not holder size. On Aave 18.5% of sUSDe whales have used a flashloan against 4.16% of the holder base; on Morpho none of the 41 whales ever has, and flashloan volume there comes from a separate population of non-holding addresses.

7 Machine Learning Analysis

This section presents the machine learning analysis and addresses RQ1–RQ3. We first apply unsupervised clustering to the daily feature panel from Section 5 to identify recurring leverage and looping strategy archetypes (RQ1, RQ2), and then train supervised models on the same panel to test how far publicly observable on-chain state variables can predict short-horizon risk rebalancing and stressed risk states (RQ3). Treating both within a single analysis allows the unsupervised archetypes to be cross-checked against the supervised stress label, since the two are derived independently of one another.

7.1 Unsupervised Strategy Archetypes

This part addresses RQ1 and RQ2: what recurring leverage and looping strategy archetypes can be identified among large yield-bearing token holders, and which observable risk-management characteristics differentiate them? We cluster the daily feature panel from Section 5 with HDBSCAN to group looper-days by behaviour rather than by wallet, and characterise the resulting archetypes by their leverage and risk profiles.

7.1.1 Data and Features The analysis starts from the full combined daily panel: 2,506 looper-day observations across all 23 loopers (L-1 to L-16 on Morpho, L-17 to L-23 on Aave). We restrict it to the 2,069 days on which a looper holds an open, actively looped position, that is, days with a non-missing `ltv`, `carry_spread`, and `carry_spread_levered`; days with no open debt carry no looping behaviour to cluster. Each retained day is described by twelve features spanning leverage (`ltv`, `leverage`, `distance_to_liq`), leverage dynamics (`ltv_vol_7d`, `ltv_vol_14d`, `ltv_lag_7`), risk-management cadence (`risk_rebal_freq_7d`, `risk_rebal_freq_30d`, `days_since_risk_rebalance`), capital movement (`capital_flow_freq_30d`), and carry (`carry_spread`, `carry_spread_levered`). Cadence is measured from `risk_rebalance` rather than from the broad activity flag, so that the archetypes separate on risk management rather than on how often capital happens to move. Missing values are imputed with feature medians and all features are standardised so that HDBSCAN does not treat a high-variance feature as more important purely because of its scale.

The active days are spread very unevenly across loopers. Two loopers dominate the raw counts (L-11 with 622 active days and L-10 with 253), while several contribute only a handful (L-21 has 6, L-19 has 7, and L-15 and L-22 have 8 each). As Figure 25 shows, the loopers also operate at very different distances from liquidation: most sit at a mean health factor below 1.10 with minima at or near 1.00, while L-11, the looper contributing the most observations, keeps a mean close to 2.0. The most represented wallet is therefore also the least typical one.

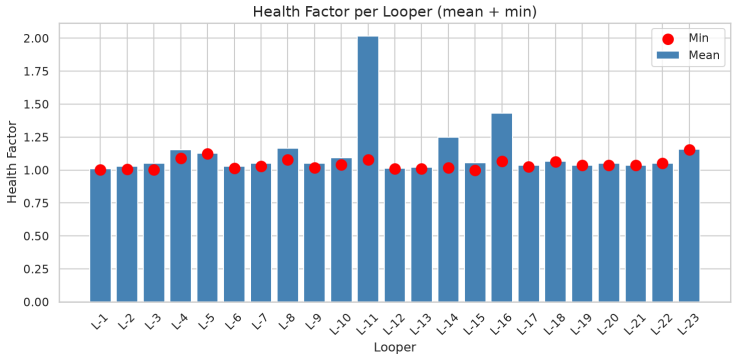


Fig. 25. Mean and minimum health factor per looper on days with an open debt position. Most loopers hold a mean health factor below 1.10 and touch a minimum at or near 1.00, meaning they sit permanently at the liquidation boundary. Only L-11 (≈ 2.0), L-16 and L-14 keep a real safety buffer.

Main findings. Risk profiles differ sharply even within the same market. L-1 remains near the liquidation boundary throughout its panel. In the sUSDS/USDT market, L-3 alternates between near-liquidation and safer health-factor levels,

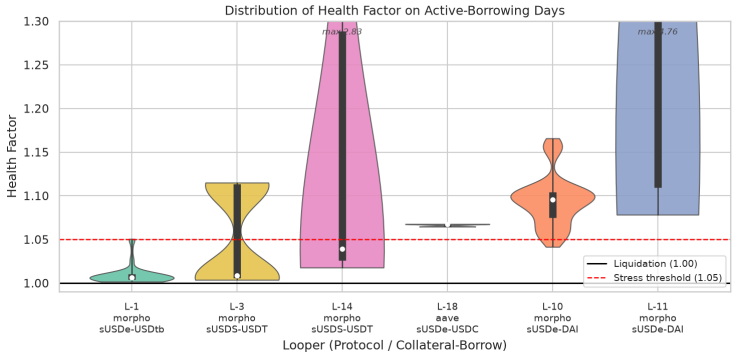


Fig. 26. Distribution of daily health factor for six representative loopers, restricted to days with an active borrow position. Loopers are labeled with their protocol and collateral-borrow market. The dashed red line marks the stress threshold (1.05); the solid black line marks the liquidation boundary (1.00).

Table 12. Summary statistics of health factor (HF) per looper, computed on active-borrowing days. *Market* is protocol / collateral-borrow pair; *N loopers* is the number of shortlisted loopers sharing that market; *Active days* is the number of days used to compute the statistics; *Stressed %* is the share of active days with HF ≤ 1.05.

Looper	Market	N loopers	Active days	Median HF	Min HF	Max HF	Stressed %
L-1	morpho, sUSDe-USDtb	1	77	1.006	1.002	1.051	97.4
L-3	morpho, sUSDS-USDT	2	94	1.009	1.004	1.115	56.4
L-14	morpho, sUSDS-USDT	2	110	1.039	1.017	2.833	70.0
L-18	aave, sUSDe-USDC	1	98	1.067	1.065	1.068	0.0
L-10	morpho, sUSDe-DAI	2	253	1.096	1.041	1.166	4.3
L-11	morpho, sUSDe-DAI	2	336	1.721	1.078	4.757	0.0

whereas L-14 stays close to the stress threshold with occasional spikes to 2.83. L-18 is the most stable holder and never crosses the stress threshold. In the sUSDe/DAI market, L-10 maintains a moderate buffer while L-11 ranges up to a health factor of 4.76. Because same-market pairs (L-3/L-14 and L-10/L-11) diverge this strongly, individual leverage choices appear to dominate market identity as a driver of risk, although a larger sample of loopers per market would be required to confirm this.

Table 13 reports the full open-debt LTV range for each looper (min, median, max). Markets are those in Table 6; the table below only adds the LTV distribution.

A per-looper mean hides how leverage moves within a position. Figure 27 plots zoomed LTV trajectories for three representative loopers on open-debt days only, with each market's liquidation LTV (LLTV) shown as a dashed line. L-12 holds a near-constant target just under its LLTV of 0.965 (about 0.95). L-11 is the opposite extreme: its LTV steps down from about 0.8 to about 0.2 over

Table 13. Open-debt LTV distribution by looper. Statistics use days with borrow balance at least $1. Market is collateral/borrow. Protocol and addresses are in Table 6.

ID	Market	Min	Median	Max	Open-debt days
L-1	sUSDe/USDtb	0.87	0.91	0.91	77
L-2	sUSDS/USDT	0.92	0.94	0.96	124
L-3	sUSDS/USDT	0.87	0.96	0.96	94
L-4	sUSDe/PYUSD	0.62	0.84	0.84	26
L-5	syrupUSDC/PYUSD	0.77	0.81	0.82	34
L-6	sUSDS/USDT	0.93	0.94	0.95	38
L-7	sUSDe/PYUSD	0.79	0.88	0.89	58
L-8	sUSDe/DAI	0.75	0.82	0.85	19
L-9	sUSDe/DAI	0.78	0.88	0.90	23
L-10	sUSDe/DAI	0.78	0.84	0.88	253
L-11	sUSDe/DAI	0.18	0.50	0.80	336
L-12	sUSDS/USDT	0.95	0.95	0.95	70
L-13	sUSDS/USDT	0.92	0.94	0.96	88
L-14	sUSDS/USDT	0.34	0.93	0.95	110
L-15	syrupUSDC/USDC	0.79	0.87	0.92	8
L-16	syrupUSDC/USDC	0.31	0.75	0.86	20
L-17	sUSDe/USDT	0.83	0.89	0.90	87
L-18	sUSDe/USDC	0.86	0.86	0.86	98
L-19	sUSDe/USDe	0.89	0.89	0.89	7
L-20	sUSDe/USDe	0.84	0.89	0.89	37
L-21	sUSDe/USDe	0.89	0.89	0.89	6
L-22	sUSDe/USDC	0.87	0.87	0.88	8
L-23	sUSDe/USDC	0.79	0.79	0.80	77

a year, well below an LLTV of 0.86. L-3 sits on the ceiling near 0.96 for weeks and then deleverages in one step to about 0.87. Within an open position, LTV is typically held near a fixed target rather than steered gradually; large moves are discrete adjustments. That pattern is what the leverage-dynamics features (`ltv_vol_7d`, `days_since_risk_rebalance`) later pick up as a main axis separating the archetypes.

7.1.2 Baseline Fit and Why It Fragments We use HDBSCAN, a density-based clustering algorithm that groups points in dense regions of feature space and labels sparse points as noise rather than forcing every observation into a cluster. This suits our setting, where many looper-days fall into a few recurring behavioural modes while transitional or idle days do not. We visualise the clusters by projecting the twelve standardised features onto their first two principal components (PCA), which is used only for display and not for the clustering itself.

A baseline fit on the full panel (`min_cluster_size`= 15, `min_samples`= 5) returns 40 small clusters together with 34.5% noise. The problem is that these clusters are largely looper-specific: because some loopers contribute many more days than others, the baseline structure reflects *who* has the most days as much

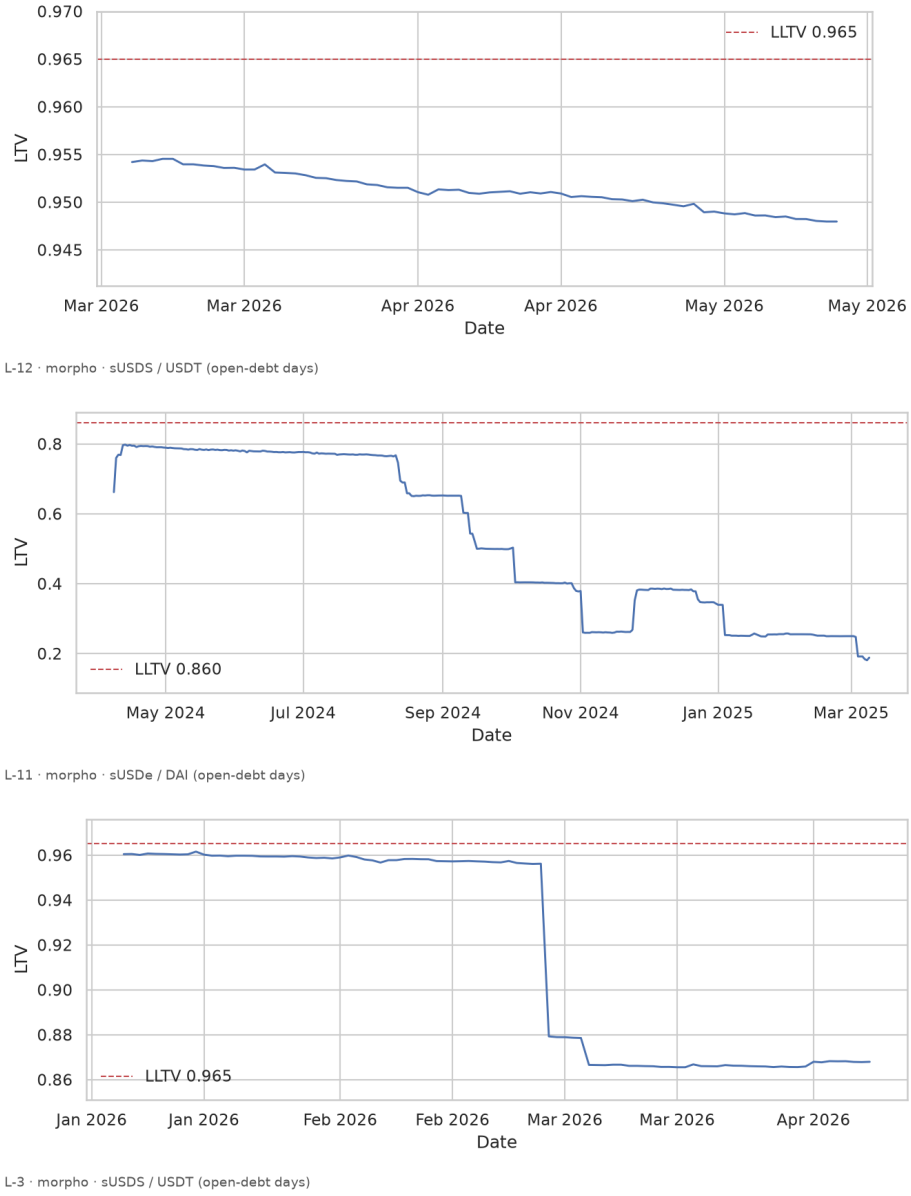


Fig. 27. Zoomed LTV over time for three representative loopers (open-debt days only). The dashed red line is the market LLTV. Top: L-12 (Morpho sUSDS/USDT) holds near 0.95 under LLTV 0.965. Middle: L-11 (Morpho sUSDe/DAI) steps down from about 0.8 to 0.2 under LLTV 0.86. Bottom: L-3 (Morpho sUSDS/USDT) sits near the ceiling then drops to about 0.87. Y-axes are zoomed to each looper's LTV band, not 0–1.

as *how* they behave. Colouring the same PCA projection by wallet makes this clear, with the high-day loopers (L-11 in particular) forming their own groups (Figure 28). Forty tiny clusters, several of them a single looper on its own, are too fragmented to read as clean strategy archetypes.

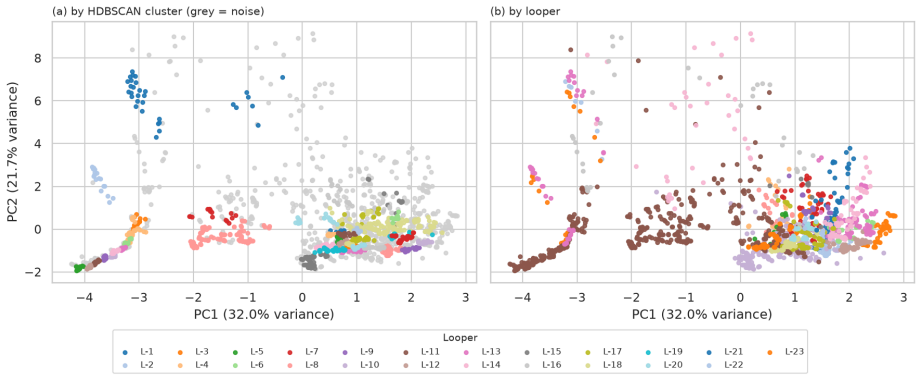


Fig. 28. PCA projection of looper-days from the baseline (unbalanced) fit, coloured by HDBSCAN cluster (left) and by looper (right). The cluster boundaries largely coincide with wallet identity, which shows that the unbalanced fit recovers looper-specific groups rather than shared strategy archetypes and motivates the balanced fit in Section 7.1.3.

7.1.3 Balanced Fit and Final Clusters To recover shared strategy archetypes rather than per-wallet groups, we give each looper equal weight by capping its contribution at 40 days. Loopers with 40 or fewer active days keep all of them, while the two high-day loopers (L-11 and L-10) are trimmed to 40. This keeps most of the real data while stopping the largest loopers from dominating the fit, and leaves a balanced sample of 723 days.

We then sweep the HDBSCAN parameters on this balanced sample. Small `min_cluster_size` values (8 to 12) reproduce the earlier fragmentation, giving 16 to 30 tiny clusters. Raising it to the range 60 to 90 collapses the structure to 3 to 5 clusters. At `min_cluster_size`= 60 two of the five clusters describe almost the same risk state, split in two; at 75 and 90 we instead get three clean, clearly distinct clusters. We adopt `min_cluster_size`= 75 and `min_samples`= 1, which gives three clusters at 46% noise, the lowest noise among the settings that still produce three clean groups.

To label every day rather than only the balanced sample, we take the centroid of each of the three clusters, assign all 2,069 full-panel days to their nearest centroid, and flag the furthest 5% of days as outliers. This yields 1,148 stale low-leverage days, 512 active looping days, 305 max-leverage-hold days, and 104 outliers, named from their feature profiles rather than fixed cluster IDs.

7.1.4 Identified Archetypes The four archetypes and their mean profiles are reported in Table 14 and visualised in Figure 31. The per-looper composition of each archetype is given in Table 15.

Answering RQ2, the three looping archetypes are differentiated primarily by leverage level and by the cadence with which leverage is adjusted. Levered carry is a secondary signal that largely tracks leverage rather than separating the groups independently. Unless otherwise noted, the figures below are means over the looper-days assigned to each archetype (Table 14).

The stale, low-leverage archetype has a mean LTV of 0.54, a mean leverage of 4.1×, and a mean liquidation buffer of 0.36. Positions in this group are rarely risk-rebalanced, with a mean risk-rebalancing frequency of 0.01 events per week. This archetype generates the lowest levered carry and represents a relatively conservative, passively managed strategy. It is also the largest archetype, comprising 1,148 position-days across 16 of the 23 loopers.

The active-looping archetype has a mean LTV of 0.78 and a mean leverage of 8.1×. These positions are actively managed, with a mean risk-rebalancing frequency of 1.4 events per week and 3.2 events per 30-day period. This pattern indicates that borrowers maintain relatively high leverage while frequently adjusting their positions to manage liquidation risk. Active looping is the second-largest archetype, comprising 512 position-days (22 loopers have at least one such day).

The maximum-leverage-hold archetype has a mean LTV of 0.94, a mean leverage of 18.8×, and a mean liquidation buffer of 0.02. Despite remaining close to the liquidation threshold, these positions are risk-rebalanced infrequently, with a mean frequency of 0.1 events per week. This archetype therefore represents an aggressive but largely passive looping strategy in which borrowers maintain near-maximal leverage for extended periods (305 position-days; 6 loopers).

The outlier group comprises the 104 days furthest from all three centroids. These observations are not parked positions: they have a low mean LTV (0.23) and a wide buffer, yet exhibit the highest short-horizon LTV volatility and risk-rebalance about as often as active looping (1.8 versus 1.4 events per week). Outlier days are therefore best interpreted as unsettled transitions between strategies rather than as a maintenance mode (7 loopers).

Separating deliberate risk rebalancing from other position activity sharpens the comparison. Every archetype records more `position_activity` days than risk-rebalance days (1.1, 2.1, and 0.5 touches per week for maximum-leverage hold, active looping, and stale low leverage, respectively, against 0.1, 1.4, and 0.01 risk rebalances), while capital flows remain similarly infrequent across groups (`capital_flow_freq_30d` of 0.68, 0.73, and 0.27). The archetypes are therefore distinguished by risk management, not by how often capital enters or leaves the position.

Cadence features in the clustering input are `risk_rebal_freq_7d`, `risk_rebal_freq_30d`, `days_since_risk_rebalance`, and `capital_flow_freq_30d`. These underlie Figures 31 and 29; Figure 32 uses mean `days_since_risk_rebalance` on the horizontal axis.

We report cadence as event counts rather than as the mean of `days_since_risk_rebalance`, which is a poor summary here: the feature is 0 on the days it counts, and a handful of long-dormant days skew the mean upward. For outlier days the mean is 9.3 while the median is 3, because 19 of the 104 days exceed two weeks.

Table 14. Strategy archetypes: day counts and mean risk / cadence profiles. Every numeric profile column is the *mean over the looper-days assigned to that archetype* (2,069 assigned days in total), not a median and not an average of per-looper means. *Loopers* counts distinct wallets with at least one day in the archetype (a looper may appear in more than one row). *Lev.* is leverage, $1/(1 - \text{LTV})$; *Buffer* is the remaining LTV distance to liquidation; *Carry* is `carry_spread_levered`. *Risk/7d* and *Risk/30d* are the means of `risk_rebal_freq_7d` and `risk_rebal_freq_30d`, rolling counts of days on which the looper risk-rebalanced; *Touch/7d* is `activity_freq_7d`, the broader count of any supply/withdraw/borrow/repay day. Cadence is reported as event counts rather than mean days since the last risk rebalance, which is skewed by same-day zeros and by a few long-dormant outliers.

Archetype	Days	%	Loopers	LTV	Lev.	Buffer	Risk/7d	Risk/30d	Touch/7d	Carry
stale, low leverage	1,148	55	16	0.54	4.1	0.36	0.01	0.7	0.5	2.7
active looping	512	25	22	0.78	8.1	0.14	1.4	3.2	2.1	11.4
max leverage hold	305	15	6	0.94	18.8	0.02	0.1	1.1	1.1	8.1
outlier	104	5	7	0.23	1.6	0.70	1.8	5.3	2.3	−5.7

The *Loopers* column in Table 14 shows how many of the 23 wallets contribute days to each mode. Active looping and stale low leverage are widely shared; max leverage hold is concentrated in six wallets.

Figure 29 presents the same membership split as a stacked bar per looper. Pure strategists appear as bars dominated by a single colour: L-12 is entirely maximum-leverage hold, and L-15, L-19, L-21 and L-22 are entirely active looping. Mixed strategists are distributed across two or more archetypes, including L-1, L-3, L-8 and L-14. A few loopers concentrate in one mode without meeting the pure threshold; L-6, for example, spends most of its days in maximum-leverage hold but retains a block of active looping. Overall, most loopers settle into one primary behaviour, while a smaller group moves between several.

The mix chart reports proportions but not temporal order. Figure 30 plots the same assigned days chronologically, one row per looper, so that strategy evolution is visible. Mode switches are typically one-directional rather than oscillatory: L-3 opens in maximum-leverage hold in early 2026 and settles into stale low leverage; L-11 spends most of a year at low leverage with intermittent bursts of active looping before winding down; and L-2's maximum-leverage phase in 2025 ends in a single transition out. The timeline also shows that the archetypes are not calendar artefacts: in any given month, several loopers occupy different modes side by side.

Table 15. Archetype membership as a percentage of each looper's days. Rows may not sum to exactly 100 due to rounding.

Looper	active looping	max leverage hold	stale, low lev.	outlier
L-1	70.1	0.0	29.9	0.0
L-2	1.1	89.8	1.1	8.0
L-3	9.4	38.1	46.8	5.8
L-4	50.0	0.0	50.0	0.0
L-5	20.6	0.0	79.4	0.0
L-6	50.0	50.0	0.0	0.0
L-7	87.9	0.0	12.1	0.0
L-8	78.9	0.0	21.1	0.0
L-9	91.3	0.0	8.7	0.0
L-10	11.1	0.0	87.0	2.0
L-11	15.8	0.0	80.1	4.2
L-12	0.0	100.0	0.0	0.0
L-13	25.7	33.3	25.0	16.0
L-14	49.6	31.9	5.3	13.3
L-15	100.0	0.0	0.0	0.0
L-16	45.9	0.0	0.0	54.1
L-17	23.0	0.0	77.0	0.0
L-18	7.1	0.0	92.9	0.0
L-19	100.0	0.0	0.0	0.0
L-20	51.4	0.0	48.6	0.0
L-21	100.0	0.0	0.0	0.0
L-22	100.0	0.0	0.0	0.0
L-23	9.1	0.0	90.9	0.0



Fig. 29. Archetype mix by looper. Each bar is one looper's active days split by archetype. Pure strategists are dominated by a single archetype, while mixed strategists spread across several.

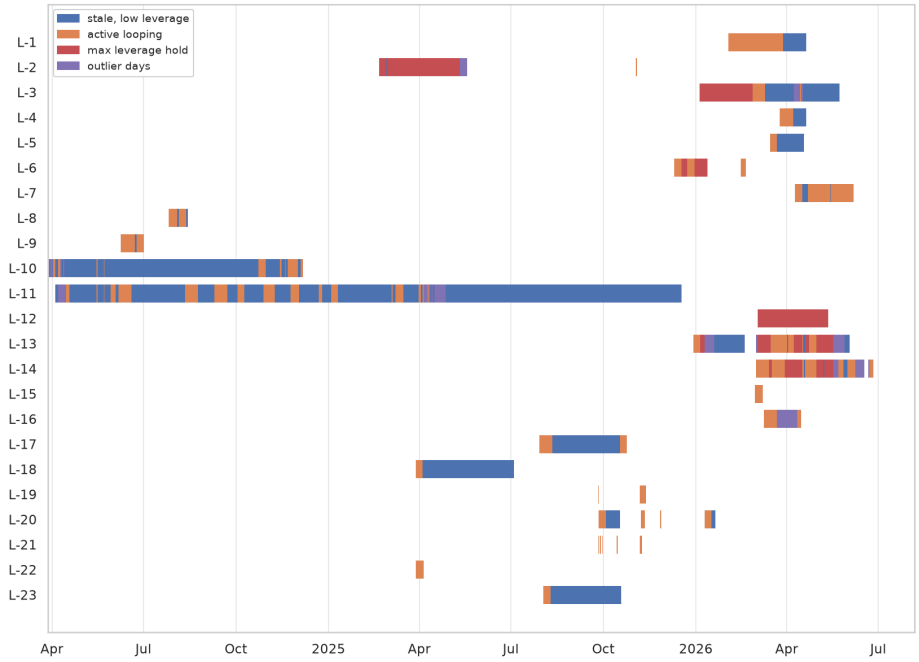


Fig. 30. Archetype assignment over calendar time, one row per looper (active days only; gaps are days with no open looped position). Colours match the four archetypes. Reading along a row shows how a looper's strategy evolves; reading down a column shows that modes coexist across loopers at any point in time.

7.1.5 Cross-check against the Supervised Stress Label The archetypes above are built purely from behavioural features and never see the supervised `stressed_state` label of Section 7.2. Laying that label back over the assigned days is a simple check of whether the behavioural clusters line up with realised risk. All 2,069 archetype days match a label, and the overall stressed share is 29.0%.

Figure 32 plots each archetype's stressed share against how stale it is, measured as the mean number of days since the last risk rebalance. The stressed share tracks leverage, not staleness. *Max leverage hold* and *stale, low leverage* are both stale (about 30 versus 72 days since the last risk rebalance), yet their stressed rates are 100% versus 11%. The former is expected: positions running at 19× leverage sit so close to the liquidation threshold that they are stressed by definition. The latter shows that a long quiet spell alone does not imply risk. A long gap since the last risk rebalance is therefore not itself a risk signal; what matters is whether the stale position is also highly levered. This mirrors the supervised stress model of Section 7.2, where `ltv` had to be excluded for leakage precisely because a high LTV almost perfectly separates stressed days.

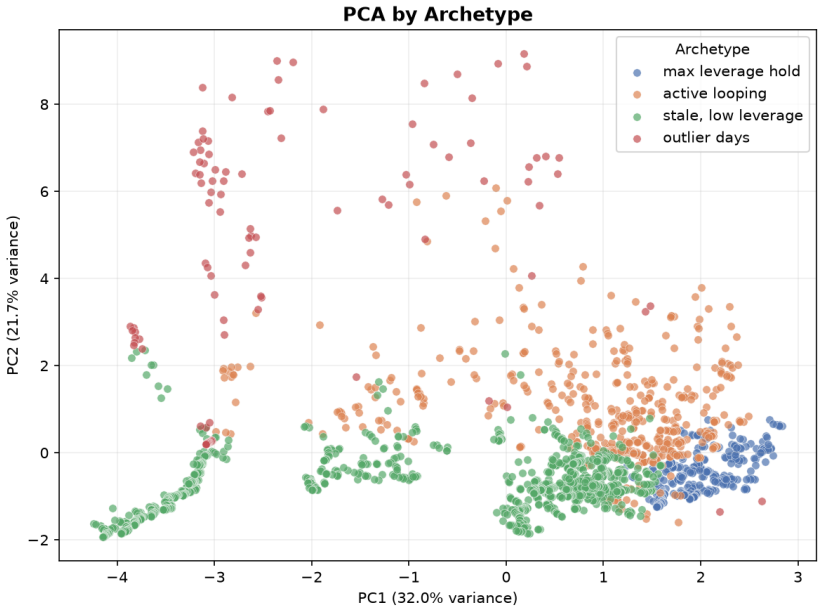


Fig. 31. PCA projection of looper-days coloured by assigned archetype. The first two components explain 53% of the variance (32.0% plus 21.0%). The three dense looping archetypes separate on leverage level and the cadence of risk rebalances; the outlier group collects unwound or transitional days.

7.1.6 Findings The clustering addresses RQ1 and RQ2: a small number of recurring looping modes can be identified, and they are separated mainly by leverage level and by how often leverage is adjusted.

Four modes, not twenty-three looper types. After balancing across the 23 loopers, the days fall into three dense clusters plus an outlier group. These modes describe *how* a position is managed, not *which* wallet is observed. The Aave loopers (L-17 to L-23) do not form a protocol-specific cluster; they mix into the stale low-leverage and active-looping modes alongside the Morpho loopers.

About half the loopers adhere to a single mode. Twelve of the 23 loopers are *pure*, in the sense that at least 80% of their days fall in one archetype: L-2 and L-12 (maximum-leverage hold); L-7, L-9, L-15, L-19, L-21 and L-22 (active looping); and L-10, L-11, L-18 and L-23 (stale low leverage). The remaining 11 are *mixed*. L-13 is the clearest mixed case, distributed fairly evenly across all four archetypes (33% maximum-leverage hold, 26% active looping, 25% stale low leverage, 16% outlier). Strategy is therefore better characterised as switching between shared modes over time than as a fixed type per wallet.

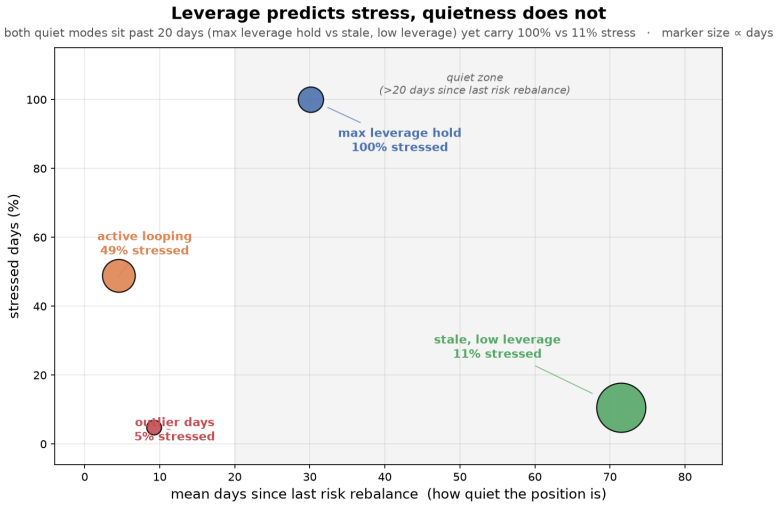


Fig. 32. Stressed-day share by archetype plotted against staleness (mean days since the last risk rebalance), using the supervised `stressed_state` label (health factor ≤ 1.05); marker area is proportional to the number of days. Max leverage positions are stressed by definition (near liquidation), while stale low-leverage positions carry only 11% stress despite also being stale, showing that a long quiet spell alone does not predict risk.

Three loopers changed dominant archetype under the strict cadence. A looper's dominant archetype is the mode with the plurality of its days. We compare that plurality before and after replacing any-touch cadence with `risk_rebalance` cadence (frequency and recency features) in the clustering input; Figure 29 shows the post-change mix. Under that change, 382 of 2,069 assigned days (18.5%) moved between archetypes, but only three wallets changed which archetype held the plurality of their days: L-3 (active looping → stale, low leverage), L-4 (stale, low leverage → active looping), and L-16 (active looping → outlier days). All three remain mixed strategists. L-11 moved from mixed to pure stale low leverage (80% of its days). The remaining wallets retained the same dominant mode; for example, L-12 remains entirely maximum-leverage hold and L-15, L-19, L-21 and L-22 remain entirely active looping.

Half of recorded position activity is not risk rebalancing. Loopers exhibit `position_activity` on 14.8% of looper-days but risk-rebalance on only 7.2%; the remainder consists of capital flows (2.0%) and `minor_activity` (5.6%). Table 16 reports the same counts by looper. The gap is widest for holders that appear busiest under the broad activity flag: L-3 is active on 47.5% of its days yet changes leverage on only 3.6% (66 activity days contain five risk rebalances), and L-11 on 15.8% versus 3.5%. Every looper risk-rebalances at least twice; none is a pure capital mover. The panel therefore separates wallets that manage a leveraged position

(L-14, L-15, L-16, at 24.6% to 44.7% of their days) from those that park a position and only occasionally move capital.

Table 16. Per-looper activity taxonomy (looper-days). *Protocol* and *Market* (collateral/borrow) identify the position from Table 6; *Days* is the length of that looper's panel; *Touch* is `position_activity`; *Risk* is `risk_rebalance`; *Periods* counts risk-rebalance episodes: a run of consecutive flagged days is one period, and a gap of at least one calendar day starts a new one; *Lev. adj.* is `leverage_adjust`; *Capital* is `capital_flow`; *Lev. scale* is `levered_scaling`; *Minor* is `minor_activity`; *Risk %* is risk days as a share of that looper's days. Counts are looper-days: several on-chain transactions on the same day count as one flagged day, never as multiple events.

Looper	Protocol	Market	Days	Touch	Risk	Periods	Lev. adj.	Capital	Lev. scale	Minor	Risk %
L-1	Morpho	sUSDe/USDtb	78	15	11	5	5	3	6	1	14.1
L-2	Morpho	sUSDS/USDT	300	11	10	7	3	1	7	0	3.3
L-3	Morpho	sUSDS/USDT	139	66	5	5	1	7	4	54	3.6
L-4	Morpho	sUSDe/PYUSD	27	5	4	4	0	1	4	0	14.8
L-5	Morpho	syrupUSDC/PYUSD	35	4	2	2	0	2	2	0	5.7
L-6	Morpho	sUSDS/USDT	72	7	6	6	1	0	5	1	8.3
L-7	Morpho	sUSDe/PYUSD	58	26	10	8	4	5	6	11	17.2
L-8	Morpho	sUSDe/DAI	20	4	4	3	1	0	3	0	20.0
L-9	Morpho	sUSDe/DAI	24	7	7	6	2	0	5	0	29.2
L-10	Morpho	sUSDe/DAI	254	4	4	4	0	0	4	0	1.6
L-11	Morpho	sUSDe/DAI	622	98	22	17	8	11	14	65	3.5
L-12	Morpho	sUSDS/USDT	71	2	2	2	0	0	2	0	2.8
L-13	Morpho	sUSDS/USDT	156	19	10	8	5	7	5	2	6.4
L-14	Morpho	sUSDS/USDT	118	34	29	14	4	2	25	3	24.6
L-15	Morpho	syrupUSDC/USDC	9	5	4	2	1	1	3	0	44.4
L-16	Morpho	syrupUSDC/USDC	38	20	17	4	7	3	10	0	44.7
L-17	Aave	sUSDe/USDT	88	6	5	4	0	0	5	1	5.7
L-18	Aave	sUSDe/USDC	99	3	2	2	0	1	2	0	2.0
L-19	Aave	sUSDe/USDe	49	5	4	3	0	1	4	0	8.2
L-20	Aave	sUSDe/USDe	117	8	8	7	0	0	8	0	6.8
L-21	Aave	sUSDe/USDe	45	12	10	4	0	1	10	1	22.2
L-22	Aave	sUSDe/USDC	9	5	2	2	0	3	2	0	22.2
L-23	Aave	sUSDe/USDC	78	4	2	2	0	1	2	1	2.6

Day counts overstate multi-day campaigns. The *Periods* column of Table 16 merges runs of consecutive risk-rebalance days into single episodes. Across the panel, the 180 risk-rebalance days collapse to 121 episodes (370 activity days collapse to 208), so 59 of 180 risk-rebalance days (33%) continue an adjustment begun the day before rather than initiating a new decision. The gap between the two counts separates two management styles. One-shot actors adjust within a single day: L-3's five risk-rebalance days are five isolated episodes. Multi-day campaigners spread one decision over consecutive days, presumably to limit market impact: L-16's 17 risk-rebalance days form only four episodes, one of which is a three-week staged wind-down of the entire position, and L-14's 29 days form 14 episodes. Day counts alone therefore overstate decision frequency for such loopers, whereas the episode count better reflects the number of distinct adjustments. Figure 33 makes the periods visible in time: each looper's LTV trajectory

is overlaid with its risk-rebalance days, and runs of consecutive days are shaded as one period.

A stale position is not risky on its own. A long gap since the last risk rebalance appears in two very different modes. Stale low leverage is quiet because it is comparatively safe: low LTV, a wide buffer, and little need to adjust. Maximum-leverage hold is also quiet, but while running high leverage and a thin buffer, which is the risky case. A long quiet spell by itself is therefore not informative about risk; what matters is a stale position held at high leverage, and that is the combination that warrants monitoring.

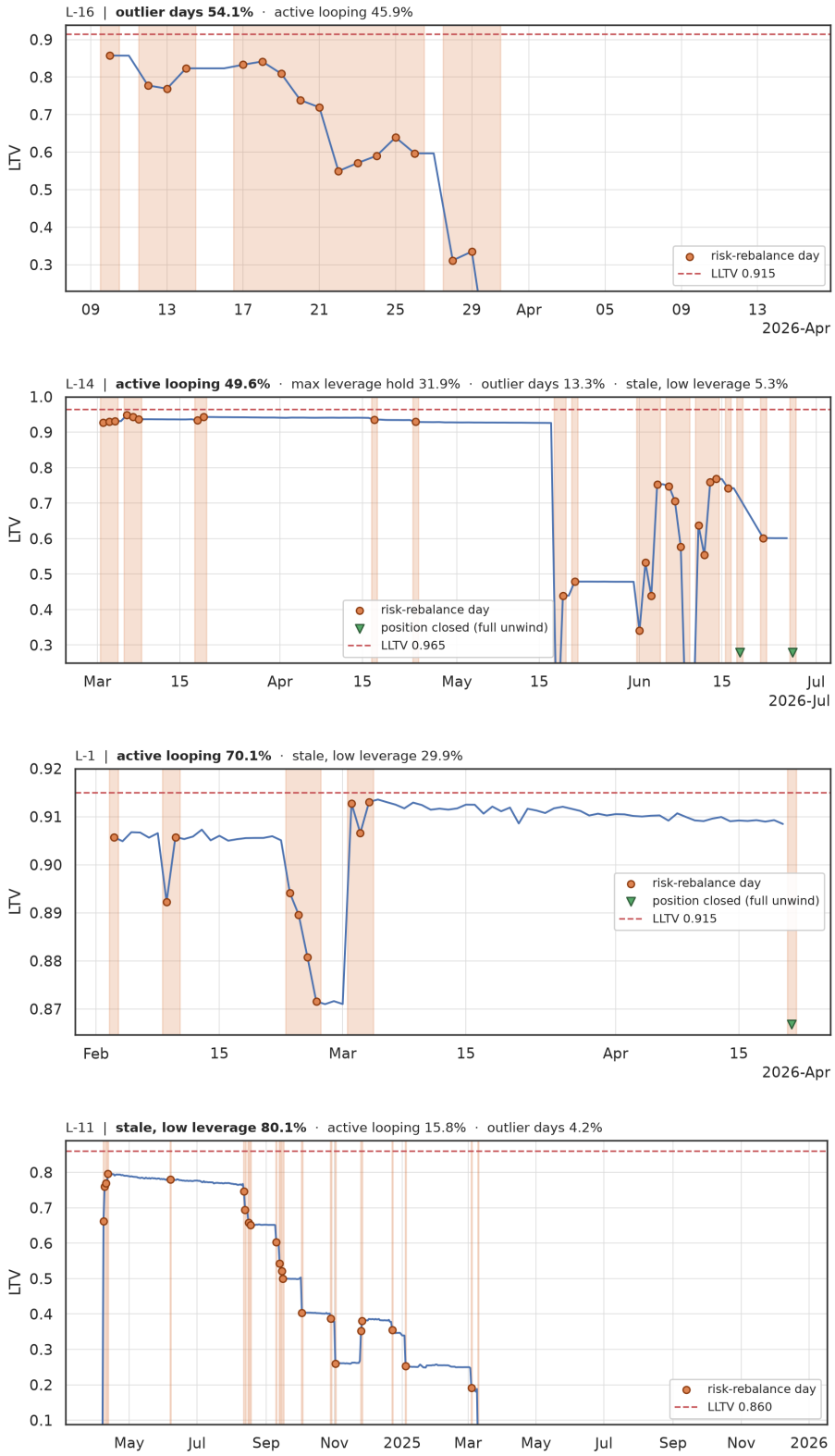


Fig. 33. Chronology of risk rebalancing for four loopers with multi-day periods. The line is the daily LTV (zoomed to each looper's band), the dashed line is the market LLTV, dots mark risk-rebalance days, and shaded bands merge consecutive days into one period. Each panel's header states the looper's archetype mix with its dominant archetype in bold. Green triangles mark full unwinds: the position is closed that day, so LTV is undefined and the line cannot reach the shaded period.

7.2 Supervised Prediction of Risk Rebalancing and Stress

This section addresses RQ3: to what extent can on-chain state variables predict short-horizon risk rebalancing and stressed risk states? We train supervised models on the daily feature panel constructed in Section 5 in order to assess whether publicly observable on-chain signals carry sufficient information to anticipate looper behaviour and position risk. The problem is formulated as two independent binary classification tasks: whether a looper changes its leverage on a given day (Model A), and whether a looper occupies a stressed risk state on a given day (Model B).

7.2.1 Data and Labels The analysis draws on the daily feature panel of 2,506 looper-day observations across 23 loopers (L-1 through L-23), spanning 2024-03-29 to 2026-06-27. Each observation carries the leverage, risk, behavioural, and carry features defined in Section 5.

The target for Model A is `risk_rebalance`, the union of days on which leverage moved materially (`leverage_adjust` or `levered_scaling`; see Section 5.3). This target is deliberately narrower than the broad `position_activity` flag, which registers any supply, withdraw, borrow, or repay event and therefore conflates capital movement and `minor_activity` with deliberate risk management. Risk-rebalance days constitute the minority class, accounting for 7.2% of observations (180 of 2,506 days), compared with 14.8% under the broad activity flag.

The target for Model B, `stressed_state`, is defined by a fixed absolute criterion: a looper-day is labelled as stressed if its health factor falls to 1.05 or below. A fixed threshold is preferred to a per-looper percentile because health factor carries a fixed economic interpretation, with 1.0 denoting the liquidation boundary. A per-looper percentile criterion (for example, marking each looper's worst 30% of days) would evaluate each looper only relative to itself, so that a holder operating permanently near 1.0 would appear no riskier than one that remains consistently above 1.4, since both would exhibit comparable stressed rates. The fixed threshold ensures that the label reflects genuine proximity to liquidation and remains comparable across all loopers.

7.2.2 Exploratory Analysis Prior to model estimation, we examine the distributions of health factor, distance to liquidation, stressed rate, and position activity across all 23 loopers.

Health factor and distance to liquidation. The majority of loopers cluster tightly, exhibiting median health factors between 1.0 and 1.1 with minima close to 1.0. Several holders deviate markedly from this pattern. L-11 is the most pronounced outlier, with a median health factor of 2.0 and a maximum exceeding 4.7; its position is predominantly conservative but subject to occasional sharp contractions in buffer. L-16 and L-14 likewise operate above the sample average, with medians of 1.4 and 1.25 respectively. At the opposite end of the distribution,

L-1 and L-12 maintain median health factors close to 1.01, indicating that they operate with almost no buffer above the liquidation boundary at any point in their panels.

Distance to liquidation. The distance-to-liquidation metric exhibits the same broad structure. L-11 and L-16 record the highest median buffers, both above 0.5, although their minima still fall to 0.06 and 0.08. L-3 and L-13 display a combination of high median and very low minimum: their median buffers are 0.34 and 0.39, while their minima approach zero, indicating alternation between comparatively safe periods and abrupt episodes of proximity to liquidation. L-1 occupies the opposite extreme, with both median and minimum close to zero.

Across both metrics the sample separates into three broad groups: holders combining high average safety with occasional risky episodes (L-11, L-16, L-3, L-13); consistently conservative holders that never approach liquidation (L-4, L-5, L-8, L-23); and holders that remain close to liquidation almost continuously (L-1, L-12).

Stressed rate under the fixed threshold. Under the 1.05 threshold, the aggregate stressed rate is 28.9% (725 of 2,506 days), with substantial heterogeneity across holders. L-12 and L-1 are stressed on more than 95% of their days, while L-17 and L-7 record rates of 90.9% and 81.0% respectively. At the other extreme, eight loopers (L-4, L-5, L-8, L-11, L-16, L-18, L-22, L-23) never reach or fall below the threshold. The remaining holders lie between these bounds, with stressed rates ranging from 13% to 71%.

Realised liquidations. No liquidation events are observed across the 23 loopers over the full sample period. Even the most aggressively geared holders remained unliquidated: L-1 and L-12 maintained health factors close to 1.01 on the majority of their days, retaining almost no buffer above the liquidation boundary, yet neither position was closed by a liquidator. This finding qualifies the interpretation of the stressed label. Although 28.9% of looper-days fall at or below the 1.05 threshold, none of these days resulted in liquidation. Within this dataset a stressed state therefore denotes a position that approached the liquidation boundary and subsequently recovered, rather than one that failed. When holders approached this boundary they responded by supplying additional collateral, repaying debt, or awaiting a favourable market movement, a pattern consistent with the active risk management documented below.

Position activity relative to risk rebalancing. Loopers register position activity on 370 of 2,506 days (14.8%) but adjust leverage on only 180 (7.2%). Ranking holders by contact frequency therefore misrepresents which of them actively manage risk. L-15 and L-22 are active on 55.6% of their days and L-3 on 47.5%, yet L-3 risk-rebalances on only 3.6% of its days (66 activity days contain five risk rebalances). The most active risk managers under the narrower criterion are L-16 (44.7%), L-15 (44.4%), L-9 (29.2%), and L-14 (24.6%). Model A is estimated against this narrower signal.

7.2.3 Models and Setup For each task the data are partitioned into stratified training and test sets in a 70/30 ratio, and two classifiers are estimated. A logistic regression provides an interpretable linear baseline, and a gradient-boosted tree model (XGBoost) serves as the principal nonlinear specification. This partition estimates predictive signal within the panel; held-out-looper validation, in which the model is trained on one subset of holders and evaluated on another, is identified as a direction for future robustness work. XGBoost constructs an ensemble of shallow decision trees fitted sequentially, each correcting the residual errors of its predecessors, and is well suited to imbalanced tabular data of the kind considered here. Model interpretation relies on SHAP (SHapley Additive exPlanations), which attributes each prediction to its constituent features and quantifies the direction and magnitude of each feature's contribution. This attribution is what permits a transition from a binary prediction to a substantive statement regarding *which* on-chain variables drive looper behaviour and to what extent, as required by RQ3.

To address class imbalance in the risk-rebalance task, in which the positive base rate is 6% within the modelling set, the positive class is weighted using XGBoost's `scale_pos_weight` parameter (16.49). For the stressed-state task the decision threshold is lowered from the default 0.5 to 0.3; both thresholds are evaluated for both classifiers, and this choice is justified in Section 7.2.5.

Avoiding data leakage. Any feature that encodes the quantity it is intended to predict is excluded from the corresponding specification. Model A omits the cadence and action-delta features derived from its target (`risk_rebal_freq_7d`, `risk_rebal_freq_30d`, `days_since_risk_rebalance`, `delta_leverage_rel`, `delta_equity_rel`).

Model B required a more extensive exclusion procedure. The initial feature set comprised `ltv` and its lagged variants, `health_factor`, `distance_to_liq`. The first estimation run returned a ROC-AUC of 1.000, indicating leakage. Inspection of the underlying values identified the cause: an `ltv` above 0.87 separated stressed from non-stressed days almost perfectly. Both features were removed, together with `ltv_lag_1`, `ltv_lag_3`, and `ltv_lag_7`, since `health_factor`, which defines `stressed_state`, is derived directly from `ltv`. A second and more subtle leak was subsequently identified: the inclusion of `collateral_balance` and `borrow_balance` allowed XGBoost to attain a ROC-AUC of 0.993, an improvement over logistic regression larger than would be expected on substantive grounds. Tree-based models can exploit implicit cut-points in raw magnitudes more readily than linear specifications, so a leak invisible to logistic regression remained detectable in XGBoost. Removing these two balance features returned XGBoost to a plausible 0.977, close to the logistic regression benchmark of 0.971, confirming that the leakage had been resolved.

The final exclusion list for Model B comprises: `ltv`, `ltv_lag_1`, `ltv_lag_3`, `ltv_lag_7`, `health_factor`, `distance_to_liq`, `collateral_balance_usd`, `borrow_balance_usd`, `liquidation_threshold`, `collateral_balance`, and `borrow_balance`.

7.2.4 Model A: Predicting a Leverage Change Model A predicts whether a looper changes its leverage on a given day using 11 features: current and lagged LTV (`ltv`, `ltv_lag_1`, `ltv_lag_3`, `ltv_lag_7`), LTV volatility (`ltv_vol_7d`, `ltv_vol_14d`), distance to liquidation, health factor, carry spread in raw and levered form, and leverage. After removal of observations with missing features, the modelling set comprises 1,475 rows; a stratified 70/30 partition yields 1,032 training and 443 test observations, of which 26 are positive, corresponding to a base rate of 5.9%. Results are reported in Table 17.

Table 17. Model A: risk-rebalance prediction (`risk_rebalance`).

Model	ROC-AUC	Recall	Precision
Logistic Regression	0.881	0.62	0.42
XGBoost	0.901	0.81	0.26

Accuracy is uninformative at this base rate, since a trivial classifier predicting the negative class throughout would attain 94%. XGBoost identifies 21 of the 26 leverage change, against 16 for the linear baseline, but generates 61 false positives against 22. XGBoost is therefore preferable where the cost of a false negative exceeds that of a false positive, whereas the baseline is better calibrated under symmetric costs. Given only 26 positive test observations, these differences should be interpreted with caution. Both specifications separate the classes more effectively than the broad-activity target employed in earlier estimation (ROC-AUC 0.820 and 0.877 respectively), despite the narrower target being rarer, which indicates that deliberate risk management is more learnable than undifferentiated contact with a position.

Interpretation of the nonlinear model proceeds through its feature importance (Figure 34) and the SHAP summary of its predictions (Figure 35). `ltv_vol_7d` is dominant (split weight 52), followed by `carry_spread_levered` (18), `ltv_vol_14d` (17), `carry_spread` (14), and `ltv_lag_1` (13). Contemporaneous leverage level (`ltv`, `leverage`) contributes little, and `health_factor` is never selected by any tree.

The prominence of the carry features under this target constitutes the clearest evidence that the activity taxonomy split was warranted: once capital movements and `minor_activity` are excluded, the economics of the position become predictive. The SHAP summary corroborates the direction of these effects. Stable leverage contributes strongly against a predicted leverage change, whereas volatile leverage removes this negative contribution rather than predicting a change in its own right. A weak carry spread likewise pushes a day away from a predicted leverage change, indicating that positions which are not earning tend to be left unmanaged.

7.2.5 Model B: Predicting Stressed States Model B predicts whether a looper occupies a stressed risk state on a given day using 8 features: LTV volatil-

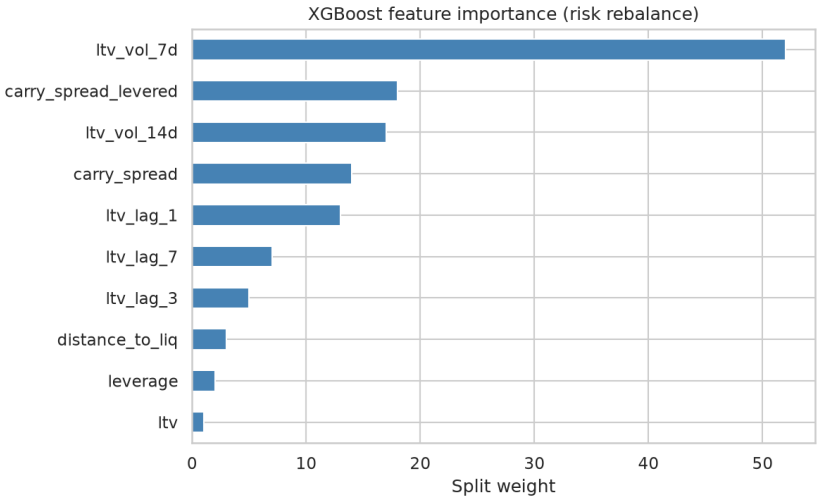


Fig. 34. Feature importance (split weight) for the XGBoost risk-rebalance model. Bars report the frequency with which each feature is used across all trees. LTV volatility (`ltv_vol_7d`) is dominant, and the carry features rank near the top, in contrast to their position under the broad activity flag. `health_factor` is never selected by any tree and therefore does not appear.

ity (`ltv_vol_7d`, `ltv_vol_14d`), carry spread in raw and levered form, yield and interest rate variables (`ybt_apy`, `borrow_rate`), and token prices (`collateral_price`, `borrow_price`). The feature set departs from that of Model A because several variables admissible when predicting a leverage change, notably `ltv`, `health_factor`, and `leverage`, are too closely tied to the definition of `stressed_state` and would introduce leakage, as set out in Section 7.2.3.

After removal of observations with missing features, the modelling set comprises 2,088 rows; a stratified 70/30 partition yields 1,460 training and 628 test observations. Both classifiers are evaluated at two decision thresholds, the default 0.5 and a lowered 0.3, and all four configurations are reported in Table 18.

Table 18. Model B: stressed-state prediction across both classifiers and both decision thresholds. ROC-AUC is threshold-independent and therefore repeats within each model. The selected configuration (XGBoost at $t = 0.3$) is shown in bold.

Model	t	ROC-AUC	Recall	Precision	Accuracy
Logistic Regression	0.5	0.971	0.95	0.75	0.89
Logistic Regression	0.3	0.971	0.98	0.67	0.84
XGBoost	0.5	0.977	0.82	0.93	0.92
XGBoost	**0.3**	**0.977**	**0.90**	**0.81**	**0.90**

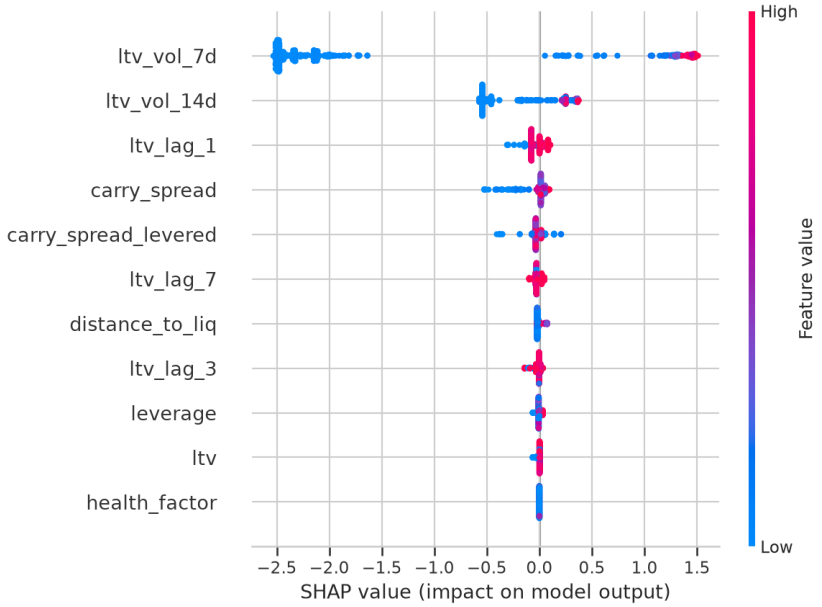


Fig. 35. SHAP summary for the XGBoost risk-rebalance model. Each point represents a looper-day; its horizontal position gives the feature's signed contribution to the predicted log-odds of a leverage change, and colour encodes the feature value (red high, blue low). Features are ordered by median absolute SHAP value. Stable short-window LTV volatility contributes strongly against a predicted leverage change, while carry economics constitute a secondary signal.

Choice of decision threshold. Under the default configuration a classifier labels a day as stressed only where the predicted probability exceeds 0.5. This threshold is lowered to 0.3, so that a flag is raised whenever the model assigns at least 30% confidence, thereby capturing a greater proportion of stressed days at the cost of additional false positives. The trade-off is deliberate: in a risk-monitoring application the cost of a false negative is substantially higher than that of a false positive, since an unflagged stressed position may proceed to liquidation whereas a false alarm entails only an unnecessary review. Lowering the threshold accordingly prioritises recall, which is the quantity of principal interest for this task.

Choice of model. The two classifiers behave differently at the lower threshold. Logistic regression at $t = 0.3$ attains the highest recall (0.98) but at a substantial cost in precision: only 0.67 of its positive predictions are correct, and accuracy falls to 0.84. XGBoost at $t = 0.3$ attains a recall of 0.90, identifying 180 of 199 stressed days and omitting 19, while retaining a precision of 0.81, corresponding to 42 false positives out of 429 non-stressed days, an accuracy of 0.90, and a higher ROC-AUC (0.977 against 0.971). XGBoost therefore identifies stressed

states from their underlying structure rather than by indiscriminately widening its positive predictions, and it is adopted at $t = 0.3$ as the final stressed-state specification.

With this configuration fixed, the gain-based feature importance (Figure 36) and the SHAP summary (Figure 37) indicate that stress is driven principally by yield and interest rate conditions rather than by leverage alone. `ybt_apy` (77.2) ranks first, followed closely by `borrow_rate` (74.3), so that the two components of the carry spread occupy the leading positions. LTV volatility follows (`ltv_vol_7d` at 57.5 and `ltv_vol_14d` at 56.4), ahead of `collateral_price` (52.1). The composite carry measures rank lower than their constituents (`carry_spread` at 43.6 and `carry_spread_levered` at 31.3), which is consistent with the trees recovering the spread directly from the two underlying rate series rather than from the derived variable. `borrow_price` (25.3) ranks lowest, although it is not close to zero. Notably, no single feature dominates in the manner of `ltv` in the leaking specifications, which supports the interpretation of this result as substantive rather than artefactual.

The SHAP summary corroborates the direction of these effects. `ybt_apy` exhibits the widest dispersion, with low values contributing toward a stressed classification and high values contributing away from it, consistent with the proposition that declining collateral yields render a position more vulnerable. `carry_spread_levered` shows the opposite orientation, with high values contributing toward stress, reflecting the fact that the levered spread scales with leverage and is therefore largest for the most highly geared positions. The volatility features exhibit smaller contributions, indicating that they carry some signal but less than the yield and rate features.

7.3 Behavioural Patterns and Stress Events

The results support two substantive behavioural findings, concerning the timing of leverage change and the incidence of stress across the sample.

The timing of leverage change. Loopers do not change leverage immediately upon an increase in risk. Model A indicates that the strongest predictor is the magnitude of leverage movement over the preceding week, with carry economics second, rather than the contemporaneous leverage level. This pattern is consistent with an observation period preceding action rather than an immediate reaction to changing conditions. Separating capital movement from risk management also revises which holders qualify as active managers: L-3 appears to be a frequent manager under the broad activity flag but is close to passive once leverage changes alone are counted.

The clustering results likewise identify four recurring behavioural modes: stale low leverage (conservative and infrequently touched), active looping (moderate leverage with frequent adjustment), maximum-leverage hold (very high leverage and infrequent adjustment), and outlier days (brief transitions between modes). Some holders adhere consistently to a single mode, while others transition between modes over the course of the sample.

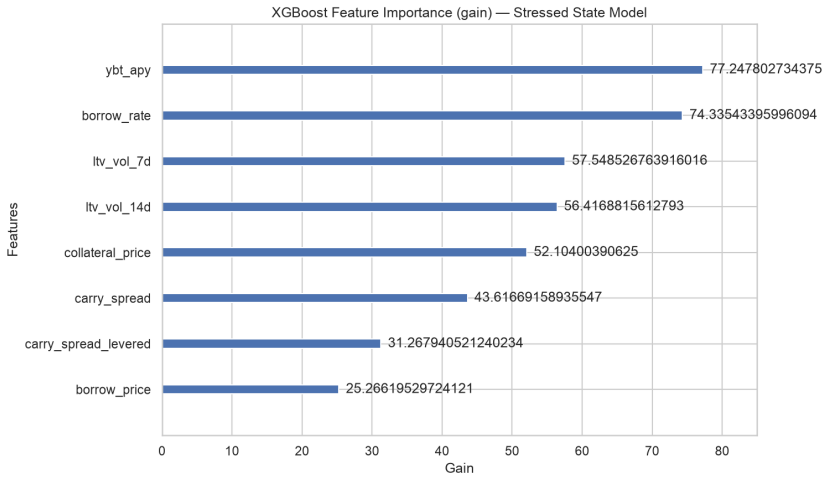


Fig. 36. Gain-based feature importance for the XGBoost stressed-state model. Bars report each feature's total gain summed across all trees. The yield and borrow rate features (`ybt_apy`, `borrow_rate`) rank highest and `borrow_price` ranks lowest. No single feature dominates, and the label-defining features (`health_factor`, `distance_to_liq`), together with the remaining leaking variables, are excluded.



Fig. 37. SHAP summary for the XGBoost stressed-state model. Each point represents a looper-day; its horizontal position gives the feature's signed contribution to the predicted log-odds of being stressed, and colour encodes the feature value (red high, blue low). Features are ordered by median absolute SHAP value. `ybt_apy` exhibits the widest dispersion, with low values contributing toward a stressed classification.

The incidence of stress. A day is classified as stressed where the health factor falls to 1.05 or below, a level proximate to liquidation. In aggregate, 28.9% of days (725 of 2,506) meet this criterion, with pronounced heterogeneity across holders. L-1 and L-12 are stressed on more than 95% of their days, whereas eight loopers are never stressed. Figure 38 makes this concentration explicit: stress is confined to a subset of holders rather than constituting a market-wide condition.

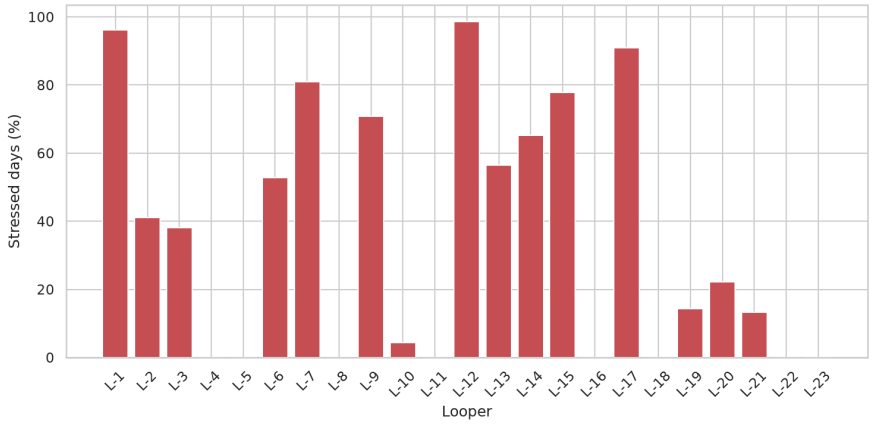


Fig. 38. Share of each looper's active days spent in a stressed state (health factor $\leq$ 1.05), ordered L-1 to L-23. The dashed line marks the overall median of 28.9%. Eight loopers are never stressed, while a small number (L-1, L-12, L-17) are stressed on almost every active day, so that the aggregate stress rate is driven by a small group of persistently high-leverage holders.

Model B indicates that stress is attributable principally to yield and rate conditions rather than to leverage alone. The strongest signals are the collateral token yield (`ybt_apy`) and the borrow rate, implying that a holder enters a stressed state when the yield earned converges toward the borrowing cost, rather than simply when leverage is high.

7.3.1 Findings Considered jointly, the two specifications provide a direct answer to RQ3: on-chain state variables carry predictive information for both risk rebalancing and stress, but through distinct mechanisms.

Leverage change constitutes a delayed response. Loopers do not change leverage at the moment risk increases. The strongest predictor is LTV volatility over the preceding week, with carry economics second, rather than the contemporaneous leverage level. Leverage change therefore has the character of a considered decision rather than a reactive one. Approximately half of what the broad activity

flag recorded as rebalancing consists of capital movement or `minor_activity`, and the carry features emerge as predictors only once these are excluded.

Stress tracks yield and rate conditions. The stressed-state model achieves strong detection performance (ROC-AUC of 0.977 and recall of 0.90 at a threshold of 0.3) using only yield, rate, volatility, and price features, without recourse to direct leverage or structural variables. The dominant predictors are `ybt_apy` and `borrow_rate`, the two components of the carry spread. This suggests that a holder is more likely to occupy a stressed state when the yield earned declines toward the borrowing cost than simply when leverage is high. Stress within this dataset is therefore better characterised as the position ceasing to compensate for its risk than as leverage becoming excessive.

Leverage volatility is common to both models but plays distinct roles. In the risk-rebalance model, `ltv_vol_7d` is the single most important feature. In the stressed-state model it remains informative but ranks below the yield and rate features. Volatility therefore functions primarily as a trigger for action, in the form of leverage change, rather than as an indicator of danger, in the form of stress.

Absence of realised liquidations. No looper in the sample was liquidated over the period studied. The closest approach was that of L-15, whose health factor briefly reached 0.998 on a single day (1 March 2026) without triggering liquidation. Stressed states in these data therefore denote positions that recovered rather than positions that failed, and the risk quantified here is proximity to liquidation rather than realised loss.

Interpretation. Loopers appear to observe leverage movement over time before undertaking a leverage change, whereas their stressed state on any given day is shaped more by yield and rate conditions than by leverage dynamics alone. On-chain data capture both patterns clearly, which indicates that DeFi leverage risk can plausibly be monitored, and to some extent anticipated, in real time using only publicly available blockchain data. This finding bears directly on on-chain risk monitoring and on protocol parameter calibration.

8 Discussion

8.1 Price Stability and the Probability of the Stress Scenario

The stress tests show that a 3 to 5% decline in the sUSDe price would push cumulative liquidated Aave collateral from $40M to over $510M, yet no position in the panel was ever liquidated. The two results are not in tension once the price behaviour of the collateral is considered.

A yield-bearing stablecoin is not pegged to one dollar; its exchange rate accretes upward as yield compounds. What matters for liquidation is therefore the size of realistic *downside* deviations, not stability in absolute terms. Over the year

to September 2026, syrupUSDC rose steadily from approximately $1.12 to $1.18 with no material drawdown, while Bitcoin moved between $125,000 and $59,000 with single-day declines of 5% or more a recurring occurrence (Figure 39). The same holds within our panel: over the life of L-1's position the sUSDe price rose from 1.218 to 1.228, with no intervening fall approaching the stress scenario.

The shock applied in these tests is therefore a tail event for the collateral actually studied. Positions sitting permanently near a health factor of 1.01 survive because the event that would liquidate them is rare, not because they are conservatively managed. This does not make the exposure benign: a move of that size would most likely originate in a failure at the issuer rather than in ordinary market volatility, and the concentration documented means the resulting liquidations would fall on a few very large positions at once. The risk is better characterised as low-probability and high-severity than as absent.

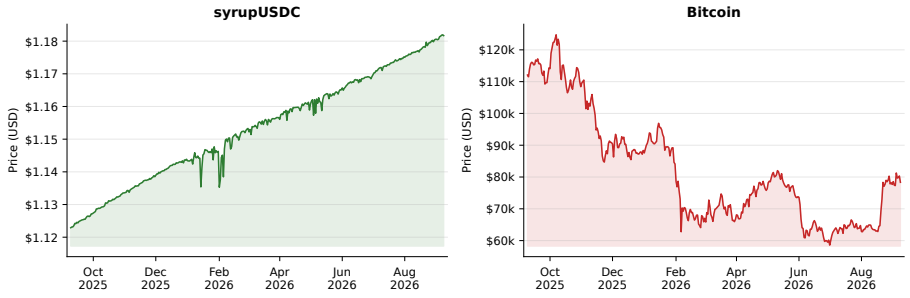


Fig. 39. One-year daily price history of syrupUSDC (left) and Bitcoin (right) in USD, 9 September 2025 to 8 September 2026. Source: CoinGecko API.

8.2 PT Looping: A Related Strategy Variant

During looper identification, several addresses were found to execute a structurally related but analytically distinct strategy using Pendle Principal Tokens (PT-sUSDe) as collateral rather than sUSDe directly. This variant is documented here for completeness and is excluded from the core quantitative analysis.

The Pendle Mechanism. Pendle Finance decomposes a yield-bearing token into two constituent components: a Principal Token (PT), redeemable for the underlying asset at a fixed maturity date, and a Yield Token (YT), which captures the floating yield accruing until maturity. As the PT is stripped of its yield entitlement, it trades at a discount to the underlying asset, thereby implying a fixed annualised return for a holder who retains it to maturity.

The PT Looping Mechanism. A PT looper acquires sUSDe and decomposes it via the Pendle Router into PT-sUSDe and YT-sUSDe. The PT-sUSDe is subsequently supplied as collateral on Morpho Blue, against which a stablecoin

is borrowed; the borrowed stablecoin is then converted back to sUSDe, and the cycle is repeated. The resulting carry spread is given by the PT's implied fixed rate less the borrowing rate, and is consequently fixed at entry rather than floating on a daily basis, in contrast to the mechanism examined in the remainder of this paper.

Rationale for Exclusion. Table 19 summarises the principal differences between the two strategies.

Table 19. Comparison of YBT looping versus PT looping.

Dimension	YBT Looping (this study)	PT Looping (excluded)
Collateral	Floating-rate YBT	Fixed-rate PT
Carry spread	Changes daily	Locked at entry
Price trajectory	Varies with yield	Predetermined accretion to par
Liquidation driver	YBT depeg or rate spike	PT price deviation from fair value
Panel structure	Continuous	Breaks at maturity

The discontinuity introduced by PT maturity constitutes the principal methodological obstacle to inclusion. Each PT expiry necessitates bespoke panel handling that does not arise for floating-rate positions, as a matured PT position cannot be represented within the continuous daily panel employed elsewhere in this study. This requirement, together with the need for Pendle AMM implied-rate data obtained from a separate API, constitutes a non-trivial expansion of scope beyond that of the present work.

Two PT loopers were identified during screening: an institutional-scale holder (`0x1b82b4f4...`), executing transactions of $28–34M in value across only six total transactions, and an iterative looper (`0xbfb77183...`), executing 34 transactions concentrated within a two-day interval. Both addresses are excluded from the core sample but establish that looping extends beyond floating-rate YBTs into fixed-rate Pendle derivatives.

PT looping occupies a distinct position within the risk-return space, in that the holder exchanges floating carry exposure for a predetermined fixed return. This characteristic situates PT looping in closer proximity to fixed-income arbitrage than to the floating-yield strategies examined in this paper, and constitutes a natural direction for future research.

9 Conclusions

Lending against yield-bearing stablecoins is concentrated among a small number of large holders who run leverage close to the liquidation threshold the protocol permits. The proof-of-concept bot established that the mechanism works as described, building 10× leverage from 10 USDC of own capital in a single flashloan-funded transaction on Morpho Blue, with an iterative variant reaching

the same outcome without a smart contract (Section 3). The empirical analysis then applied this to sUSDe, sUSDS, and syrupUSDC across Aave v3 and Morpho Blue, covering both the full holder base (Section 6) and a panel of 23 loopers (Section 4.6).

Allocation is all or nothing (Section 6). Holders either keep the token in the wallet or commit it fully to a lending protocol. Wallet-only holders are 81 to 97% of each token's base and mixed allocation stays below 1% for sUSDS and syrupUSDC. Value-weighting sharpens this: the Aave-only group rises from 14.8% of sUSDe holders to 44% of sUSDe value.

Deployed collateral is invisible in wallet balances (Section 4.5, Section 6.3). A looped position sits inside the protocol, so an active looper's wallet reads near zero, which is why loopers must be identified by behaviour rather than by balance. The protocols hide it in opposite ways: on Morpho every whale position carries an active borrow, whereas on Aave 61.0% of whales and 69.4% of retail holders supply sUSDe and draw no debt at all.

Borrowers lever close to the protocol ceiling (Table 13, Section 6.5). Morpho borrowers cluster at 5× and above across all three tokens, which places most of their collateral in the high-risk health-factor band. On Aave most borrowers sit below 1.1×, yet 90.5% of stablecoin-collateralised sUSDe value ($531.1M of $587.1M) is still critical or high-risk. Risk concentration holds under both designs: whales are 22.0% of at-risk Morpho positions but 93.1% of at-risk value, and 13.1% of at-risk Aave positions but 97.0% of value. Holding under two markedly different lending designs, this looks structural rather than a product of either protocol's mechanics.

The liquidation cliff never triggered (Section 6.6, Section 8.1). Liquidation would arrive abruptly: a 3 to 5% sUSDe decline takes cumulative liquidated Aave collateral from $40M to over $510M, and Morpho syrupUSDC liquidates over $130M within a 12 to 18% band. No liquidation was realised anywhere in the sample, the closest being a health factor of 0.998 (Section 7.1.5). A move of that size is a rare event for an asset engineered to hold its peg, so the risk quantified here is proximity to liquidation, not realised loss.

Four modes, not twenty-three strategies (RQ1 and RQ2; Section 7.1.4, Section 7.1.6). Clustering recovers four recurring modes rather than a strategy per wallet: stale low leverage (1,148 looper-days), active looping (512), maximum-leverage hold (305), and outlier transitions (104). They separate on leverage level and rebalancing cadence, with levered carry a secondary signal. Twelve of 23 holders spend at least 80% of their days in one mode; the rest move between modes, so strategy is better described as switching between shared modes than as a fixed wallet type. Separating risk management from capital movement is what makes this interpretable: activity occurs on 14.8% of looper-days but leverage moves on only 7.2%.

Behaviour and stress are predictable, by different signals (RQ3; Section 7.2.4 and 7.2.5). Risk rebalancing is best predicted by short-window LTV volatility, with carry economics second and current leverage contributing little (ROC-AUC 0.901, recall 0.81), consistent with a considered rather than reflexive response. Stress is predicted almost entirely by yield and rate conditions (ROC-AUC 0.977, recall 0.90 at $t = 0.3$) without any direct leverage variable, so stress is better read as the position ceasing to compensate for its risk than as leverage becoming excessive. Profitability is correspondingly mixed: 3 of 8 Morpho and 4 of 7 Aave markets sampled carry a negative spread.

Flashloans separate the two protocols (Section 6.4). Usage tracks protocol, not holder size. On Aave 18.5% of sUSDe whales have used a flashloan against 4.16% of the holder base; on Morpho none of the 41 whales ever has. Morpho whales build levered positions through ordinary multi-transaction borrowing, while a minority of Aave whales hold a tool that can unwind exposure in one transaction, exactly the price move that would otherwise liquidate them.

Limitations and future work. The panel is modest and uneven: 23 wallets, with a few high-activity holders contributing most days, so archetype and wallet identity remain partially confounded and the risk ordering between archetypes is provisional. With no realised liquidation in the sample, the 1.05 stress threshold is a chosen buffer rather than a calibrated one. Three extensions follow: held-out-looper validation to test generalisation across holders, calibration of the stress threshold against realised liquidations, and treatment of the Pendle Principal Token variant (Section 8.1) as fixed-income arbitrage rather than as an extension of the strategies studied here.

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A Glossary and Addresses

Looper Addresses

ID	Address
L-1	0x080ccea8654fd4ff80b81e46cff2278689d88fb0
L-2	0x97bb0e08b664f65f05e0cff03b68bc1238b2e21e
L-3	0xcb1effb6f6e531137ccc1508a8d4982e873aee58
L-4	0x0d4f7b204626e8233c5b42b1269a78e236e9a06b
L-5	0x24385a793f725328d7f6224430e48b4236326717
L-6	0xfafc2c5f6a0d05718f3e50d0765a79fc69cdde13
L-7	0x32f07401eb177f2c0fc4f95f3928050d88dae7ed
L-8	0x218b5b98805ad5193432999406e17bceb51770aa
L-9	0x309e161a7056610f0af12c608b3a31f0ffbb146c
L-10	0x3363aca0a6bd542d146408246b5d1e3743b977d6
L-11	0xb48ac9c5585e5f3c88c63cf9bcbaedc921f76df2
L-12	0x9bcb2ee24bd11046c0dcdfebd608eeb360a64495
L-13	0x206777f4396817398ee61e7ac5e62af311e79836
L-14	0xc05855cbe2a68f9a28a4d85a3a497ebe3adc9a1c
L-15	0x59b6c75eeb7dfe417505919c3e5a919b7c1b0773
L-16	0xf28dc8b6ded7e45f8cf84b9972487c8e1857a442
L-17	0x029469ca80e6c94436af010a369f54935a711080
L-18	0x5D876B100Dd38D0701725035B97632553eaa1836
L-19	0x5D876B100Dd38D0701725035B97632553eaa1836
L-20	0x852689F44a4CA94a0E424EAd14A1Bb63719314db
L-21	0x24201ECD8b3f062215640B8c0740C4e892c091E3
L-22	0x23270C161c42fFd0636fe786Cc7c518A0181234F
L-23	0xF3AdD38c51924221Dcf30767f69c6E57D77D9566

Wallet to Market Mapping

ID	Protocol	Morpho Market (Collateral/Debt)	Aave Pool / Debt Token
L-1	Morpho	sUSDe/USDtb	–
L-2	Morpho	sUSDS/USDT	–
L-3	Morpho	sUSDS/USDT	–
L-4	Morpho	sUSDe/PYUSD	–
L-5	Morpho	syrupUSDC/PYUSD	–
L-6	Morpho	sUSDS/USDT	–
L-7	Morpho	sUSDe/PYUSD	–
L-8	Morpho	sUSDe/DAI	–
L-9	Morpho	sUSDe/DAI	–
L-10	Morpho	sUSDe/DAI	–
L-11	Morpho	sUSDe/DAI	–
L-12	Morpho	sUSDS/USDT	–
L-13	Morpho	sUSDS/USDT	–
L-14	Morpho	sUSDS/USDT	–
L-15	Morpho	syrupUSDC/USDC	–
L-16	Morpho	syrupUSDC/USDC	–
L-17	Aave v3	–	sUSDe (E-Mode) / USDT
L-18	Aave v3	–	sUSDe (E-Mode) / USDC
L-19	Aave v3	–	sUSDe (E-Mode) / USDe
L-20	Aave v3	–	sUSDe (E-Mode) / USDe
L-21	Aave v3	–	sUSDe (E-Mode) / USDe
L-22	Aave v3	–	sUSDe (E-Mode) / USDC
L-23	Aave v3	–	sUSDe (E-Mode) / USDC

Glossary

Term	Definition
YBT	Yield-Bearing Token. A token that automatically earns yield (e.g., sUSDe, sUSDS, syrupUSDC).
Leverage	Using borrowed funds to increase exposure beyond one's own capital.
Looping	Specific form of leverage (also called recursive leverage): deposit collateral, borrow, buy more collateral, repeat.
LTV	Loan-to-Value ratio. Borrowed amount divided by collateral value.
LLTV	Liquidation LTV. The threshold where a position gets liquidated.
Flashloan	Borrow and repay in the same transaction with no collateral required.
Morpho Blue	A lending protocol with isolated markets for each collateral-borrow pair.
Aave v3	A pooled lending protocol with shared liquidity across assets.
Gnosis Safe	Multi-signature wallet requiring M-of-N signers to approve transactions.
ERC-4626	Standard interface for yield-bearing vaults defining deposit, redeem, and share conversion.
The Graph	Decentralised indexing protocol for querying blockchain events via GraphQL.
Dune Analytics	SQL-based blockchain analytics platform.
Whale	Large holder *by balance*: an address holding at least 1,000,000 YBS tokens (≈$1M) across its wallet, Morpho Blue and Aave v3. Used only in the full holder panel (Section 6).
Looper	Address that actively runs the recursive leverage strategy, identified *by behaviour*: flagged as looper or flashloaner by the heuristics of Section 4.5. The 23 selected loopers L-1 to L-23 of the detailed analysis additionally clear $500k lifetime inflow and a manual review; they are not necessarily whales.
Flashloaner	Address that executed flashloans in at least three separate transactions on a given protocol.
EOA	Externally Owned Account. A wallet controlled by a private key (vs smart contract).
Carry spread	Yield earned minus borrow cost. Positive means profitable loop.
Position activity	Any protocol event on a position that day (supply, withdraw, borrow, repay); informally, a touch.
Risk rebalance	A day on which position leverage moved materially: `leverage_adjust` or `levered_scaling`. Subset of position activity.

B Smart Contract Code

The following is the complete `LeverageBot.sol` contract deployed on Base mainnet.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "./ILeverageBot.sol";

contract LeverageBot is IMorphoFlashLoanCallback {
    address public constant MORPHO = 0xBBBBBbbBBb9cC5e90e3b3Af64bdAF62C37EEFFCb;
    address public constant USDC = 0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913;
    address public constant YO_USD = 0x0000000f2eB9f69274678c76222B35eEc7588a65;

    address public constant MARKET_ORACLE = 0x77f59C5C09e7ABEF6991B456170930324FE36222;
    address public constant MARKET_IRM = 0x46415998764C29aB2a25CbeA6254146D50D22687;
    uint256 public constant MARKET_LLTV = 915000000000000000;
    bytes32 public constant MARKET_ID =
        0x1a3e69d0109bb1be42b80e11034bb6ee98fc466721f26845dc83b2aa8d979137;

    uint256 public constant OWN_CAPITAL = 10_000_000;      // 10 USDC
    uint256 public constant FLASHLOAN_AMOUNT = 90_000_000; // 90 USDC

    uint8 private constant ACTION_LEVERAGE = 1;
    uint8 private constant ACTION_UNWIND = 2;

    address public owner;
    uint256 private _unwindUsdcReceived;

    modifier onlyOwner() {
        require(msg.sender == owner, "LeverageBot: not owner");
        _;
    }

    modifier onlyMorpho() {
        require(msg.sender == MORPHO, "LeverageBot: caller not Morpho");
        _;
    }

    constructor() {
        owner = msg.sender;
    }

    function execute() external onlyOwner {
        IERC20(USDC).transferFrom(msg.sender, address(this), OWN_CAPITAL);
```

```
        bytes memory data = abi.encode(ACTION_LEVERAGE, OWN_CAPITAL, FLASHLOAN_AMOUNT);
        IMorpho(MORPHO).flashLoan(USDC, FLASHLOAN_AMOUNT, data);
    }

    function onMorphoFlashLoan(
        uint256 flashloanAmount,
        bytes calldata data
    ) external override onlyMorpho {
        uint8 action = abi.decode(data, (uint8));
        if (action == ACTION_LEVERAGE) {
            _onLeverage(flashloanAmount, data);
        } else if (action == ACTION_UNWIND) {
            _onUnwind(flashloanAmount, data);
        }
        IERC20(USDC).approve(MORPHO, flashloanAmount);
    }

    function unwind() external onlyOwner {
        Position memory pos = IMorpho(MORPHO).position(MARKET_ID, address(this));
        require(pos.borrowShares > 0 || pos.collateral > 0,
            "LeverageBot: no position");
        uint256 owedAssets = _debtSharesToAssets(pos.borrowShares);
        uint256 flashAmt = (owedAssets * 10100) / 10000 + 1;
        _runUnwindFlashloan(flashAmt, pos);
        uint256 netToOwner = IERC20(USDC).balanceOf(address(this));
        if (netToOwner > 0) {
            IERC20(USDC).transfer(owner, netToOwner);
        }
    }

    function transferOwnership(address newOwner) external onlyOwner {
        require(newOwner != address(0), "LeverageBot: zero owner");
        owner = newOwner;
    }

    function _onLeverage(uint256 flashloanAmount, bytes calldata data) private {
        (, uint256 ownCapital, ) = abi.decode(data, (uint8, uint256, uint256));
        uint256 totalUsdc = ownCapital + flashloanAmount;
        IERC20(USDC).approve(YO_USD, totalUsdc);
        uint256 yoUsdReceived = IERC4626(YO_USD).deposit(totalUsdc, address(this));
        MarketParams memory mp = _getMarketParams();
        IERC20(YO_USD).approve(MORPHO, yoUsdReceived);
        IMorpho(MORPHO).supplyCollateral(mp, yoUsdReceived, address(this), "");
        IMorpho(MORPHO).borrow(mp, flashloanAmount, 0, address(this), address(this));
    }
```

```
    function _onUnwind(uint256 flashloanAmount, bytes calldata data) private {
        (, uint256 borrowShares, uint256 collateral) =
            abi.decode(data, (uint8, uint256, uint256));
        MarketParams memory mp = _getMarketParams();
        IERC20(USDC).approve(MORPHO, flashloanAmount);
        IMorpho(MORPHO).repay(mp, 0, borrowShares, address(this), "");
        IMorpho(MORPHO).withdrawCollateral(mp, collateral, address(this), address(this));
        uint256 yoUsdHeld = IERC20(YO_USD).balanceOf(address(this));
        if (yoUsdHeld > 0) {
            _unwindUsdcReceived = IERC4626(YO_USD).redeem(
                yoUsdHeld, address(this), address(this));
        }
    }

    function _getMarketParams() private pure returns (MarketParams memory) {
        return MarketParams({
            loanToken: USDC,
            collateralToken: YO_USD,
            oracle: MARKET_ORACLE,
            irm: MARKET_IRM,
            lltv: MARKET_LLTV
        });
    }
}
```

The full source code including interfaces and tests is available in the project repository at `bot/contracts/`.

C SQL Queries

The following Dune Analytics queries are used for data collection. Full SQL is available in the project repository.

Sample SQL: Raw Aave v3 Events

Full query available on Dune Analytics: 7847567.

```
/* Aave events: all reserves */
/* Wallet: 0x23270C161c42fFd0636fe786Cc7c518A0181234F */

WITH events AS (
    SELECT 'supply' AS event_type, evt_block_time, evt_block_number, evt_tx_hash, reserve, a
    FROM aave_v3_ethereum.Pool_evt_Supply
    WHERE "onBehalfOf" = 0x23270C161c42fFd0636fe786Cc7c518A0181234F
```

```
    UNION ALL

    SELECT 'withdraw', evt_block_time, evt_block_number, evt_tx_hash, reserve, amount
    FROM aave_v3_ethereum.Pool_evt_Withdraw
    WHERE "user" = 0x23270C161c42fFd0636fe786Cc7c518A0181234F

    UNION ALL

    SELECT 'borrow', evt_block_time, evt_block_number, evt_tx_hash, reserve, amount
    FROM aave_v3_ethereum.Pool_evt_Borrow
    WHERE "onBehalfOf" = 0x23270C161c42fFd0636fe786Cc7c518A0181234F

    UNION ALL

    SELECT 'repay', evt_block_time, evt_block_number, evt_tx_hash, reserve, amount
    FROM aave_v3_ethereum.Pool_evt_Repay
    WHERE "user" = 0x23270C161c42fFd0636fe786Cc7c518A0181234F
)

SELECT
    e.event_type,
    e.evt_block_time AS timestamp,
    e.evt_block_number AS block_number,
    e.evt_tx_hash AS tx_hash,
    e.reserve,
    CAST(e.amount AS DOUBLE) / POWER(10, COALESCE(t.decimals, 18)) AS amount
FROM events e
LEFT JOIN tokens.erc20 t
    ON t.contract_address = e.reserve AND t.blockchain = 'ethereum'
ORDER BY timestamp ASC
```

Sample SQL: Looper Classification Query (sUSDS / Morpho Blue)

```
WITH balances AS (
    SELECT
        t.address,
        SUM(CASE WHEN t.amount > 0 THEN t.amount ELSE 0 END)
            / POWER(10, 18) AS total_received,
        SUM(t.amount) / POWER(10, 18) AS current_balance
    FROM (
        SELECT "to" AS address, CAST(value AS DOUBLE) AS amount
        FROM erc20_ethereum.evt_Transfer
        WHERE contract_address = 0xa3931d71877C0E7a3148CB7Eb4463524FEc27fbD
        UNION ALL
        SELECT "from" AS address, -CAST(value AS DOUBLE) AS amount
```

```
        FROM erc20_ethereum.evt_Transfer
        WHERE contract_address = 0xa3931d71877C0E7a3148CB7Eb4463524FEc27fbD
    ) t
    INNER JOIN ( -- candidate loopers and flashloaners, see below
        SELECT address FROM ( /* ... */ ) candidates
    ) c ON t.address = c.address
    GROUP BY t.address
    HAVING SUM(CASE WHEN t.amount > 0 THEN t.amount ELSE 0 END) > 0
),
loopers AS (
    SELECT s.caller AS address
    FROM morpho_blue_ethereum.MorphoBlue_evt_SupplyCollateral s
    INNER JOIN morpho_blue_ethereum.MorphoBlue_evt_Borrow b
        ON s.caller = b.caller AND b.id IN (/* market ids */)
    WHERE s.id IN (/* market ids */)
    GROUP BY s.caller
    HAVING COUNT(DISTINCT s.evt_tx_hash) >= 1
       AND COUNT(DISTINCT b.evt_tx_hash) >= 3
),
flashloaners AS (
    SELECT caller AS address
    FROM morpho_blue_ethereum.MorphoBlue_evt_FlashLoan
    GROUP BY caller
    HAVING COUNT(DISTINCT evt_tx_hash) >= 3
)
SELECT
    b.address,
    b.current_balance,
    b.total_received,
    ROW_NUMBER() OVER (ORDER BY b.total_received DESC) AS rank,
    CASE WHEN l.address IS NOT NULL THEN true ELSE false END AS is_looper,
    CASE WHEN f.address IS NOT NULL THEN true ELSE false END AS is_flashloaner,
    CASE
        WHEN l.address IS NOT NULL AND f.address IS NOT NULL THEN 'both'
        WHEN l.address IS NOT NULL THEN 'looper'
        WHEN f.address IS NOT NULL THEN 'flashloaner'
    END AS strategy_type
FROM balances b
LEFT JOIN loopers      l ON b.address = l.address
LEFT JOIN flashloaners f ON b.address = f.address
WHERE (l.address IS NOT NULL OR f.address IS NOT NULL)
ORDER BY b.total_received DESC
LIMIT 40
```

Looper Identification Queries

Token	Protocol	Dune Query ID
sUSDe	Aave v3	7349560
sUSDe	Morpho Blue	7344848
sUSDS	Morpho Blue	7367663
syrupUSDC	Morpho Blue	7367806

Liquidation Threshold and LLTV Queries

Description	Looper(s)	Dune Query ID
Aave v3 Ethereum Liquidation Threshold, sUSDe	L-3	7514590
Morpho Blue LLTV by Market ID	L-1, L-4, L-5	7514579

Yield and Interest Rate Queries

Description	Dune Query ID
YBT Depeg Monitor	7359293
Aave sUSDe Supply Rate	7751804
Aave USDe Borrow Rate	7751790
Morpho Blue Daily Borrow Rates (sUSDe/sUSDS/syrupUSDC)	7748824
Aave v3 Ethereum Daily USDT Borrow Rate	7748882
Aave USDC Borrow Rate	7847721

Per-Looper and Token Event Queries

Description	Looper(s)	Dune Query ID
sUSDe Aave v3 Events	–	7359302
YBT Morpho Blue Events	–	7359324
syrupUSDC/PYUSD Morpho Blue Events	L-5	7514403
sUSDS/USDT Morpho Events	L-2	7564610
sUSDS/USDT Morpho Events	L-3	7564654
Aave v3 Events, sUSDe Collateral / USDT Debt	L-6	7524498
sUSDS/USDT Morpho Events	L-6	7839723
Event Query	L-7	7564945
Aave Events, sUSDe / USDC	L-19	7750912
Aave Events, sUSDe / USDe	L-20	7847504
Aave Events, sUSDe / USDe	L-21	7847525
Aave Events, sUSDe / USDC	L-23	7847567
Aave Events, sUSDe/USDC and sUSDe/USDe	L-24	7847582